Biometrika (2024), **xx**, x, *pp*. 1–75
Printed in Great Britain

Supplementary material for "Two-Sample Distribution Tests in High Dimensions via Max-Sliced Wasserstein Distance and Bootstrapping"

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S.1. ALGORITHMS

Recall that both test statistic and bootstrapped statistics are related to the s-sparse max-sliced Wasserstein distances between two discrete distributions. More specifically, for the test statistic, the two discrete distributions are $\mu_1 = \sum_{i=1}^{n_1} \delta_{X_i} n_1^{-1}$ and $\mu_2 = \sum_{i=1}^{n_2} \delta_{Y_i} n_2^{-1}$, and for the bootstrapped statistics, the discrete distributions are $\mu_1 = \sum_{i=1}^{n_1} \delta_{X_i}\pi_i + \sum_{i=n_1+1}^{n} \delta_{Y_{i-n_1}}\pi_i$ and $\mu_2 = \sum_{i=1}^{n_1} \delta_{X_i}\pi_i' + \sum_{i=n_1+1}^{n} \delta_{Y_{i-n_1}}\pi_i'$, where $\pi_i = M_i/(2n_1)$ for $i = 1, \ldots, n_1$, $\pi_{n_1+j} = 1/(2n_2)$ for $j = 1, \ldots, n_2$, and $\pi_i' = 1/(2n_1)$ for $i = 1, \ldots, n_1$, $\pi_{n_1+j}' = M_j'/(2n_2)$ for $j = 1, \ldots, n_2$. In this section, we focus on the computational algorithms for computing these statistics.

To deal with the non-convex ℓ_0 constraint, we propose a sequential ℓ_1 approximation strategy. First, given a sequence of ℓ_1 sparsity parameters $1 \le \lambda_j \le s^{1/2}$, we use Algorithm 2, to be presented later, to compute the ℓ_1-constrained max-sliced Wasserstein distances under $\mathcal{V}_{1,\lambda_j} = \{v \in \mathbb{R}^p : \|v\|_1 \le \lambda_j, \|v\|_2 = 1\}$, $j = 1, \ldots, m$. Then, we select the direction that achieves the largest empirical max-sliced Wasserstein distance among all solutions satisfying the ℓ_0 constraint. Let $\mathcal{S}$ be the set of the coordinates corresponding to the non-zero entries of the selected direction. Then, we recalculate the max-sliced Wasserstein distance between the joint distributions of the variables indexed by $\mathcal{S}$. The recalculated value is treated as the final estimate of the s-sparse max-sliced Wasserstein distance; see Algorithm 1. Moreover, by setting $\mu_1 = \sum_{i=1}^{n_1} \delta_{X_i} n_1^{-1}$ and $\mu_2 = \sum_{i=1}^{n_2} \delta_{Y_i} n_2^{-1}$ in Algorithm 1, we can obtain the optimal $\hat{v}$ in (3).

It remains to develop an algorithm for computing the ℓ_1-constrained max-sliced Wasserstein distance. For this we utilize the following equivalent formula

$$W_1(U_1, U_2) = \int_0^1 |F_1^{-1}(u) - F_2^{-1}(v)| du$$

of $W_1(\cdot, \cdot)$ for univariate distributions, where $F_1(t)$ and $F_2(t)$ are the distribution functions of U_1 and U_2, respectively, and $F_1^{-1}(u) = \inf\{t \in \mathbb{R} : F_1(t) \ge u\}$. This representation with quantile functions facilitates the computation, as illustrated below.

To present the algorithm, we introduce the generic notation for two discrete probability measures on $\mathbb{R}^p$, $\mu_1 = \sum_{i=1}^{m_1} \delta_{U_i} a_i$ and $\mu_2 = \sum_{j=1}^{m_2} \delta_{U_j'} a_j'$. Given a direction v, the corresponding univariate distributions are $\mu_{1,v} = \sum_{i=1}^{m_1} \delta_{U_{v,i}} a_i$ and $\mu_{2,v} = \sum_{j=1}^{m_2} \delta_{U_{v,j}'} a_j'$, where $U_{v,i} = v^\top U_i$ and $U_{v,j}' = v^\top U_j'$. Denote the ordered statistics by $U_{v,(1)} \le \ldots U_{v,(m_1)}$ and $U_{v,(1)}' \le \cdots \le U_{v,(m_2)}'$, and let $b_{v,i} = \sum_{j=1}^{i} a_{v,(j)}$ and $b_{v,i}' = \sum_{j=1}^{i} a_{v,(j)}'$, where $a_{v,(j)}$ and $a_{v,(j)}'$ are the probability mass of $U_{v,(j)}$ and $U_{v,(j)}'$, respectively. The ℓ_1-constrained max-sliced Wasserstein distance corresponds to

$$\max_{v \in \mathcal{V}_{1,\lambda}} \int_0^1 |\mu_{1,v}^{-1}(u) - \mu_{2,v}^{-1}(u)| du, \tag{S1}$$

where $\mu_{1,v}^{-1}(u)$ and $\mu_{2,v}^{-1}(u)$ are quantile functions of $\mu_{1,v}$ and $\mu_{2,v}$, respectively. Let m_v be the number of unique elements in $\{b_{v,1}, \ldots, b_{v,m_1}, b_{v,1}', \ldots, b_{v,m_2}'\}$, and the unique elements are denoted by $\check{b}_{v,1} \le \cdots \le \check{b}_{v,m_v}$. Find the indices $i_{v,k}$ and $j_{v,k}$ such that $\mu_{1,v}^{-1}(\check{b}_{v,k}) = v^\top U_{i_{v,k}}$ and $\mu_{2,v}^{-1}(\check{b}_{v,k}) = v^\top U'_{j_{v,k}}$ for $k = 1, \ldots, m_v$. Consequently,

$$\begin{aligned}\int_0^1 |\mu_{1,v}^{-1}(u) - \mu_{2,v}^{-1}(u)|du &= \sum_{k=1}^{m_v} |\mu_{1,v}^{-1}(\check{b}_{v,k}) - \mu_{2,v}^{-1}(\check{b}_{v,k})|(\check{b}_{v,k} - \check{b}_{v,k-1}) \\ &= \sum_{k=1}^{m_v} |v^\top U_{i_{v,k}} - v^\top U'_{j_{v,k}}|(\check{b}_{v,k} - \check{b}_{v,k-1}),\end{aligned}$$

where $\check{b}_{v,0} = 0$ for all $v \in \mathcal{V}_{1,\lambda}$. The problem (S1) is then equivalently formulated as

$$\max_{v \in \mathcal{V}_{1,\lambda}} h(v), \tag{S2}$$

where $h(v) = \sum_{k=1}^{m_v} |v^\top U_{i_{v,k}} - v^\top U'_{j_{v,k}}|(\check{b}_{v,k} - \check{b}_{v,k-1})$. We use the projected subgradient method to solve (S2), which involves a nontrivial projection onto the intersection of an ℓ_1 ball and the ℓ_2 unit sphere (Liu et al., 2020). The details about the algorithm are presented below. Note that to compute the subgradient, the order statistics $U_{v,(i)}$ and $U'_{v,(j)}$ are readily evaluated by sorting algorithms with $O(n \log n)$ complexity.

We first introduce the notations used in the projection algorithm. For any $v = (v_1, \ldots, v_p)^\top$, v^+ denotes the vector such that $v_j^+ = \max(v_j, 0)$, and $|v| = (|v|_1, \ldots, |v|_p)^\top = (|v_1|, \ldots, |v_p|)^\top$. Let $|v|_{(j)}$ represent the j-th largest value in $|v|$, that is, $|v|_{(1)} \ge \cdots \ge |v|_{(p)}$. For a given ϱ, denote the index set $I_\varrho = \{j : |v_j| \ge \varrho, j = 1, \ldots, p\}$ and $R_\varrho = |I_\varrho|$. Also, let $\bar{v}_\varrho = \sum_{j \in I_\varrho} |v_j|$ and $w_\varrho = \sum_{j \in I_\varrho} |v_j|^2$. Given j, set $\breve{I}_j = I_{|v|_{(j)}}$ and $\breve{R}_j = R_{|v|_{(j)}}$. Then, for $\varrho \in (|v|_{(j+1)}, |v|_{(j)}]$, we have $I_\varrho = \breve{I}_j = \{l : |v_l| \ge |v|_{(j)}, l = 1, \ldots, p\}$. Moreover, given $\lambda \in [1, p^{1/2}]$, define $\phi(\varrho) = \|(|v| - \varrho 1_p)^+\|_1^2 - \lambda^2 \|(|v| - \varrho 1_p)^+\|_2^2$, where 1_p is a vector in $\mathbb{R}^p$ with all elements equal to 1. Given $l < r$, let

$$\dot{\varrho} = r - \frac{l - r}{\phi(l) - \phi(r)}\phi(r) \ \text{ and } \ \ddot{\varrho} = \frac{1}{R_l}\left\{\bar{v}_l - \lambda\left(\frac{R_l w_l - \bar{v}_l^2}{R_l - \lambda^2}\right)^{1/2}\right\}.$$

Let $\mathcal{P}_{\mathcal{V}_{1,\lambda}}(v)$ the projection of v onto $\mathcal{V}_{1,\lambda}$. With these preparations, we provide the projected subgradient descent algorithm in Algorithm 2. The algorithms about projection onto the intersection of the ℓ_1 ball and the ℓ_2 unit sphere are further outlined in Algorithms 3 and 4. For more algorithmic details, see Liu et al. (2020).

The worst-case computational complexity of each iteration in Algorithm 2 is $O(n \log n + np + p^2)$. This includes subgradient calculation with a complexity of $O(n \log n + np)$ and projection with a complexity of $O(p^2)$. Note that the observed complexity is actually $O(n \log n + np)$ in practice, because the complexity for the projection algorithm has the observed complexity $O(p)$ instead of $O(p^2)$ as demonstrated in Liu et al. (2020). When calculating the overall computational cost of the proposed test, the computational complexity $O(n \log n + np)$ need be multiplied by the number B of bootstrap replicates. However, thanks to modern parallel computing platforms, the bootstrap replicates can be executed in parallel to accelerate the computation. In comparison, MMD has a computational complexity of $O(n^2 p)$ but does not require iterations; note that the standardized MMD (Yan & Zhang, 2023; Gao & Shao, 2023) may not require bootstrapping or permutation either. Therefore, MMD is computationally more efficient for small or moderate sample sizes, while our algorithm may scale more efficiently with larger sample sizes.

Remark S.1. As the optimization problem is non-convex, we suggest using different initial values v_0 and selecting the one that leads to the largest D_n to construct the test statistic. To reduce the computational cost, we may also reuse the chosen direction as the initial value for the algorithm in the bootstrapping process.

Algorithm 1. Compute the s-sparse max-sliced Wasserstein distance.

Input: probability measures μ_1 and μ_2, the sparsity parameter s, a sequence $\lambda_1, \dots, \lambda_m$.
for $j = 1, \dots, m$ **do**
  Obtain the direction $\hat{v}_j$ and the corresponding distance $\hat{D}_j$ with the ℓ_1 penalty parameter λ_j by using Algorithm 2.
end for
Find $j_\star$ such that $j_\star = \arg\max_{j:\|\hat{v}_j\|_0 \le s} \hat{D}_j$.
Compute the max-sliced Wasserstein distance $\hat{D}$ and the optimal direction $\hat{v}_\star$ between distributions restricted to the subset $\mathcal{S}$ by setting $\lambda = |\mathcal{S}|^{1/2}$ in (S1) or (S2) and using Algorithm 2, where $\mathcal{S}$ comprises the locations of non-zero values of $\hat{v}_{j_\star}$.
Output $\hat{D}$, $\hat{v}_\star$.

Algorithm 2. Projected subgradient descent for solving (S2).

Input: probability measures μ_1 and μ_2, initial value $v_0 \in \mathcal{V}_{1,\lambda}$, step size γ_0, penalty λ, tolerance ε.
for $t = 0, 1, \dots$ **do**
  Compute the subgradient

$$\partial h(v_t) = \sum_{k=1}^{m_{v_t}} (\check{b}_{v_t,k} - \check{b}_{v_t,k-1}) sign(v^\top U_{i_{v_t,k}} - v^\top U'_{j_{v_t,k}})(U_{i_{v_t,k}} - U'_{j_{v_t,k}}).$$

  Decrease the step size by setting $\gamma_{t+1} = \gamma_t/(t+1)^{1/2}$.
  Update $v_{t+1} = \mathcal{P}_{\mathcal{V}_{1,\lambda}}(v_t + \gamma_{t+1}\partial h(v_t))$ by using Algorithm 3.
  Stop the iteration when $\|v_{t+1} - v_t\|/(1 + \|v_t\|) \le \varepsilon$.
end for
Output v_{t+1}.

Algorithm 3. Projection algorithm.

Input: v, λ.
If $\|v\|_0 \leq s\|v\|_2$, then $\breve{v} = \frac{|v|}{\|v\|_2}$.
Otherwise, compute $\breve{I}_1$ and $\breve{R}_1$.
If $\breve{R}_1 < \lambda^2$, then use Algorithm 4 to obtain ϱ, and set $\breve{v} = \frac{(|v|-\varrho 1_p)^+}{\|(|v|-\varrho 1_p)^+\|_2}$.
If $\breve{R}_1 = \lambda^2$, then $\breve{v}_j = \breve{R}_1^{-1/2}$ for $j \in \breve{I}_1$ and $\breve{v}_j = 0$ for $j \notin \breve{I}_1$.
If $\breve{R}_1 > \lambda^2$, then $\breve{v}_j = 0$ for $j \notin \breve{I}_1$. Pick some $j_\circ \in \breve{I}_1$ and for $j \in \breve{I}_1$ with $j \neq j_\circ$, set

$$\breve{v}_j = \frac{\lambda(\breve{R}_1 - 1) - (\breve{R}_1 - 1)^{1/2}(\breve{R}_1 - \lambda^2)^{1/2}}{\breve{R}_1(\breve{R}_1 - 1)}.$$

For any $j \in \breve{I}_1$ but $j \neq j_\circ$, set

$$\breve{v}_{j_\circ} = \lambda - (\breve{R}_1 - 1)\breve{v}_j.$$

Compute $v = sign(v) \otimes \breve{v}$, where "$\otimes$" denotes the element-wise multiplication.
Output v.

Algorithm 4. Quadratic approximation secant bisection method.

Input: v, λ, tolerance ε, and r and l satisfying $\phi(l) > 0$ and $\phi(r) < 0$.
while $r - l > \varepsilon$ and $|\phi(\varrho)| > \varepsilon$ **do**
  Compute $\dot{\varrho}$, $\phi(\dot{\varrho})$, $\ddot{\varrho}$, $\phi(\ddot{\varrho})$, $\varrho = (\dot{\varrho} + \ddot{\varrho})/2$, $\phi(\varrho)$.
  If there is no $|v_i|$ in $(l, \ddot{\varrho})$, then set $\varrho = \ddot{\varrho}$ and terminate.
  If $\phi(\varrho) > 0$, set $l = \varrho$, $r = \dot{\varrho}$; otherwise, set $l = \ddot{\varrho}$, $r = \varrho$.
end while
Output ϱ.

S.2. Additional Simulation Results

S.2.1. Sizes of the test under different regimes of p and n

We examine the empirical sizes of the proposed under different regimes of p and n by using two null models, namely Gaussian (Model A or B, $\beta = 0$) and Gaussian mixture (Model C or D, $\beta = 0$). We present empirical sizes under various combinations of $p = 60, 200, 500$ and $n = 100, 250$ in Table S1. The empirical sizes of the proposed method are fairly close to the nominal level $\alpha = 0.05$, indicating a successful control of the type I error across all examined situations. Other methods with permutation also lead to valid tests.

S.2.2. Effect of the sparsity tuning parameter

We investigate the effect of the sparsity tuning parameter s on the size and power performance. For this purpose, we record empirical sizes or powers under different models with various choices of s, i.e., $s \in \{1, 5, 10, 20, 50, 100, 200, 500\}$. As shown in Figure S1, although Theorems 1 and 2 suggest that s should not be very large to guarantee approximation accuracy, the type I errors

Table S1. *The empirical sizes under different model settings*

Model	n_1 (n_2)	p	Proposed	MMD-G	MMD-L	ED$_2$	BG
Gaussian	100	60	0.063	0.053	0.057	0.053	0.050
		200	0.057	0.040	0.043	0.043	0.053
		500	0.063	0.050	0.050	0.057	0.033
	250	60	0.067	0.047	0.043	0.037	0.057
		200	0.067	0.070	0.070	0.070	0.077
		500	0.043	0.067	0.067	0.063	0.077
Gaussian mixture	100	60	0.073	0.053	0.057	0.060	0.057
		200	0.057	0.037	0.037	0.037	0.050
		500	0.067	0.057	0.060	0.057	0.050
	250	60	0.060	0.033	0.037	0.037	0.043
		200	0.071	0.057	0.053	0.057	0.060
		500	0.060	0.033	0.033	0.033	0.027

n_1 (n_2), sample sizes; p, dimension.

remain well controlled even for relatively large s. It indicates that the proposed method remains valid over a wide range of sparsity parameters. Regarding power performance, we present the results under Models A-D with $\beta = 0.6$ in Figure S2 for illustration; the results with other values of β are similar and thus not presented. As the signal of the alternative hypotheses in Models A and B is fast decaying and thus relatively sparse, the proposed method achieves the highest power when s is small, e.g., when $s = 1$. For Models C and D, where the distributional differences are relatively dense (especially for Model D), a moderately large value of s leads to optimal performance. In all of these models, an excessive large s has negative impact on the power and should be avoided.

In summary, if the tuning parameter s is too small, it may not be able to capture sufficiently large distributional discrepancies. Conversely, an overly large s introduces additional estimation variability that compromises power performance. Thus, it is crucial to select an appropriate value for the sparsity parameter s, for which we have proposed a cross-validation procedure in the main text. Our numerical results suggest that the proposed procedure is a reliable tool to select an appropriate value of s yielding satisfactory performance.

S.2.3. Effect of the sparsity level in the underlying distributional differences

In this section, we investigate the performance of the proposed method across various sparsity levels in the underlying differences between distributions. To this end, we consider two additional models, Models A2 and B2, with discrepancies lying in mean and variance, respectively.

- Model A2. Let $X \sim N(0_p, \Sigma)$ and $Y \sim N(\mu, \Sigma)$, where $\Sigma = (r_{ij})_{i,j=1}^p$ with $r_{ij} = 0.5^{|i-j|}$, and $\mu = (\mu_1, \ldots, \mu_p)^\top$ with $\mu_j = 0.8 s_0^{-0.5}$ for $j = 1, \ldots, s_0$ and $\mu_j = 0$ for $j = s_0 + 1, \ldots, p$.
- Model B2. Let $X \sim N(0_p, \Sigma)$ and $Y \sim N(0_p, \Sigma')$, where Σ is defined in Model A and Σ' is defined by replacing the diagonal entries of Σ with $\Sigma'_{jj} = 1 + 8 s_0^{-0.5}$ for $j = 1, \ldots, s_0$ and $\Sigma'_{jj} = 1$ for $j = s_0 + 1, \ldots, p$.

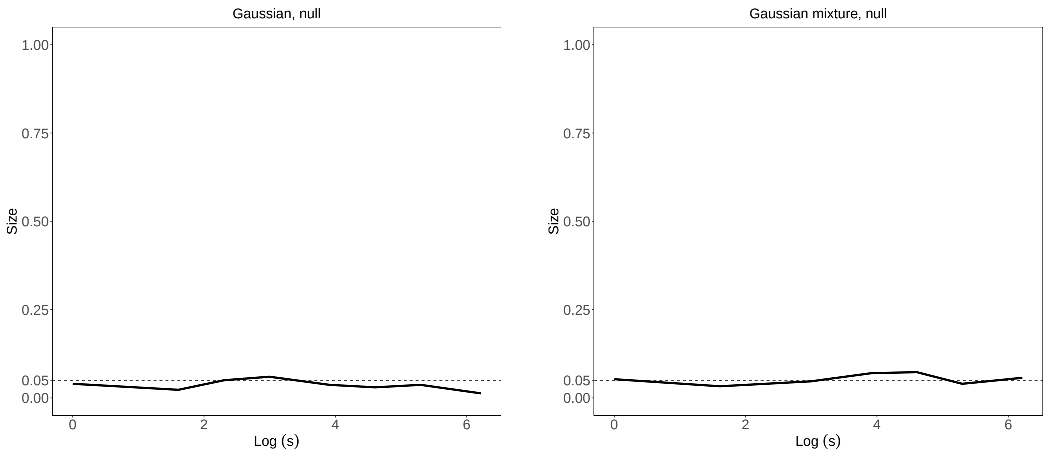

Fig. S1. The size of the proposed method with different sparsity parameters s under $n_1 = n_2 = 250, p = 500$. The Gaussian null model corresponds to $\beta = 0$ in Model A or B, while the Gaussian mixture null model corresponds to $\beta = 0$ in Model C or D.

In these models, the quantity s_0 signifies the number of coordinates on which the distributions are different, depicting the sparsity level of the model. The larger s_0 is, the denser the model becomes. We consider various values between $s_0 = 1$ and $s_0 = p$, where $s_0 = p$ corresponds to the densest models. To reveal the effect of the sparsity level, the signal strength, quantified by $\|\mu\|_2$ for mean differences and by $\|\Sigma - \Sigma'\|_F$ for variance differences, stays constant across varying s_0.

As shown in Tables S2 and S3, the proposed method exhibits significant advantages against sparse alternatives, which is consistent with theoretical findings. In comparison, the methods MMD-G, MMD-L, ED_2 and SWD (sliced Wasserstein distance) yield relatively robust performance under Model A2. In Model B2, these methods have limited power when the model is sparse, i.e., when $s_0 = 1, 5, 10$, and begin to show their strengths when the model becomes dense. The findings align with the expectation that dense models favor MMD-G, MMD-L, ED_2 and SWD, as they average differences over a range of directions. The method BG is capable of detecting variance differences, while showing limitations for identifying mean disparities. It is evident that no single method excels universally across all scenarios. The proposed method is well-suited for applications where the signal is likely sparse, given the prevalence of the sparsity principle in high-dimensional contexts. However, if there is evidence or prior knowledge indicating a dense and uniform signal across all directions, average-based methods such as MMD and SWD could serve as viable alternatives.

As revealed by Yan & Zhang (2023); Gao & Shao (2023); Zhu & Shao (2021), the kernel and interpoint distance based methods, such as MMD-G, MMD-L, ED_2, mainly target the discrepancy lying on the means and the traces of covariance matrices. It means that these methods may encounter power loss if the sum of componentwise variances is the same. To demonstrate

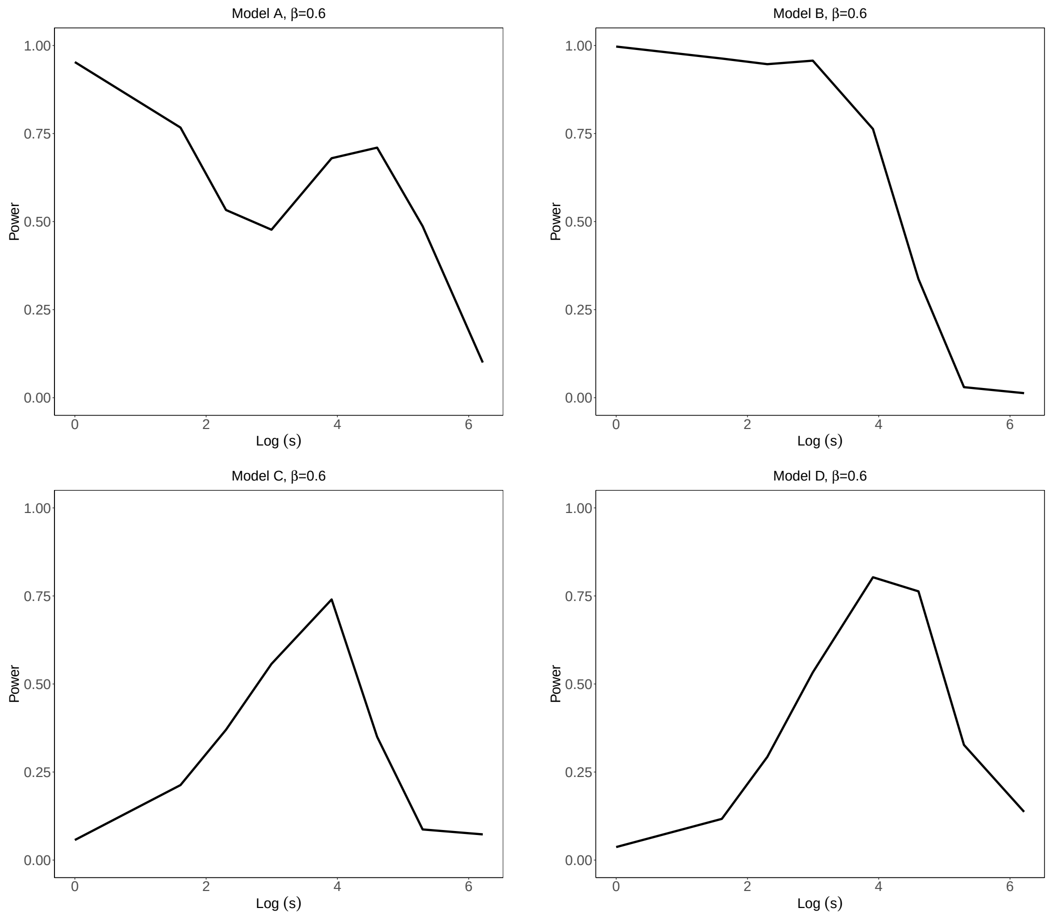

Fig. S2. The power performance of the proposed method using different sparsity parameters s under various models with $n_1 = n_2 = 250, p = 500$. Here Log denotes the natural logarithm with base e.

the performance of the proposed method in such scenarios, we consider Model B3, where the variances differ but their sum is identical.

- Model B3. Let $X \sim N(0_p, \Sigma)$ and $Y \sim N(0_p, \Sigma')$, where $\Sigma_{ij} = \Sigma'_{ij} = 0.5^{|i-j|}$ for $1 \leq i, j \leq p$. Then we replace the diagonal entries with $\Sigma_{jj} = 1 + 8s_0^{-0.5}$ for $j = \lfloor s_0/2 \rfloor + 1, \ldots, s_0$, and $\Sigma'_{jj} = 1 + 8s_0^{-0.5}$ for $j = 1, \ldots, \lfloor s_0/2 \rfloor$.

The results in Table S4 suggest that, the proposed method remains powerful against relatively sparse alternatives. In contrast, all other methods have difficulty in detecting the distributional differences regardless of the sparsity level, which shows the limitations of these methods in this context and is consistent with the discoveries of Yan & Zhang (2023); Gao & Shao (2023); Zhu & Shao (2021) about the kernel and interpoint distance based methods.

Table S2. *The empirical powers under Model A2 with* $n_1 = n_2 = 250, p = 500$ *across different* s_0

	Proposed	MMD-G	MMD-L	ED_2	BG	SWD
$s_0 = 1$	1.000	0.553	0.557	0.560	0.033	0.380
$s_0 = 5$	0.877	0.530	0.530	0.537	0.083	0.433
$s_0 = 10$	0.657	0.587	0.590	0.607	0.050	0.420
$s_0 = 30$	0.477	0.587	0.583	0.590	0.063	0.353
$s_0 = 60$	0.380	0.567	0.567	0.573	0.047	0.383
$s_0 = 120$	0.247	0.530	0.530	0.533	0.067	0.433
$s_0 = 240$	0.210	0.563	0.563	0.560	0.070	0.457
$s_0 = 500$	0.267	0.530	0.530	0.527	0.057	0.427

Table S3. *The empirical powers under Model B2 with* $n_1 = n_2 = 250, p = 500$ *across different* s_0

	Proposed	MMD-G	MMD-L	ED_2	BG	SWD
$s_0 = 1$	1.000	0.063	0.060	0.050	0.543	0.053
$s_0 = 5$	0.987	0.183	0.180	0.117	1.000	0.107
$s_0 = 10$	0.917	0.300	0.297	0.117	1.000	0.167
$s_0 = 30$	0.747	0.853	0.853	0.430	1.000	0.423
$s_0 = 60$	0.703	1.000	1.000	0.837	1.000	0.840
$s_0 = 120$	0.570	1.000	1.000	1.000	1.000	1.000
$s_0 = 240$	0.193	1.000	1.000	1.000	1.000	1.000
$s_0 = 500$	0.083	1.000	1.000	1.000	1.000	1.000

Table S4. *The empirical powers under Model B3 with* $n_1 = n_2 = 250, p = 500$ *across different* s_0

	Proposed	MMD-G	MMD-L	ED_2	BG	SWD
$s_0 = 2$	1.000	0.057	0.057	0.050	0.067	0.063
$s_0 = 4$	0.987	0.057	0.057	0.057	0.047	0.070
$s_0 = 10$	0.907	0.063	0.063	0.060	0.053	0.067
$s_0 = 30$	0.750	0.080	0.080	0.077	0.073	0.077
$s_0 = 60$	0.730	0.073	0.073	0.060	0.037	0.073
$s_0 = 120$	0.627	0.070	0.070	0.057	0.050	0.063
$s_0 = 240$	0.220	0.053	0.053	0.053	0.063	0.057
$s_0 = 500$	0.080	0.083	0.083	0.073	0.030	0.063

S.2.4. Performance under high-order moment discrepancies

In this section, we aim to highlight the efficacy of the proposed method for detecting high-order moment discrepancies, by using the following model adapted from Yan & Zhang (2023).

- Model E. Let $X \sim N(1_p, I_p)$ and $Y_j \overset{i.i.d.}{\sim} \text{Poisson}(1)$ for $j = 1, \ldots, s_0$ and $Y_j \overset{i.i.d.}{\sim} N(1, 1)$ for $j = s_0 + 1, \ldots, p$. Set $s_0 = \lfloor 100\beta \rfloor$.

In this model, the difference between distributions of X and Y appears at the third-order moment. We examine several signal levels, represented by β. The results in Table S5 show that the proposed method indeed exhibits significant advantages in such challenging scenarios, providing numeric support for our claim about the efficacy of the proposed method in identifying high-order moment discrepancies. By contrast, the other methods are unable to detect such distributional differences.

Table S5. *The empirical sizes* $(\beta = 0)$ *and powers* $\beta > 0$ *under Model E with* $n_1 = n_2 = 250, p = 500$

	Proposed	MMD-G	MMD-L	ED_2	BG	SWD
$\beta = 0$	0.050	0.057	0.057	0.053	0.037	0.030
$\beta = 0.2$	0.707	0.053	0.053	0.053	0.057	0.053
$\beta = 0.4$	0.863	0.043	0.043	0.043	0.073	0.050
$\beta = 0.6$	0.913	0.050	0.050	0.047	0.027	0.027
$\beta = 0.8$	0.937	0.080	0.080	0.080	0.057	0.073
$\beta = 1.0$	0.923	0.053	0.053	0.053	0.033	0.047

S.2.5. Simultaneous confidence intervals

Beyond testing the global null hypothesis, practitioners may also be interested in investigating if there are significant distributional differences along some meaningful directions. As discussed in Sections S.4.1 and S.4.3, the SCIs can provide such additional testing information, including identifying significant directions and marginal coordinates. Furthermore, the SCIs are valuable not only for hypothesis testing but also in their own right, as they provide quantitative information about the separation of projected distributions, which is often of practical interest in various applications.

For illustration, Figure S3 presents the one-sided lower bounds for simultaneous confidence intervals of Wasserstein distances across p univariate marginal distributions, i.e., $W_1(v^\top X, v^\top Y)$ with v chosen from the canonical unit vectors $e_1, \ldots, e_p$. Results are provided under two null models, as well as Models A and B, for brevity. Under the null models, the lower bounds between univariate marginal distributions fall below zero, suggesting no significant differences among these univariate marginal distributions. In Model A, a significant difference is observed for the distribution of the first coordinate. In Model B, both the first and second coordinates show significant distributional differences, with a larger Wasserstein distance observed between the distributions of the first coordinate. Therefore, the simultaneous confidence intervals not only help test for significant directional differences but also offer quantitative insights into the degree of distributional separation.

S.3. PROOFS OF THEOREMS 1, 2 AND 4

We prove Theorems 1 and 2 via establishing Berry–Esseen type bounds on the Gaussian and bootstrap approximations, namely, in Equations (S20) and (S26); these bounds directly imply the convergence stated in the theorems. It is noteworthy that the Lipschitz class is not VC-type; the metric entropy of the Lipschitz class $\mathcal{F}_n$ under consideration is of the order $1/\epsilon$, rather than $\log(1/\epsilon)$ that is usually encountered in the literature. Consequently, techniques and results for VC-type classes, such as Corollary 2.2 of Chernozhukov et al. (2014) and the theories of Chernozhukov et al. (2016), cannot be directly employed. This often poses challenges in the theoretical analysis and contributes to a relatively slow rate in n. For instance, in a bound

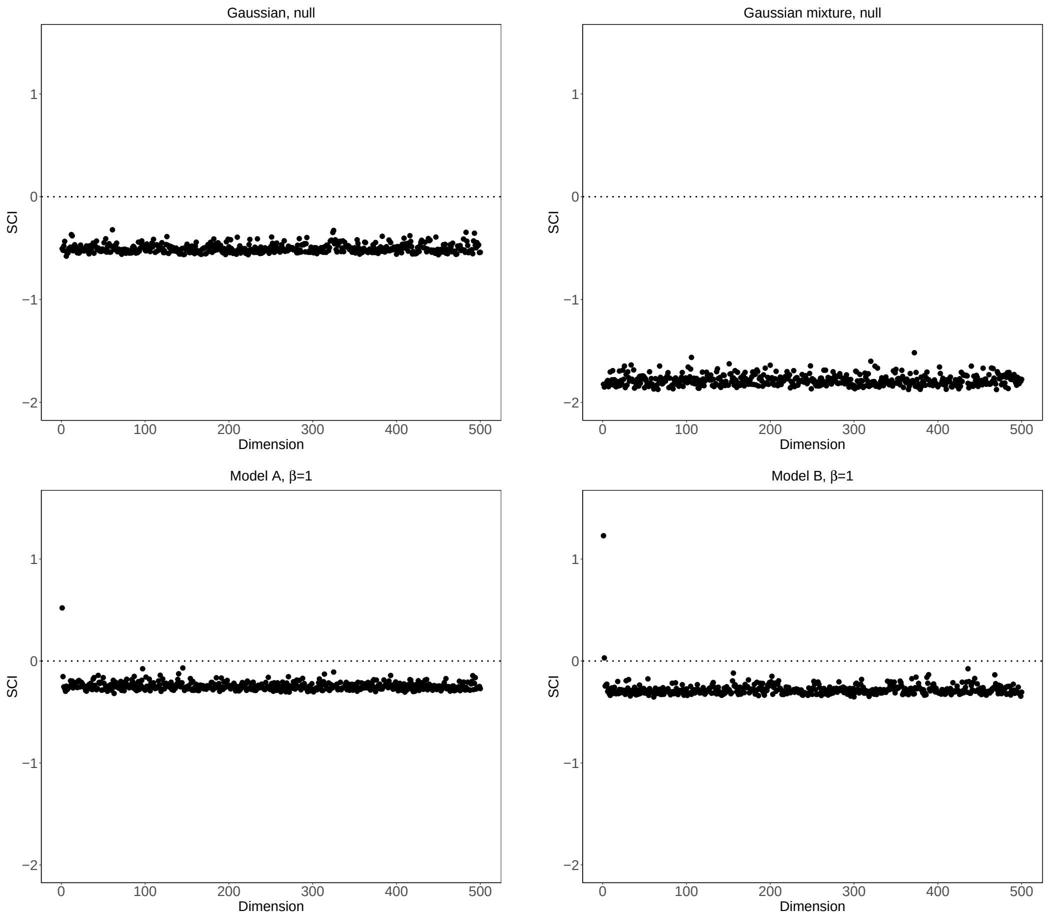

Fig. S3. One-sided lower bounds for simultaneous confidence intervals of Wasserstein distances across p univariate marginal distributions with $n_1 = n_2 = 250, p = 500$.

from Theorem 3 of Sadhu et al. (2022) for bootstrapping the one-sample smoothed Wasserstein distance, the rate in n is at best $n^{-1/12}$.

S.3.1. Preliminaries on proofs under the ℓ_0 sparsity constraint

Notation and Convention. Unless otherwise stated, $c, c_0, c_1, \ldots$ denote positive constants that may depend on the constants (e.g., ν_1, ν_2) in Assumptions 1–3, but not on p or n. In addition, we allow the value of c to vary from place to place.

In addition to the notations introduced in Section 3, we will use the following notations. For a deterministic vector $v = (v_1, \ldots, v_p) \in \mathbb{R}^p$, the ℓ_q norm is $\|v\|_q = \left(\sum_{j=1}^p |v_j|^q\right)^{1/q}$ for $q > 0$ and $\|v\|_\infty = \max_{j=1,\ldots,p} |v_j|$. Similarly, $\|W\|_\infty = \max_{j,k} |w_{jk}|$ for a matrix W of elements w_{jk}. For a real-valued random variable U, let $\|U\|_q = \{E(|U|^q)\}^{1/q}$ for $1 \le q < \infty$. For a random variable U following a probability distribution m, write $E_m(U) = \int U dm$. In addition, we write $\hat{E}_n$ for the empirical expectation, for example, $\hat{E}_n\{f(X)\} = n^{-1}\sum_{i=1}^n f(X_i)$. Let P_n be the empirical distribution that assigns probability n_1^{-1} to each X_i; Q_n is defined in a similar fashion. For a set $\mathcal{A}$, $|\mathcal{A}|$ denotes the cardinality of $\mathcal{A}$. Recall that, for a general probability measure m on $\mathbb{R}^p$, we write $e_m(f,g) = [\int_{\mathbb{R}^p}\{f(x)-g(x)\}^2 dm(x)]^{1/2}$; note that $\|g\|_{m,2} = e_m(g,0)$. In the proofs, we will use e_P, e_Q, e_{P_n}, e_{Q_n} and $e_{(P+Q)/2}$.

Recall that the empirical process $\mathbb{G}_n$ indexed by g is defined by

$$\mathbb{G}_n g = \left(\frac{n_1 n_2}{n_1+n_2}\right)^{1/2}\left(\frac{1}{n_1}\sum_{i=1}^{n_1}[g(X_i) - E\{g(X)\}] - \frac{1}{n_2}\sum_{j=1}^{n_2}[g(Y_j) - E\{g(Y)\}]\right).$$

Recall that Lip_1 denotes the collection of 1-Lipschitz functions. Also recall that

$$\mathcal{F}_n = \{f \in Lip_1 : f(0) = 0, f(x) = f(-\kappa_n) \text{ if } x < -\kappa_n, f(x) = f(\kappa_n) \text{ if } x > \kappa_n\},$$

$$\mathcal{V}_{0,s} = \{v : \|v\|_2 = 1, \|v\|_0 \le s\},$$

and

$$\mathcal{G}_n = \{f \circ v : f \in \mathcal{F}_n, v \in \mathcal{V}_{0,s}\}.$$

The event Ω_n. Define the event

$$\Omega_n = \{|X_{ij}| \le \kappa_n/s^{1/2} \text{ for all } i = 1,\dots,n_1 \text{ and } |Y_{ij}| \le \kappa_n/s^{1/2} \text{ for all } i = 1,\dots,n_2\}. \quad \text{(S3)}$$

Under Ω_n, $\sup_{v\in\mathcal{V}_{0,s}} |v^\top X_i| \le \sup_{v\in\mathcal{V}_{0,s}} \|v\|_1 \max_{1\le j\le p}|X_{ij}| \le \kappa_n$ and $\sup_{v\in\mathcal{V}_{0,s}} |v^\top Y_i| \le \kappa_n$.

The Processes $\mathbb{Z}$, $\mathbb{X}_{n_1}$, $\mathbb{X}^\star_{n_1}$, $\mathbb{Y}_{n_2}$, $\mathbb{Y}^\star_{n_2}$, $\mathbb{V}^\star_n$ and $\mathbb{Z}^\star_n$. Recall the centered Gaussian process $\mathbb{Z}$ indexed by $g \in \mathcal{G}_n$ with the covariance

$$E\{\mathbb{Z}(f)\mathbb{Z}(g)\} = (n_1+n_2)^{-1}[n_2 \mathrm{cov}\{f(X), g(X)\} + n_1 \mathrm{cov}\{f(Y), g(Y)\}].$$

Let $\xi_1,\dots,\xi_n$ be independent standard Gaussian random variables independent of everything else. Define the multiplier bootstrap process,

$$\mathbb{Z}^\star_n g = \frac{1}{n^{1/2}}\left(\sum_{i=1}^{n_1}(n_2/n_1)^{1/2}\xi_i[g(X_i) - \hat{E}_{n_1}\{g(X)\}] - \sum_{i=1}^{n_2}(n_1/n_2)^{1/2}\xi_{n_1+i}[g(Y_i) - \hat{E}_{n_2}\{g(Y)\}]\right) \quad \text{(S4)}$$

indexed by $g \in \mathcal{G}_n$. In the proofs the following processes are often involved:

$$\mathbb{X}_{n_1}(g) = n_1^{-1/2}\sum_{i=1}^{n_1}[g(X_i) - E\{g(X)\}] \quad \text{and} \quad \mathbb{Y}_{n_2}(g) = n_2^{-1/2}\sum_{j=1}^{n_2}[g(Y_j) - E\{g(Y)\}]. \quad \text{(S5)}$$

$$\mathbb{X}^\star_{n_1}(g) = n_1^{-1/2}\sum_{i=1}^{n_1}[g(X^\star_i) - \hat{E}_{n_1}\{g(X_i)\}] \quad \text{and} \quad \mathbb{Y}^\star_{n_2}(g) = n_2^{-1/2}\sum_{j=1}^{n_2}[g(Y^\star_j) - \hat{E}_{n_2}\{g(Y)\}], \quad \text{(S6)}$$

$$\mathbb{V}_n^\star g = \frac{1}{n^{1/2}}\left(\sum_{i=1}^{n_1}(n_2/n_1)^{1/2}[g(X_i^\star) - \hat{E}_{n_1}\{g(X)\}] - \sum_{i=1}^{n_2}(n_1/n_2)^{1/2}[g(Y_i^\star) - \hat{E}_{n_2}\{g(Y)\}]\right), \tag{S7}$$

for $g \in \mathcal{G}$, where $X_1^\star, \ldots, X_{n_1}^\star$ are i.i.d. drawn from the empirical distribution of $X_1, \ldots, X_{n_1}$, and $Y_1^\star, \ldots, Y_{n_2}^\star$ are i.i.d. drawn from the empirical distribution of $Y_1, \ldots, Y_{n_2}$, $\hat{E}_{n_1}\{g(X)\} = n_1^{-1}\sum_{i=1}^{n_1} g(X_i)$ and $\hat{E}_{n_2}\{g(Y)\} = n_2^{-1}\sum_{i=1}^{n_2} g(Y_i)$.

Discretize $\mathbb{G}_n$ and $\mathbb{Z}$. To establish the theorems we discretize the empirical process $\mathbb{G}_n$ and the Gaussian processes $\mathbb{Z}$. For $\epsilon > 0$, let

$$\mathcal{I} = \{g_1, \ldots, g_N\} \text{ be an } \epsilon\|G\|_{(P+Q)/2,2}\text{-net of } \mathcal{G}_n \text{ under the metric } e_{(P+Q)/2}, \tag{S8}$$

which is given in Lemma S1. Here, N is the cardinality of $\mathcal{I}$ and is dependent on ϵ. Define

$$\mathcal{G}_\epsilon = \left\{f - g : f, g \in \mathcal{G}_n, e_{(P+Q)/2}(f,g) \leq \epsilon\|G\|_{(P+Q)/2,2}\right\}. \tag{S9}$$

Note that

$$0 \leq \sup_{g\in\mathcal{G}_n} \mathbb{G}_n g - \max_{j=1,\ldots,N} \mathbb{G}_n g_j \leq \|\mathbb{G}_n\|_{\mathcal{G}_\epsilon},$$

$$0 \leq \sup_{g\in\mathcal{G}_n} \mathbb{Z} g - \max_{j=1,\ldots,N} \mathbb{Z} g_j \leq \|\mathbb{Z}\|_{\mathcal{G}_\epsilon}.$$

Together with a simple calculation, these imply

$$\begin{aligned}
&\sup_{t\in\mathbb{R}}\left|\mathrm{pr}(\mathbb{G}_n\mathcal{G}_n \leq t) - \mathrm{pr}(\mathbb{Z}\mathcal{G}_n \leq t)\right| \\
\leq& 3\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\big(\mathbb{G}_n\mathcal{I} \leq t\big) - \mathrm{pr}\big(\mathbb{Z}\mathcal{I} \leq t\big)\right| + 2\sup_{t\in\mathbb{R}}\mathrm{pr}\big(\left|\mathbb{Z}\mathcal{I} - t\right| \leq \eta\big) \\
&+ \mathrm{pr}\big(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon} \geq \eta\big) + \mathrm{pr}\big(\|\mathbb{G}_n\|_{\mathcal{G}_\epsilon} \geq \eta\big).
\end{aligned}$$

If there were a lower bound on the variance $\mathrm{var}(\mathbb{Z}g)$ for $g \in \mathcal{I}$, then we could use anti-concentration inequalities to bound the first two terms. Since such a lower bound does not exist, we further divide $\mathcal{I}$ into two parts, namely,

$$\mathcal{J} = \{g \in \mathcal{I} : E\{(\mathbb{Z}g)^2\} \geq \tau_n^2\} \tag{S10}$$

and the complement $\mathcal{J}^c$ of $\mathcal{J}$ in $\mathcal{I}$, where we recall $\tau_n \asymp \log^{-2}(pn)s^{-1/2}$. Let

$$r = |\mathcal{J}| \leq N. \tag{S11}$$

Some influential constants/rates. Let $\sigma^2 = \max\left\{\sup_{g\in\mathcal{G}_n} E\{g^2(X)\}, \sup_{g\in\mathcal{G}_n} E\{g^2(Y)\}\right\} \leq c$. By Dudley's maximal inequality, Lemma S5 and Lemma S23, we have

$$\begin{aligned}
E(\mathbb{Z}\mathcal{G}_n) &\leq c\int_0^\sigma \{s\log(ep/s) + s\log(1+8\nu/\epsilon)\}^{1/2}d\epsilon + c\int_0^\sigma \kappa_n^{1/2}\epsilon^{-1/2}d\epsilon \\
&\leq c\kappa_n^{1/2} + c\{s\log(ep/s)\}^{1/2} = A_n.
\end{aligned} \tag{S12}$$

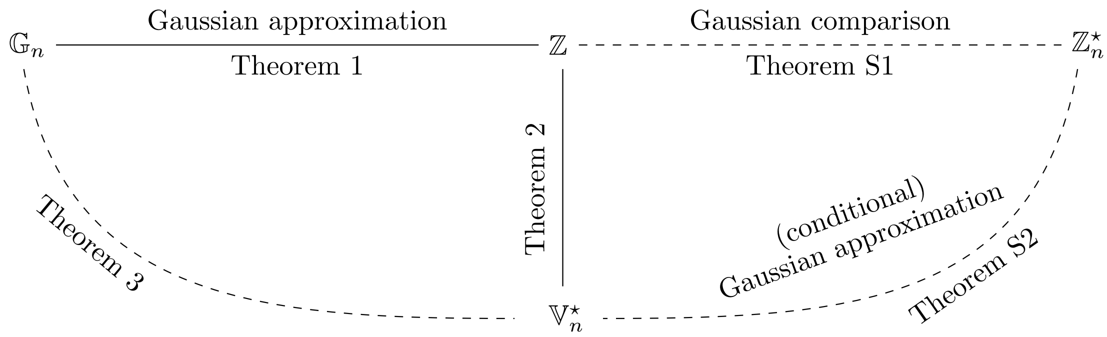

Fig. S4. Overview of the proofs.

Define

$$B_n \asymp \kappa_n + n^{-1/2}\kappa_n^4, \tag{S13}$$

and

$$J(\varepsilon) = c\int_0^\varepsilon \{1 + s\log(ep/s) + s\log\big(1+4/\delta\big) + 1/\delta\}^{1/2} d\delta. \tag{S14}$$

Overview of the proofs. Figure S4 provides an overview of the proofs, where the solid lines represent results provided in the main paper. The quantity $\mathbb{Z}_n^\star$ is introduced to bridge $\mathbb{Z}$ and $\mathbb{V}_n^\star$ via Theorems S1 and S2.

S.3.2. Proof of Theorem 1

Proof. Recall that $\kappa_n = 2s^{1/2}\nu\log(pn)$. We first claim

$$d_K\big(\mathcal{L}(\mathbb{G}_n\mathcal{G}), \mathcal{L}(\mathbb{Z}\mathcal{G}_n)\big) \le c\left[\frac{\eta}{\tau_n^2}\{A_n + 1 \vee \log^{1/2}(\tau_n/\eta)\} + \left(\frac{A_n + B_n\tau_n}{n^{1/4}\tau_n^2}\right)(\log r)^{3/4} + \frac{1}{n}\right], \tag{S15}$$

with the conditions

$$\eta \ge c_3[\epsilon\kappa_n s^{1/2}\log^{1/2}(pn) + \epsilon^{1/2}\kappa_n + \kappa_n n^{-1/2}\{\epsilon^{-1} + s\log(pn)\}], \tag{S16}$$

$$\tau_n \le c_3^{-1}\log^{-2}(pn)s^{-1/2}, \tag{S17}$$

$$\tau_n \ge c_3\{n^{-1/3}\kappa_n + n^{-1/2}\kappa_n s^{1/2}\log^{1/2}(pn)\}, \tag{S18}$$

$$\log r \le c_3^{-1} n\kappa_n^{-4},$$

$$\epsilon\kappa_n \le c_3^{-1}(\log n)^{-1}, \tag{S19}$$

for some sufficiently large constant $c_3 > 0$. If we take $\tau_n = c\log^{-2}(pn)s^{-1/2}$ and $\eta = c\epsilon^{1/2}\kappa_n$, and choose ϵ so that $cn^{-1/3} \le \epsilon \le cs^{-1}\log^{-2}(pn)$, then under the assumption $s^3 \le cn\{\log(pn)\}^{-9}$, all the above conditions can be satisfied. In this case, the rate of (S15) becomes

$$d_K\big(\mathcal{L}(\mathbb{G}_n\mathcal{G}), \mathcal{L}(\mathbb{Z}\mathcal{G}_n)\big) \le c\frac{\{s\log(pn)\}^{1/2}}{\tau_n^2}\left(\epsilon^{1/2}\kappa_n + n^{-1/4}\epsilon^{-3/4}\right).$$

Taking $\epsilon \asymp (n\kappa_n^4)^{-1/5}$ yields the rate

$$d_K\big(\mathcal{L}(\mathbb{G}_n\mathcal{G}), \mathcal{L}(\mathbb{Z}\mathcal{G}_n)\big) \le cs^{9/5}\{\log(pn)\}^{51/10}n^{-1/10} \le cs^{9/5}\{\log(pn)\}^{6}n^{-1/10}. \quad \text{(S20)}$$

This Berry–Esseen type bound implies the convergence stated in the theorem.

It remains to establish the claim (S15). Observe that

$$\begin{aligned} d_K\big(\mathcal{L}(\mathbb{G}_n\mathcal{G}), \mathcal{L}(\mathbb{Z}\mathcal{G}_n)\big) &= \sup_{t\in\mathbb{R}} |\mathrm{pr}(\mathbb{G}_n\mathcal{G} \le t) - \mathrm{pr}(\mathbb{Z}\mathcal{G}_n \le t)| \\ &\le \sup_{t\in\mathbb{R}} |\mathrm{pr}(\mathbb{G}_n\mathcal{G}_n \le t) - \mathrm{pr}(\mathbb{Z}\mathcal{G}_n \le t)| \\ &\quad + \sup_{t\in\mathbb{R}} |\mathrm{pr}(\mathbb{G}_n\mathcal{G} \le t) - \mathrm{pr}(\mathbb{G}_n\mathcal{G}_n \le t)|. \end{aligned}$$

From Lemma 2, we have

$$\sup_{t\in\mathbb{R}} |\mathrm{pr}(\mathbb{G}_n\mathcal{G} \le t) - \mathrm{pr}(\mathbb{G}_n\mathcal{G}_n \le t)| \le \frac{2}{n}.$$

Below we complete the proof by quantifying the error between $\mathbb{G}_n\mathcal{G}_n$ and $\mathbb{Z}\mathcal{G}_n$, specifically in (S21) and the bounds following it.

Discretize the empirical process $\mathbb{G}_n$ and the Gaussian process $\mathbb{Z}$. Let $\mathcal{I}$, $\mathcal{G}_\epsilon$ and $\mathcal{J}$ be defined in (S8), (S9) and (S10). Note that

$$0 \le \sup_{g\in\mathcal{G}_n} \mathbb{G}_n g - \max_{j=1,\ldots,N} \mathbb{G}_n g_j \le \|\mathbb{G}_n\|_{\mathcal{G}_\epsilon},$$

$$0 \le \sup_{g\in\mathcal{G}_n} \mathbb{Z} g - \max_{j=1,\ldots,N} \mathbb{Z} g_j \le \|\mathbb{Z}\|_{\mathcal{G}_\epsilon}.$$

These imply, for any $\eta > 0$,

$$\begin{aligned} \sup_{t\in\mathbb{R}} |\mathrm{pr}(\mathbb{G}_n\mathcal{G}_n \le t) - \mathrm{pr}(\mathbb{G}_n\mathcal{I} \le t)| &\le \sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{G}_n\mathcal{I} - t| \le \eta) + \mathrm{pr}(\|\mathbb{G}_n\|_{\mathcal{G}_\epsilon} \ge \eta) \\ \sup_{t\in\mathbb{R}} |\mathrm{pr}(\mathbb{Z}\mathcal{G}_n \le t) - \mathrm{pr}(\mathbb{Z}\mathcal{I} \le t)| &\le \sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}\mathcal{I} - t| \le \eta) + \mathrm{pr}(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon} \ge \eta). \end{aligned}$$

In addition, we observe

$$\begin{aligned} &\sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{G}_n\mathcal{I} - t| \le \eta) \\ &= \sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{G}_n\mathcal{I} \le t+\eta) - \mathrm{pr}(\mathbb{G}_n\mathcal{I} < t-\eta)\big| \\ &\le \sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{G}_n\mathcal{I} \le t+\eta) - \mathrm{pr}(\mathbb{Z}\mathcal{I} \le t+\eta)\big| + \sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{G}_n\mathcal{I} \le t-\eta) - \mathrm{pr}(\mathbb{Z}\mathcal{I} \le t-\eta)\big| \\ &\quad + \sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{Z}\mathcal{I} \le t+\eta) - \mathrm{pr}(\mathbb{Z}\mathcal{I} \le t-\eta)\big| \\ &\le 2\sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{G}_n\mathcal{I} \le t) - \mathrm{pr}(\mathbb{Z}\mathcal{I} \le t)\big| + \sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}\mathcal{I} - t| \le \eta) \end{aligned}$$

and

$$\sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}\mathcal{I}-t|\le\eta) \le 2\sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{Z}\mathcal{I}\le t)-\mathrm{pr}(\mathbb{Z}\mathcal{J}\le t)\big| + \sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}\mathcal{J}-t|\le\eta),$$

based on which we further deduce

$$\begin{aligned}
&d_K\big(\mathcal{L}(\mathbb{G}_n\mathcal{G}_n),\mathcal{L}(\mathbb{Z}\mathcal{G}_n)\big)\\
&=\sup_{t\in\mathbb{R}}\big|\mathrm{pr}(\mathbb{G}_n\mathcal{G}_n\le t)-\mathrm{pr}(\mathbb{G}_n\mathcal{I}\le t)+\mathrm{pr}(\mathbb{G}_n\mathcal{I}\le t)-\mathrm{pr}(\mathbb{Z}\mathcal{I}\le t)+\mathrm{pr}(\mathbb{Z}\mathcal{I}\le t)-\mathrm{pr}(\mathbb{Z}\mathcal{G}_n\le t)\big|\\
&\le d_K\big(\mathcal{L}(\mathbb{G}_n\mathcal{I}),\mathcal{L}(\mathbb{Z}\mathcal{I})\big)+\sup_{t\in\mathbb{R}}\mathrm{pr}(|\mathbb{Z}\mathcal{I}-t|\le\eta)+\mathrm{pr}(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon}\ge\eta)+\sup_{t\in\mathbb{R}}\mathrm{pr}(|\mathbb{G}_n\mathcal{I}-t|\le\eta)\\
&\quad+\mathrm{pr}(\|\mathbb{G}_n\|_{\mathcal{G}_\epsilon}\ge\eta)\\
&\le 3d_K\big(\mathcal{L}(\mathbb{G}_n\mathcal{I}),\mathcal{L}(\mathbb{Z}\mathcal{I})\big)+2\sup_{t\in\mathbb{R}}\mathrm{pr}(|\mathbb{Z}\mathcal{I}-t|\le\eta)+\mathrm{pr}(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon}\ge\eta)+\mathrm{pr}(\|\mathbb{G}_n\|_{\mathcal{G}_\epsilon}\ge\eta)\\
&\le 6d_K\big(\mathcal{L}(\mathbb{G}_n\mathcal{J}),\mathcal{L}(\mathbb{Z}\mathcal{J})\big)+2\sup_{t\in\mathbb{R}}\mathrm{pr}(|\mathbb{Z}\mathcal{J}-t|\le\eta)+\mathrm{pr}(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon}\ge\eta)+\mathrm{pr}(\|\mathbb{G}_n\|_{\mathcal{G}_\epsilon}\ge\eta)+cn^{-1},
\end{aligned}\tag{S21}$$

where the last inequality is due to Lemma S2. Below we address each term of (S21) separately, with the choice of η satisfying (S16) for sufficiently large constant $c_3>0$. Combining the bounds of these terms then completes the proof.

Bound $d_K\big(\mathcal{L}(\mathbb{G}_n\mathcal{J}),\mathcal{L}(\mathbb{Z}\mathcal{J})\big)$. Recall that $r=|\mathcal{J}|\le N$. Without loss of generality, we simply write $\mathcal{J}=\{g_1,\ldots,g_r\}$. To apply Lemma S7, we write

$$\begin{aligned}
U_{ij}&=(n_2n_1^{-1})^{1/2}[g_j(X_i)-E\{g_j(X_i)\}]\text{ for }i=1,\ldots,n_1,\\
U_{(n_1+i)j}&=-(n_1n_2^{-1})^{1/2}[g_j(Y_i)-E\{g_j(Y_i)\}]\text{ for }i=1,\ldots,n_2,\\
U_i&=(U_{i1},\ldots,U_{ir})\text{ for }i=1,\ldots,n.
\end{aligned}$$

Define the centered Gaussian version $\tilde U_i=(\tilde U_{i1},\ldots,\tilde U_{ir})$ of U_i with the covariance structure

$$\begin{aligned}
\mathrm{cov}(\tilde U_{ij},\tilde U_{il})&=\mathrm{cov}(U_{ij},U_{ik})=n_2n_1^{-1}\mathrm{cov}\{g_j(X_i),g_k(X_i)\},\ \text{for }i=1,\ldots,n_1,\\
\mathrm{cov}(\tilde U_{(n_1+i)j},\tilde U_{(n_1+i)l})&=\mathrm{cov}(U_{ij},U_{ik})=n_1n_2^{-1}\mathrm{cov}\{g_j(Y_i),g_k(Y_i)\},\ \text{for }i=1,\ldots,n_2.
\end{aligned}$$

Also, $\tilde U_1,\ldots,\tilde U_n$ are constructed to be independent. By construction, we observe

$$\mathbb{G}_ng_j=n^{-1/2}\sum_{i=1}^n U_{ij}\quad\text{and}\quad \mathbb{Z}g_j\overset{d}{=}n^{-1/2}\sum_{i=1}^n\tilde U_{ij},\qquad\text{for }j=1,\ldots,r.$$

Below we prepare some results for applying Lemma S7. By (S12) and Lemma S19, we deduce

$$\sup_{t\in\mathbb{R}}\mathrm{pr}\left(\left|\max_{1\le j\le r}\mathbb{Z}g_j-t\right|\le 2\phi^{-1}\right)\le\frac{c\phi^{-1}}{\tau_n^2}[A_n+1\vee\{\log(\tau_n\phi)\}^{1/2}].$$

Lemma S8 yields

$$E\left[\max_{1\le j,k\le r}\left|\frac{1}{n}\sum_{i=1}^{n}\{U_{ij}U_{ik}-E(U_{ij}U_{ik})\}\right|\right]\le cn^{-1/2}B_n.$$

Assumption 1 implies

$$\max_{1\le j\le r} n^{-1}\sum_{i=1}^{n}E(U_{ij}^4)\le c.$$

Let $\phi\asymp n^{1/4}(\log r)^{-3/4}$, if $n^{-1}\kappa_n^4\log r\le c_4$ for some sufficiently small constant $c_4>0$, then from Lemma S7, we conclude that

$$d_K\big(\mathcal{L}(\mathbb{G}_n\mathcal{J}),\mathcal{L}(\mathbb{Z}\mathcal{J})\big)\le c\left[\frac{(\log r)^{3/4}}{n^{1/4}\tau_n^2}\{A_n+(\log n)^{1/2}\}+\frac{(\log r)^{3/4}B_n}{n^{1/4}\tau_n}\right],$$

where B_n is defined in (S13).

Bound $\sup_{t\in\mathbb{R}}\operatorname{pr}(|\mathbb{Z}\mathcal{J}-t|\le\eta)$**.** This term now can be handled by the anti-concentration inequality for Gaussian maxima in Lemma S19. Specifically,

$$\sup_{t\in\mathbb{R}}\operatorname{pr}\big(|\mathbb{Z}\mathcal{J}-t|\le\eta\big)\le c\frac{\eta}{\tau_n^2}\left[E(\mathbb{Z}\mathcal{J})+1\vee\{\log(\tau_n/\eta)\}^{1/2}\right].$$

With (S12), this implies

$$\sup_{t\in\mathbb{R}}\operatorname{pr}\big(|\mathbb{Z}\mathcal{J}-t|\le\eta\big)\le c\frac{\eta}{\tau_n^2}\left[A_n+1\vee\{\log(\tau_n/\eta)\}^{1/2}\right]. \tag{S22}$$

Bound $\operatorname{pr}(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon}\ge\eta)$**.** Let $R=(n_2P+n_1Q)/n$. Since $\sup_{g\in\mathcal{G}_\epsilon}\operatorname{var}(\mathbb{Z}g)\le 2\epsilon^2\|G\|^2_{(P+Q)/2,2}$, from the Borell–Sudakov–Tsirel'son inequality (van der Vaart & Wellner, 1996, Proposition A.2.1), we have

$$\operatorname{pr}\left\{\|\mathbb{Z}\|_{\mathcal{G}_\epsilon}>E(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon})+2\epsilon\|G\|_{(P+Q)/2,2}(\log n)^{1/2}\right\}\le 2n^{-1}.$$

Note that $N(\delta,\mathcal{G}_\epsilon,e_R)\le N^2(\delta/2,\mathcal{G}_n,e_R)$. By Dudley's maximal inequality (van der Vaart & Wellner, 1996, Corollary 2.2.8), we have

$$\begin{aligned}
E(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon}) &\le c\int_0^{\epsilon\|G\|_{(P+Q)/2,2}}\{1+2\log N(\delta/2,\mathcal{G}_n,e_R)\}^{1/2}\,d\delta\\
&\le c\int_0^{\epsilon\|G\|_{(P+Q)/2,2}}\{1+s\log(ep/s)+s\log(1+8\nu/\delta)\}^{1/2}d\delta+c\int_0^{\epsilon\|G\|_{(P+Q)/2,2}}\left(\frac{\kappa_n}{\delta}\right)^{1/2}d\delta\\
&\le cs^{1/2}\epsilon\|G\|_{(P+Q)/2,2}\big[\log^{1/2}\{1+1/(\epsilon\|G\|_{(P+Q)/2,2})\}+\log^{1/2}(ep/s)\big]+c(\epsilon\kappa_n)^{1/2}\|G\|^{1/2}_{(P+Q)/2,2},
\end{aligned}$$

where the second and the last inequalities respectively follow from Lemmas S5 and S23. Therefore, under $\log(1/\epsilon)\le c\log(pn)$,

$$\operatorname{pr}\Big(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon}\ge c\big[\epsilon\kappa_n\{s\log(pn)\}^{1/2}+\epsilon^{1/2}\kappa_n+\epsilon\kappa_n(\log n)^{1/2}\big]\Big)\le 2n^{-1}. \tag{S23}$$

Consequently,

$$\mathrm{pr}(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon}) \geq \eta) \leq 2n^{-1}. \tag{S24}$$

Bound $\mathrm{pr}(\|\mathbb{G}_n\|_{\mathcal{G}_\epsilon} \geq \eta)$**.** Observe that

$$\begin{aligned}
\|\mathbb{G}_n\|_{\mathcal{G}_\epsilon} &= \left(\frac{n_1 n_2}{n_1+n_2}\right)^{1/2} \sup_{g\in\mathcal{G}_\epsilon} \left| \frac{1}{n_1}\sum_{i=1}^{n_1}[g(X_i) - E\{g(X)\}] - \frac{1}{n_2}\sum_{j=1}^{n_2}[g(Y_j) - E\{g(Y)\}] \right| \\
&\leq \left(\frac{n_2}{n_1+n_2}\right)^{1/2} \|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon} + \left(\frac{n_1}{n_1+n_2}\right)^{1/2} \|\mathbb{Y}_{n_2}\|_{\mathcal{G}_\epsilon} \\
&\leq \|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon} + \|\mathbb{Y}_{n_2}\|_{\mathcal{G}_\epsilon}.
\end{aligned}$$

Since $\|g\|_\infty \leq \kappa_n$ for all $g \in \mathcal{G}_n$, we apply Lemma S20 with $b = \kappa_n$ and $t = \log n$ to deduce that, for all $a > 0$,

$$\mathrm{pr}\left\{\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon} \geq (1+a)E(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon}) + c\sigma_\epsilon(\log n)^{1/2} + \left(c + \frac{c}{a}\right)\frac{\kappa_n(\log n)}{n^{1/2}}\right\} \leq n^{-1}.$$

Noting $N(\delta, \mathcal{G}_\epsilon, e_m) \leq N^2(\delta/2, \mathcal{G}_n, e_m)$ for any probability measure m on $\mathbb{R}^p$, by an argument similar to that in the proof of Lemma S6 and observing that

$$\begin{aligned}
\sigma_\epsilon^2 = \sup_{g\in\mathcal{G}_\epsilon} E\{g^2(X)\} &= \sup_{(f,g):e_{(P+Q)/2}(f,g)\leq\epsilon\|G\|_{(P+Q)/2,2}} \{e_P(f,g)\}^2 \\
&\leq 2 \sup_{(f,g):e_{(P+Q)/2}(f,g)\leq\epsilon\|G\|_{(P+Q)/2,2}} \{e_{(P+Q)/2}(f,g)\}^2 \leq 2\epsilon^2\kappa_n^2,
\end{aligned}$$

we deduce

$$\begin{aligned}
E(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon}) &\leq cJ(\epsilon)\kappa_n + c\frac{\kappa_n J^2(\epsilon)}{\epsilon^2 n^{1/2}} \\
&\leq c[\kappa_n\epsilon\{s\log(pn)\}^{1/2} + \kappa_n\epsilon^{1/2} + n^{-1/2}\kappa_n s\log(pn) + \kappa_n\epsilon^{-1}n^{-1/2}],
\end{aligned}$$

where $J(\epsilon)$ is defined in (S14). Consequently,

$$\mathrm{pr}(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon} \geq \eta/2) \leq n^{-1}.$$

Similarly, $\mathrm{pr}(\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_\epsilon} \geq \eta/2) \leq n^{-1}$. Therefore,

$$\begin{aligned}
\mathrm{pr}(\|\mathbb{G}_n\|_{\mathcal{G}_\epsilon} \geq \eta) &\leq \mathrm{pr}(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon} + \|\mathbb{Y}_{n_2}\|_{\mathcal{G}_\epsilon} \geq \eta) \\
&\leq \mathrm{pr}(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon} \geq \eta/2) + \mathrm{pr}(\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_\epsilon} \geq \eta/2) \\
&\leq 2n^{-1},
\end{aligned}$$

which completes the proof. □

S.3.3. *Proof of Theorem 2*

Proof. First, we claim that with probability at least $1 - cn^{-1}$,

$$d_K\big(\mathcal{L}(T^\star \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{G}_n)\big) \le c\frac{A_n + B_n\tau_n + (A_nB_n\tau_n)^{1/2}}{n^{1/4}\tau_n^2}(\log r)^{3/4} + c\frac{\eta}{\tau_n^2}\left[A_n + 1 \vee \{\log(\tau_n/\eta)\}^{1/2}\right], \tag{S25}$$

under the conditions (S16), (S17), (S18), (S19), and

$$n^{-1}\kappa_n^4(\log r + \log n) \lesssim 1,\ n^{-1/2}B_n \lesssim \tau_n^2,\ n^{-1/2}\epsilon^{-3/2} \lesssim 1,\ n^{-1/2}\epsilon^{-1}\{s\log(pn)\}^{1/2} \lesssim 1.$$

If we take $\eta \asymp \epsilon^{1/2}\kappa_n$ and $\tau_n \asymp \log^{-2}(pn)s^{-1/2}$, and choose ϵ so that $cn^{-1/3} \le \epsilon \le cs^{-1}\log^{-2}(pn)$, then under the assumption $s^3 \le cn\log^{-10}(pn)$, all the above conditions can be satisfied. In this case, the rate becomes

$$\sup_{t\in\mathbb{R}} |\mathrm{pr}(T^\star \le t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\mathcal{G}_n \le t)| \le c\frac{\{s\log(pn)\}^{1/2}}{\tau_n^2}(n^{-1/4}\epsilon^{-3/4} + \epsilon^{1/2}\kappa_n).$$

Taking $\epsilon \asymp (n\kappa_n^4)^{-1/5}$ yields that, with probablity at least $1 - c/n$,

$$d_K\big(\mathcal{L}(T^\star \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{G}_n)\big) \le cs^{9/5}\{\log(pn)\}^{51/10}n^{-1/10} \le cs^{9/5}\{\log(pn)\}^{6}n^{-1/10}. \tag{S26}$$

This Berry–Esseen type bound implies the convergence stated in the theorem.

It remains to establish (S25). Recall $\mathscr{D} = \{X_1, \ldots, X_{n_1}, Y_1, \ldots, Y_{n_2}\}$ denotes the observed data and $n = n_1 + n_2$. Also recall that the event Ω_n is defined in (S3). Let $\mathbb{V}_n^\star$ and $\mathbb{Z}_n^\star$ be defined in (S7) and (S4), respectively.

From the definition of $T^\star$, we see that $T^\star$ and $\mathbb{V}_n^\star\mathcal{G}$ have the same distribution. We first observe

$$\begin{aligned}\sup_{t\in\mathbb{R}} |\mathrm{pr}(T^\star \le t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\mathcal{G}_n \le t)| \le{}& \sup_{t\in\mathbb{R}} |\mathrm{pr}(\mathbb{V}_n^\star\mathcal{G} \le t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{V}_n^\star\mathcal{G}_n \le t \mid \mathscr{D})| \\ &+ \sup_{t\in\mathbb{R}} \big|\mathrm{pr}\big(\mathbb{Z}_n^\star\mathcal{G}_n \le t \mid \mathscr{D}\big) - \mathrm{pr}(\mathbb{Z}\mathcal{G}_n \le t)\big| \\ &+ \sup_{t\in\mathbb{R}} \big|\mathrm{pr}\big(\mathbb{V}_n^\star\mathcal{G}_n \le t \mid \mathscr{D}\big) - \mathrm{pr}\big(\mathbb{Z}_n^\star\mathcal{G}_n \le t \mid \mathscr{D}\big)\big|\end{aligned}$$

With the similar arguments for Lemma 2, we can show that, with probability at least $1 - cn^{-1}$, the first term is 0. The second and third terms are handled in Theorem S1 and Theorem S2, respectively. □

THEOREM S1. *Suppose Assumptions 1–3 hold. If η satisfies (S16), τ_n satisfies (S17) and (S18), and*

$$c_3\epsilon\kappa_n(\log n) \le 1, \quad c_3n^{-1/3} \le \epsilon, \quad c_3\epsilon^{-1}\{s\log(pn)\}^{1/2} \le n^{1/2},$$

then we have with probability at least $1 - cn^{-1}$,

$$d_K\big(\mathcal{L}(\mathbb{Z}_n^\star \mathcal{G}_n \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{G}_n)\big) \le c\frac{\eta}{\tau_n^2}\big[A_n + 1 \vee \{\log(\tau_n/\eta)\}^{1/2}\big] + c\frac{(A_n B_n)^{1/2}(\log r)^{3/4}}{n^{1/4}\tau_n^{3/2}},$$

where A_n *and* B_n *are respectively defined in* (S12) *and* (S13)*, and* r *is defined in* (S11).

Proof. We have

$$0 \le \mathbb{Z}_n^\star \mathcal{G}_n - \mathbb{Z}_n^\star \mathcal{I} \le \|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon},$$

$$0 \le \mathbb{Z}\mathcal{G}_n - \mathbb{Z}\mathcal{I} \le \|\mathbb{Z}\|_{\mathcal{G}_\epsilon}.$$

By similar arguments in the proof of Theorem 1 and Lemma S2, we have, with probability at least $1 - cn^{-1}$,

$$\begin{aligned}
&d_K\big(\mathcal{L}(\mathbb{Z}_n^\star \mathcal{G}_n \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{G}_n)\big) \\
&\le 3d_K\big(\mathcal{L}(\mathbb{Z}_n^\star \mathcal{I} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{I})\big) + 2\sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}\mathcal{I} - t| \le \eta) + \mathrm{pr}(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon} \ge \eta) + \mathrm{pr}(\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon} \ge \eta \mid \mathscr{D}) \\
&\le 6d_K\big(\mathcal{L}(\mathbb{Z}_n^\star \mathcal{J} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{J})\big) + 2\sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}\mathcal{J} - t| \le \eta) \\
&\quad + \mathrm{pr}(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon} \ge \eta) + \mathrm{pr}(\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon} \ge \eta \mid \mathscr{D}) + cn^{-1}.
\end{aligned} \tag{S27}$$

In the above, the third term is bounded in (S24), and the first two terms are addressed in Lemma S16 and Lemma S19, respectively. Specifically,

$$\mathrm{pr}(\|\mathbb{Z}\|_{\mathcal{G}_\epsilon} \ge \eta) \le cn^{-1}$$

and

$$\begin{aligned}
\sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}\mathcal{J} - t| \le \eta) &\le c\frac{\eta}{\tau_n^2}\big[E(\mathbb{Z}\mathcal{J}) + 1 \vee \{\log(\tau_n/\eta)\}^{1/2}\big] \\
&\le c\frac{\eta}{\tau_n^2}\big[A_n + 1 \vee \{\log(\tau_n/\eta)\}^{1/2}\big]
\end{aligned}$$

due to (S12). Applying (S45) in Lemma S8 and Lemma S16, we obtain with probability at least $1 - cn^{-1}$,

$$d_K\big(\mathcal{L}(\mathbb{Z}_n^\star \mathcal{J} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{J})\big) \le c\frac{(A_n B_n)^{1/2}(\log r)^{3/4}}{n^{1/4}\tau_n^{3/2}}. \tag{S28}$$

We claim that $\mathrm{pr}\,(\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon} \ge \eta) \le cn^{-1}$ with probability at least $1 - cn^{-1}$, with η satisfying (S16) for a sufficiently large constant $c_3 > 0$; the proof of this claim is provided in the following. Combining this claim with the above inequalities completes the proof.

Bound $\mathrm{pr}(\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon} \ge \eta \mid \mathscr{D})$**.** By the Borell–Sudakov–Tsirel'son inequality (van der Vaart & Wellner, 1996, Proposition A.2.1),

$$\mathrm{pr}\big\{\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon} > E(\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon} \mid \mathscr{D}) + \sigma_{n,\epsilon}(2\log n)^{1/2} \mid \mathscr{D}\big\} \le 2n^{-1}, \tag{S29}$$

where

$$
\begin{aligned}
\sigma_{n,\epsilon}^2 &= \sup_{g\in\mathcal{G}_\epsilon} n^{-1}\Big(\sum_{i=1}^{n_1}(n_2/n_1)[g(X_i)-\hat{E}_{n_1}\{g(X)\}]^2+\sum_{i=1}^{n_2}(n_1/n_2)[g(Y_i)-\hat{E}_{n_2}\{g(Y)\}]^2\Big)\\
&\le \sup_{g\in\mathcal{G}_\epsilon}\left\{\frac{1}{n_1}\sum_{i=1}^{n_1}g^2(X_i)+\frac{1}{n_2}\sum_{i=1}^{n_2}g^2(Y_i)\right\}\\
&\le \sup_{g\in\mathcal{G}_\epsilon}E\{g^2(X)\}+\sup_{g\in\mathcal{G}_\epsilon}E\{g^2(Y)\}+n_1^{-1/2}\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon^2}+n_2^{-1/2}\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_\epsilon^2}\\
&\le c\epsilon^2\kappa_n^2+n_1^{-1/2}\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon^2}+n_2^{-1/2}\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_\epsilon^2},
\end{aligned}
$$

where $\mathcal{G}_\epsilon^2=\{g^2: g\in\mathcal{G}_\epsilon\}$. Applying Lemma S20 with $t=\log n$, we have for $\delta>0$,

$$
\mathrm{pr}\big\{\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon^2}\ge(1+\delta)E(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon^2})+c\check{\sigma}_\epsilon(\log n)^{1/2}+(c+c\delta^{-1})\kappa_n^2(\log n)n^{-1/2}\big\}\le n^{-1},
$$

where $\check{\sigma}_\epsilon^2=\sup_{g\in\mathcal{G}_\epsilon^2}E[g(X)-E\{g(X)\}]^2\le c\kappa_n^2\epsilon^2\|G\|_{(P+Q)/2,2}^2\le c\kappa_n^4\epsilon^2$. From the fact that $N(\epsilon,\mathcal{G}_\epsilon^2,e_m)\le N^2(\epsilon/(8\kappa_n),\mathcal{G}_n,e_m)$ for any probability measure m on $\mathbb{R}^p$, and using similar arguments in Lemma S6, we deduce

$$
E(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_\epsilon^2})\le c\big[\kappa_n^2\epsilon\{s\log(pn)\}^{1/2}+\kappa_n^2\epsilon^{1/2}+n^{-1/2}\kappa_n^2 s\log(pn)+\kappa_n^2\epsilon^{-1}n^{-1/2}\big].
$$

We can establish a bound for $\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_\epsilon^2}$ in a similar fashion. Hence, with probability at least $1-cn^{-1}$, we have

$$
\begin{aligned}
\sigma_{n,\epsilon}^2\le&c\epsilon^2\kappa_n^2+cn^{-1/2}[\epsilon\kappa_n^2\{s\log(pn)\}^{1/2}+\epsilon^{1/2}\kappa_n^2]+\\
&cn^{-1}\{\kappa_n^2 s\log(pn)+\epsilon^{-1}\kappa_n^2\}.
\end{aligned}
$$

Under the conditions $n^{-1/3}\le c\epsilon$ and $n^{-1/2}\epsilon^{-1}\{s\log(pn)\}^{1/2}\le c$, we conclude

$$
\sigma_{n,\epsilon}\le c\epsilon\kappa_n \tag{S30}
$$

with probability at least $1-cn^{-1}$.

Next we bound $E(\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon}\mid\mathscr{D})$. For this, we first observe that

$$
\begin{aligned}
\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon}\le&\sup_{g\in\mathcal{G}_\epsilon}\left|n_1^{-1/2}\sum_{i=1}^{n_1}\xi_i g(X_i)\right|+\sup_{g\in\mathcal{G}_\epsilon}\left|n_2^{-1/2}\sum_{i=1}^{n_2}\xi_{n_1+i}g(Y_i)\right|+\\
&\sup_{g\in\mathcal{G}_\epsilon}\left|n_1^{-1/2}\sum_{i=1}^{n_1}\xi_i\hat{E}_{n_1}\{g(X)\}\right|+\sup_{g\in\mathcal{G}_\epsilon}\left|n_2^{-1/2}\sum_{i=1}^{n_2}\xi_{n_1+i}\hat{E}_{n_2}\{g(Y)\}\right|\\
=&\,\mathrm{I}+\mathrm{II}+\mathrm{III}+\mathrm{IV}.
\end{aligned}
$$

By Dudley's maximal inequality, we have with probability at least $1-cn^{-1}$,

$$
\begin{aligned}
E(\mathrm{I}\mid\mathscr{D})&\le c\int_0^{\epsilon\kappa_n}\big\{1+\log N(\delta,\mathcal{G}_\epsilon,e_{P_n})\big\}^{1/2}d\delta\\
&\le c\int_0^{\epsilon\kappa_n}\big\{1+\log N(\delta/2,\mathcal{G}_n,e_{P_n})\big\}^{1/2}d\delta
\end{aligned}
$$

$$\leq c\epsilon\kappa_n\{s\log(pn)\}^{1/2} + c\kappa_n\epsilon^{1/2},$$

where the last line follows from Lemma S5(2). The same bound holds for II. For the last two terms III and IV, we have with probability at least $1 - cn^{-1}$,

$$E(\mathrm{III} \mid \mathscr{D}) \leq \sup_{g\in\mathcal{G}_\epsilon} |\hat{E}_{n_1}\{g(X)\}| \cdot E\left(\left|n_1^{-1/2}\sum_{i=1}^{n_1}\xi_i\right|\right) \leq c\epsilon\kappa_n.$$

$$E(\mathrm{IV} \mid \mathscr{D}) \leq \sup_{g\in\mathcal{G}_\epsilon} |\hat{E}_{n_2}\{g(Y)\}| \cdot E\left(\left|n_2^{-1/2}\sum_{i=n_1+1}^{n}\xi_i\right|\right) \leq c\epsilon\kappa_n.$$

Therefore, with probability at least $1 - cn^{-1}$,

$$E(\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon} \mid \mathscr{D}) \leq c[\epsilon\kappa_n\{s\log(pn)\}^{1/2} + \kappa_n\epsilon^{1/2}].$$

Combining this with (S29) and the bound $\sigma_{n,\epsilon} \leq c\epsilon\kappa_n$ yields

$$\mathrm{pr}\left(\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon} \geq \eta\right) \leq cn^{-1}, \tag{S31}$$

which completes the proof. □

THEOREM S2. *Under Assumptions 1–3 and the conditions in Theorem S1, if*

$$c_3\kappa_n^4(\log r + \log n) \leq n \quad \textit{and} \quad c_3 n^{-1/2}B_n \leq \tau_n^2,$$

then we have with probability at least $1 - cn^{-1}$,

$$d_K\big(\mathcal{L}(\mathbb{V}_n^\star\mathcal{G}_n \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}_n^\star\mathcal{G}_n \mid \mathscr{D})\big) \leq c\frac{(\log r)^{3/4}\{A_n + B_n\tau_n + (A_nB_n\tau_n)^{1/2}\}}{n^{1/4}\tau_n^2} + c\frac{\eta}{\tau_n^2}\left[A_n + 1 \vee \{\log(\tau_n/\eta)\}^{1/2}\right],$$

where A_n *and* B_n *are defined in* (S12) *and* (S13), *and* r *is defined in* (S11).

Proof. Note that

$$0 \leq \mathbb{V}_n^\star\mathcal{G}_n - \mathbb{V}_n^\star\mathcal{I} \leq \|\mathbb{V}_n^\star\|_{\mathcal{G}_\epsilon},$$

$$0 \leq \mathbb{Z}_n^\star\mathcal{G}_n - \mathbb{Z}_n^\star\mathcal{I} \leq \|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon}.$$

These imply, for any $\eta > 0$,

$$\sup_{t\in\mathbb{R}} |\mathrm{pr}(\mathbb{V}_n^\star\mathcal{G}_n \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{V}_n^\star\mathcal{I} \leq t \mid \mathscr{D})| \leq \sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{V}_n^\star\mathcal{I} - t| \leq \eta \mid \mathscr{D}) + \mathrm{pr}(\|\mathbb{V}_n^\star\|_{\mathcal{G}_\epsilon} \geq \eta \mid \mathscr{D})$$

$$\sup_{t\in\mathbb{R}} |\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{G}_n \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}_n^\star\mathcal{I} \leq t \mid \mathscr{D})| \leq \sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}_n^\star\mathcal{I} - t| \leq \eta \mid \mathscr{D}) + \mathrm{pr}(\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon} \geq \eta \mid \mathscr{D}).$$

In addition, we observe

$$
\begin{aligned}
&\sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{V}_n^\star \mathcal{I} - t| \le \eta \mid \mathscr{D}) \\
&= \sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{V}_n^\star \mathcal{I} \le t+\eta \mid \mathscr{D}) - \mathrm{pr}(\mathbb{V}_n^\star \mathcal{I} < t-\eta \mid \mathscr{D})\big| \\
&\le \sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{V}_n^\star \mathcal{I} \le t+\eta \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}_n^\star \mathcal{I} \le t+\eta \mid \mathscr{D})\big| + \\
&\quad \sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{V}_n^\star \mathcal{I} \le t-\eta \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}_n^\star \mathcal{I} \le t-\eta \mid \mathscr{D})\big| + \\
&\quad \sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{Z}_n^\star \mathcal{I} \le t+\eta \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}_n^\star \mathcal{I} \le t-\eta \mid \mathscr{D})\big| \\
&\le 2\sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{V}_n^\star \mathcal{I} \le t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}_n^\star \mathcal{I} \le t \mid \mathscr{D})\big| + \sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}_n^\star \mathcal{I} - t| \le \eta \mid \mathscr{D})
\end{aligned}
$$

and similarly

$$
\sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}_n^\star \mathcal{I} - t| \le \eta \mid \mathscr{D}) \le 2\sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{Z}_n^\star \mathcal{I} \le t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\mathcal{I} \le t)\big| + \sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}\mathcal{I} - t| \le \eta).
$$

Using similar arguments in the proof of Theorem 1 and Lemma S2, we deduce that, with probability at least $1 - cn^{-1}$,

$$
\begin{aligned}
& d_K\big(\mathcal{L}(\mathbb{V}_n^\star \mathcal{G}_n \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}_n^\star \mathcal{G}_n \mid \mathscr{D})\big) \\
&\le c d_K\big(\mathcal{L}(\mathbb{V}_n^\star \mathcal{J} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}_n^\star \mathcal{J} \mid \mathscr{D})\big) + c d_K\big(\mathcal{L}(\mathbb{Z}_n^\star \mathcal{J} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{J} \mid \mathscr{D})\big) + \mathrm{pr}(\|\mathbb{Z}_n^\star\|_{\mathcal{G}_\epsilon} \ge \eta \mid \mathscr{D}) \\
&\quad + \mathrm{pr}(\|\mathbb{V}_n^\star\|_{\mathcal{G}_\epsilon} \ge \eta \mid \mathscr{D}) + c\sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}\mathcal{J} - t| \le \eta) + cn^{-1}.
\end{aligned}
$$

The second and third terms have been handled in (S28) and (S31). The fifth term is bounded by

$$
\sup_{t\in\mathbb{R}} \mathrm{pr}(|\mathbb{Z}\mathcal{J} - t| \le \eta) \le c\frac{\eta}{\tau_n^2}\big[A_n + 1 \vee \{\log(\tau_n/\eta)\}^{1/2}\big] \tag{S32}
$$

by using the same arguments that lead to (S22). To complete the proof, below we address the remaining terms with the choice of η satisfying (S16) for some sufficiently large constant $c_3 > 0$.

Bound $\mathrm{pr}(\|\mathbb{V}_n^\star\|_{\mathcal{G}_\epsilon} \ge \eta \mid \mathscr{D})$. By Talagrand's inequality in Lemma S20 with $t = \log n$, we have for all $\delta > 0$,

$$
\mathrm{pr}\{\|\mathbb{V}_n^\star\|_{\mathcal{G}_\epsilon} \ge (1+\delta)E(\|\mathbb{V}_n^\star\|_{\mathcal{G}_\epsilon} \mid \mathscr{D}) + c\sigma_{n,\epsilon}(\log n)^{1/2} + (c + c/\delta)\kappa_n(\log n)n^{-1/2} \mid \mathscr{D}\} \le n^{-1},
$$

where

$$
\sigma_{n,\epsilon}^2 = \sup_{g\in\mathcal{G}_\epsilon}\left(\frac{n_2}{n_1+n_2} n_1^{-1}\sum_{i=1}^{n_1}[g(X_i) - \hat{E}_{n_1}\{g(X)\}]^2 + \frac{n_1}{n_1+n_2} n_2^{-1}\sum_{i=1}^{n_2}[g(Y_i) - \hat{E}_{n_2}\{g(Y)\}]^2\right).
$$

According to (S30), $\sigma_{n,\epsilon} \le c\epsilon\kappa_n$ which holds with probability at least $1 - cn^{-1}$. Using similar arguments in Lemma S6 and (S14), we have with probability at least $1 - cn^{-1}$,

$$
E(\|\mathbb{V}_n^\star\|_{\mathcal{G}_\epsilon} \mid \mathscr{D}) \le E(\|\mathbb{X}_{n_1}^\star\|_{\mathcal{G}_\epsilon} \mid \mathscr{D}) + E(\|\mathbb{Y}_{n_2}^\star\|_{\mathcal{G}_\epsilon} \mid \mathscr{D})
$$

$$\le cJ(\epsilon)\kappa_n + \frac{J^2(\epsilon)\kappa_n}{\epsilon^2 n^{1/2}}$$
$$\le c\epsilon\kappa_n \{s\log(pn)\}^{1/2} + c\epsilon^{1/2}\kappa_n + cn^{-1/2}\kappa_n s\log(pn) + cn^{-1/2}\kappa_n\epsilon^{-1},$$

where $\mathbb{X}^\star_{n_1}$ and $\mathbb{Y}^\star_{n_2}$ are defined in (S6). Thus, with probability at least $1-cn^{-1}$,

$$\mathrm{pr}(\|\mathbb{V}^\star_n\|_{\mathcal{G}_\epsilon} \ge \eta \mid \mathscr{D}) \le cn^{-1}.$$

Bound $d_K\big(\mathcal{L}(\mathbb{V}^\star_n\mathcal{J} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}^\star_n\mathcal{J} \mid \mathscr{D})\big)$**.** To apply the Nazarov's inequality, we need to lower bound the variances of $\mathbb{Z}^\star_n g$ with $g \in \mathcal{J}$ conditional on $\mathscr{D}$. According to (S45) in Lemma S8, we see that

$$\frac{n_2}{n_1+n_2} n_1^{-1}\sum_{i=1}^{n_1}[g(X_i) - \hat{E}_{n_1}\{g(X)\}]^2 + \frac{n_1}{n_1+n_2} n_2^{-1}\sum_{i=1}^{n_2}[g(Y_i) - \hat{E}_{n_2}\{g(Y)\}]^2 \ge \tau_n^2/2$$

for $g \in \mathcal{J}$, which holds for all sufficiently large n since $n^{-1/2}B_n \le c\tau_n^2$.

To apply Lemma S7, let $U_i = (U_{i1}, \ldots, U_{ir})^\top$ with

$$U_{ij} = (n_2/n_1)^{1/2}[g_j(X^\star_i) - \hat{E}_{n_1}\{g_j(X)\}], i = 1, \ldots, n_1,$$

$$U_{(n_1+i)j} = -(n_1/n_2)^{1/2}[g_j(Y^\star_i) - \hat{E}_{n_2}\{g_j(Y)\}], i = 1, \ldots, n_2.$$

The Gaussian analog of $n^{-1/2}\sum_{i=1}^n U_i$ conditional on $\mathscr{D}$ is $(\mathbb{Z}^\star_n g_1, \ldots, \mathbb{Z}^\star_n g_r)^\top$.

Note that

$$\begin{aligned}
L_n &= \max_{1\le j\le r} n^{-1}\sum_{i=1}^n E(U_{ij}^4 \mid \mathscr{D}) \\
&\le c\max_{g\in\mathcal{J}} n_1^{-1}\sum_{i=1}^{n_1} E\big[|g(X^\star_i) - \hat{E}_{n_1}\{g(X)\}|^4 \mid \mathscr{D}\big] + c\max_{g\in\mathcal{J}} n_2^{-1}\sum_{i=1}^{n_2} E\big[|g(Y^\star_i) - \hat{E}_{n_2}\{g(Y)\}|^4 \mid \mathscr{D}\big] \\
&= c\max_{g\in\mathcal{J}} n_1^{-1}\sum_{i=1}^{n_1} |g(X_i) - \hat{E}_{n_1}\{g(X)\}|^4 + c\max_{g\in\mathcal{J}} n_2^{-1}\sum_{i=1}^{n_2} |g(Y_i) - \hat{E}_{n_2}\{g(Y)\}|^4 \\
&\le c\max_{g\in\mathcal{J}} n_1^{-1}\sum_{i=1}^{n_1} g^4(X_i) + c\max_{g\in\mathcal{J}} n_2^{-1}\sum_{i=1}^{n_2} g^4(Y_i).
\end{aligned}$$

By lemma S21, we have

$$\begin{aligned}
E\bigg\{\max_{g\in\mathcal{J}}\sum_{i=1}^{n_1} g^4(X_i)\bigg\} &\le c\max_{g\in\mathcal{J}} E\bigg\{\sum_{i=1}^{n_1} g^4(X_i)\bigg\} + cE(\Lambda)\log r \\
&\le cn + c\kappa_n^4\log r,
\end{aligned}$$

where $\Lambda = \max_{i=1,\ldots,n_1}\max_{g\in\mathcal{J}} g^4(X_i) \le c\kappa_n^4$. By Lemma S22 with $\beta = 1$, we have for every $t > 0$,

$$\mathrm{pr}\{L_n \ge c(1 + n^{-1}\kappa_n^4\log r + n^{-1}\kappa_n^4 t)\} \le 3e^{-t}.$$

Hence, we have $L_n \leq c$ with probability at least $1 - cn^{-1}$ since $\kappa_n^4(\log r + \log n) \leq cn$.

In addition, we observe that

$$
\begin{aligned}
&E\biggl(\max_{1\leq j,k\leq r} \biggl| n^{-1}\sum_{i=1}^{n_1} \frac{n_2}{n_1}[g_j(X_i^\star) - \hat{E}_{n_1}\{g_j(X)\}][g_k(X_i^\star) - \hat{E}_{n_1}\{g_k(X)\}] + \\
&n^{-1}\sum_{i=1}^{n_2} \frac{n_1}{n_2}[g_j(Y_i^\star) - \hat{E}_{n_2}\{g_j(Y)\}][g_k(Y_i^\star) - \hat{E}_{n_2}\{g_k(Y)\}] - \\
&n^{-1}\sum_{i=1}^{n_1} \frac{n_2}{n_1}[g_j(X_i) - \hat{E}_{n_1}\{g_j(X)\}][g_k(X_i) - \hat{E}_{n_1}\{g_k(X)\}] - \\
&n^{-1}\sum_{i=1}^{n_2} \frac{n_1}{n_2}[g_j(Y_i) - \hat{E}_{n_2}\{g_j(Y)\}][g_k(Y_i) - \hat{E}_{n_2}\{g_k(Y)\}] \biggr| \mid \mathscr{D} \biggr) \\
&\leq \frac{2}{n_1^{1/2}} E(\|\mathbb{X}_{n_1}^\star\|_{\mathcal{G}_n} \mid \mathscr{D}) \sup_{g\in\mathcal{G}_n} |\hat{E}_{n_1}\{g(X)\}| + \frac{1}{n_1^{1/2}} E(\|\mathbb{X}_{n_1}^\star\|_{\mathcal{G}_n\times\mathcal{G}_n} \mid \mathscr{D}) + \\
&\quad \frac{2}{n_2^{1/2}} E(\|\mathbb{Y}_{n_2}^\star\|_{\mathcal{G}_n} \mid \mathscr{D}) \sup_{g\in\mathcal{G}_n} |\hat{E}_{n_2}\{g(Y)\}| + \frac{1}{n_2^{1/2}} E(\|\mathbb{Y}_{n_2}^\star\|_{\mathcal{G}_n\times\mathcal{G}_n} \mid \mathscr{D}) \\
&\leq cn^{-1/2}\biggl\{ J(c/\kappa_n)\kappa_n + \frac{J^2(c/\kappa_n)\kappa_n}{n^{1/2}(c/\kappa_n)^2} + J(c/\kappa_n^2)\kappa_n^2 + \frac{J^2(c/\kappa_n^2)\kappa_n^2}{n^{1/2}(c/\kappa_n^2)^2} \biggr\} \\
&\leq cn^{-1/2}B_n,
\end{aligned}
$$

where $\mathcal{G}_n \times \mathcal{G}_n = \{f \times g : f, g \in \mathcal{G}_n\}$ with $(f \times g)(x) = f(x)g(x)$, $\mathbb{X}_{n_1}^\star$, $\mathbb{Y}_{n_2}^\star$ are defined in (S6), and the last inequality holds with probability at least $1 - cn^{-1}$ by applying similar arguments in Lemma S6 and (S14). Moreover,

$$
\begin{aligned}
&\sup_{t\in\mathbb{R}} \mathrm{pr}\bigl(|\mathbb{Z}_n^\star\mathcal{J} - t| \leq \phi^{-1} \mid \mathscr{D}\bigr) \\
&\leq \sup_{t\in\mathbb{R}} \mathrm{pr}\bigl(|\mathbb{Z}\mathcal{J} - t| \leq \phi^{-1}\bigr) + 2d_K\bigl(\mathcal{L}(\mathbb{Z}_n^\star\mathcal{J} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{J})\bigr) \\
&\leq c\frac{\eta}{\tau_n^2}\bigl[A_n + 1 \vee \{\log(\tau_n/\eta)\}^{1/2}\bigr] + 2d_K\bigl(\mathcal{L}(\mathbb{Z}_n^\star\mathcal{J} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{J})\bigr),
\end{aligned}
$$

where the last term is addressed in (S28).

Now we are ready to apply Lemma S7 by taking $\phi \asymp n^{1/4}(\log r)^{-3/4}$ to deduce the following bound: With probability at least $1 - cn^{-1}$,

$$
\begin{aligned}
&d_K\bigl(\mathcal{L}(\mathbb{V}_n^\star\mathcal{J} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}_n^\star\mathcal{J} \mid \mathscr{D})\bigr) \\
&\leq c\frac{(\log r)^{3/4}A_n}{n^{1/4}\tau_n^2} + c\frac{(\log r)^{3/4}B_n}{n^{1/4}\tau_n} + c\frac{\eta}{\tau_n^2}\bigl[A_n + 1 \vee \{\log(\tau_n/\eta)\}^{1/2}\bigr] + c\frac{(A_nB_n)^{1/2}(\log r)^{3/4}}{n^{1/4}\tau_n^{3/2}}.
\end{aligned}
$$

This finally completes the proof of Theorem S2. □

S.3.4. Proof of Theorem 4

Proof. To prove the consistency, we need to show that $\mathrm{pr}\{T_n > q_{T^\star}(1-\alpha)\} \to 1$ under the alternative for any $\alpha \in (0,1)$. Recall that $T^\star \overset{d}{=} \sup_{g\in\mathcal{G}} \mathbb{V}_n^\star g$. Lemma 2 implies that with probability at least $1 - cn^{-1}$, $\sup_{g\in\mathcal{G}} \mathbb{V}_n^\star g = \sup_{g\in\mathcal{G}_n} \mathbb{V}_n^\star g$. By the Talagrand concentration inequality,

$$\mathrm{pr}\left\{ \sup_{g\in\mathcal{G}_n} |\mathbb{V}_n^\star g| \geq cE\left(\sup_{g\in\mathcal{G}_n} |\mathbb{V}_n^\star g| \mid \mathscr{D} \right) + c\sigma_n \log^{1/2} n + cn^{-1/2}\kappa_n \log n \mid \mathscr{D} \right\} \leq n^{-1},$$

where

$$\begin{aligned}\sigma_n^2 &= \sup_{g\in\mathcal{G}_n} \left(\frac{n_2}{n_1+n_2} n_1^{-1} \sum_{i=1}^{n_1} [g(X_i) - \hat{E}_{n_1}\{g(X)\}]^2 + \frac{n_1}{n_1+n_2} n_2^{-1} \sum_{i=1}^{n_2} [g(Y_i) - \hat{E}_{n_2}\{g(Y)\}]^2 \right) \\ &\leq c \sup_{g\in\mathcal{G}_n} [E\{g^2(X)\} + E\{g^2(Y)\}] + cn^{-1/2} \|\mathbb{X}_{n_1}\|_{\mathcal{G}_n^2} + cn^{-1/2} \|\mathbb{Y}_{n_2}\|_{\mathcal{G}_n^2}.\end{aligned}$$

Together with the Talagrand concentration inequality and the arguments in Lemma S6, we conclude that with probability at least $1 - cn^{-1}$,

$$\sigma_n^2 \leq c + cn^{-1/2} B_n \leq c,$$

and

$$E\left(\sup_{g\in\mathcal{G}_n} |\mathbb{V}_n^\star g| \mid \mathscr{D} \right) \leq c\{s\log(pn)\}^{1/2} + cn^{-1/2}\kappa_n s \log(pn) \leq c\{s\log(pn)\}^{1/2},$$

under the condition $n^{-1}s^2 \log^4(pn) \lesssim 1$.

Thus, we show that

$$\mathrm{pr}\left[|T^\star| \leq c\{s\log(pn)\}^{1/2} \right] \geq 1 - cn^{-1}.$$

It implies that $q_{T^\star}(1-\alpha) \leq c\{s\log(pn)\}^{1/2}$ for $\alpha \in (0,1)$.

Let

$$\Upsilon_{n,g} = \left(\frac{n_1 n_2}{n_1+n_2} \right)^{1/2} \left\{ \frac{1}{n_1} \sum_{i=1}^{n_1} g(X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} g(Y_j) \right\}.$$

Then, $T_n = \sup_{g\in\mathcal{G}} \Upsilon_{n,g}$. By Bernstein's inequality (Theorem 2.8.1, Vershynin, 2018), for any $g \in \mathcal{G}$ and every $t \geq 0$,

$$\mathrm{pr}\left(|\Upsilon_{n,g} - E\Upsilon_{n,g}| \geq t \right) \leq 2\exp\left\{ -c\min(t^2, tn^{1/2}) \right\},$$

for some constant $c > 0$ not depending on g. Taking $t = (\log n)^{1/2}$ leads to

$$\begin{aligned}\mathrm{pr}\{T_n \leq q_{T^\star}(1-\alpha)\} &= \mathrm{pr}\left\{ \sup_{g\in\mathcal{G}} \Upsilon_{n,g} \leq q_{T^\star}(1-\alpha) \right\} \\ = \mathrm{pr}&\left(\cap_{g\in\mathcal{G}} \left\{ \Upsilon_{n,g} - E(\Upsilon_{n,g}) + \left(\frac{n_1 n_2}{n_1+n_2} \right)^{1/2} \left[E\{g(X)\} - E\{g(Y)\} \right] \leq q_{T^\star}(1-\alpha) \right\} \right)\end{aligned}$$

$$\leq 2\exp(-c\log n) + \mathrm{pr}\left(\left(\frac{n_1 n_2}{n_1+n_2}\right)^{1/2}\sup_{g\in\mathcal{G}}[E\{g(X)\}-E\{g(Y)\}] \leq q_{T^\star}(1-\alpha)+(\log n)^{1/2}\right)$$
$$\to 0$$

when $\sup_{g\in\mathcal{G}}[E\{g(X)\}-E\{g(Y)\}] \geq c_a n^{-1/2}\{s\log(pn)\}^{1/2}$ for a sufficiently large constant $c_a > 0$. □

S.3.5. Technical lemmas and proofs

Proof of Lemma 1. It is obvious that if $X \stackrel{d}{=} Y$, then $D(X,Y)=0$. Thus, it suffices to prove that if $D(X,Y)=0$, then $X \stackrel{d}{=} Y$, i.e., X and Y has the same distribution. To this end, note that

$$\begin{aligned} D(X,Y)=0 &\Longleftrightarrow W_1(v^\top X, v^\top Y)=0,\ \ \forall\, v^\top v=1,\\ &\Longleftrightarrow v^\top X \stackrel{d}{=} v^\top Y,\ \ \forall\, v^\top v=1\\ &\Longleftrightarrow E\{\exp(itv^\top X)\}=E\{\exp(itv^\top Y)\},\ \ \forall\, v^\top v=1\\ &\Longleftrightarrow X\stackrel{d}{=}Y, \end{aligned}$$

where the last line follows from the fact that $X \stackrel{d}{=} Y$ if and only if X and Y have the same characteristic function. □

Proof of Lemma 2. Let $V_i = X_i$ for $i=1,\ldots,n_1$, $V_{n_1+j}=Y_j$ for $j=1,\ldots,n_2$, and $n=n_1+n_2$. By Assumption 1,

$$\mathrm{pr}\left\{\max_{i=1,\ldots,n}\max_{j=1,\ldots,p}|V_{ij}|>2\nu\log(pn)\right\} \leq \sum_{i=1}^n\sum_{j=1}^p \mathrm{pr}\{|V_{ij}|>2\nu\log(pn)\} \leq 2(np)^{-1}.$$

Let Ω_n be the event defined in (S3). Then $\mathrm{pr}(\Omega_n)\geq 1-2/(pn)$. Under the event Ω_n that implies $-\kappa_n/s^{1/2}\leq V_{ij}\leq \kappa_n/s^{1/2}$ and further $-\kappa_n \leq v^\top V_i \leq \kappa_n$ for all i and $v\in\mathcal{V}_{0,s}$, it is straightforward to see that $\mathbb{G}_n\mathcal{G}=\mathbb{G}_n\mathcal{G}_n$. □

LEMMA S1. *Let $L=\lfloor \kappa_n/\epsilon_1\rfloor$ and consider the points in $[-\kappa_n,\kappa_n]$ given by*

$$x_j=(j-1)\epsilon_1,\ \ x_{-j}=(-j+1)\epsilon_1,\ \ j=1,\ldots,L.$$

Moreover, define the functions

$$\chi_+(x)=\begin{cases}0 & \text{for}\ \ x<0,\\ x & \text{for}\ \ x\in[0,1],\\ 1 & \text{otherwise.}\end{cases}\qquad \chi_-(x)=\begin{cases}0 & \text{for}\ \ x>0,\\ x & \text{for}\ \ x\in[-1,0],\\ -1 & \text{otherwise.}\end{cases}$$

For each $\beta=(\beta_{-L},\ldots,\beta_{-1},\beta_1,\ldots,\beta_L)^\top\in\{-1,1\}^{2L}$, it induces a function $f_\beta: [-\kappa_n,\kappa_n]\to[-\kappa_n,\kappa_n]$ via

$$f_\beta(x)=\sum_{j=1}^L \beta_j\epsilon_1\chi_+\left(\frac{x-x_j}{\epsilon_1}\right)+\sum_{j=1}^L\beta_{-j}\epsilon_1\chi_-\left(\frac{x-x_{-j}}{\epsilon_1}\right).$$

Define the collection of functions $\mathcal{E}_{\mathcal{F}_n,\epsilon_1} = \{f_\beta, \beta \in \{-1,1\}^{2L}\}$. *Let* $\mathcal{I} = \{f \circ v : f \in \mathcal{E}_{\mathcal{F}_n,\epsilon_1}, v \in \mathcal{E}_{\mathcal{V}_{0,s},\epsilon_2}\}$*, where* $\mathcal{E}_{\mathcal{V}_{0,s},\epsilon_2}$ *is a minimal* ϵ_2*-net of* $\mathcal{V}_{0,s}$ *under the* $\|\cdot\|_2$*. Let* $\epsilon_1 = \epsilon\|G\|_{(P+Q)/2,2}/2, \epsilon_2 = \epsilon\|G\|_{(P+Q)/2,2}/(4\nu)$*. Then, we have* $\mathcal{I}$ *is an* $\epsilon\|G\|_{(P+Q)/2,2}$*-net of* $\mathcal{G}_n$ *under the metric* $e_{(P+Q)/2}$ *and*

$$\log N = \log|\mathcal{I}| \le cs\log(ep/s) + cs\log\left(1 + \frac{8\nu}{\epsilon\|G\|_{(P+Q)/2,2}}\right) + \frac{c\kappa_n}{\epsilon\|G\|_{(P+Q)/2,2}}.$$

Proof. The set $\mathcal{E}_{\mathcal{F}_n,\epsilon_1}$ is an ϵ_1-net of $\mathcal{F}_n$ under $\|\cdot\|_\infty$ and satisfies $\log|\mathcal{E}_{\mathcal{F}_n,\epsilon_1}| \le c\kappa_n/\epsilon_1$; a proof can be found in Example 5.10 in Wainwright (2019) and Lemma 2.8 in Sen (2018). By construction of $\mathcal{I}$ and Lemma S5, we have

$$\log N \le \log|\mathcal{E}_{\mathcal{F}_n,\epsilon_1}| + \log N(\epsilon_2, \mathcal{V}_{0,s}, \|\cdot\|_2),$$

which completes the proof. □

LEMMA S2. *Let* $\mathcal{I}$ *and* $\mathcal{J}$ *be defined in* (S8) *and* (S10)*. Suppose* $\epsilon\kappa_n\log n \le c$*,* $n^{-1/3}\kappa_n \le c\tau_n$*,* $n^{-1/2}\kappa_n\{s\log(pn)\}^{1/2} \le c\tau_n$ *and* $\tau_n \le c(\log(pn))^{-2}s^{-1/2}$*. Then,*

$$\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\left(\mathbb{Z}\mathcal{J} \le t\right) - \mathrm{pr}\left(\mathbb{Z}\mathcal{I} \le t\right)\right| \le cn^{-1}, \tag{S33}$$

$$\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\left(\mathbb{G}_n\mathcal{I} \le t\right) - \mathrm{pr}\left(\mathbb{Z}\mathcal{I} \le t\right)\right| \le cn^{-1} + 2\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\big(\mathbb{G}_n\mathcal{J} \le t\big) - \mathrm{pr}\big(\mathbb{Z}\mathcal{J} \le t\big)\right|. \tag{S34}$$

In addition, with probability at least $1 - cn^{-1}$*,*

$$d_K\big(\mathcal{L}(\mathbb{Z}_n^\star\mathcal{I} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{I})\big) \le cn^{-1} + 2\sup_{t\in\mathbb{R}}\left|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J} \le t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\mathcal{J} \le t)\right|, \tag{S35}$$

$$\begin{aligned} d_K\big(\mathcal{L}(\mathbb{V}_n^\star\mathcal{I} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}_n^\star\mathcal{I} \mid \mathscr{D})\big) &\le cn^{-1} + 2d_K\big(\mathcal{L}(\mathbb{V}_n^\star\mathcal{J} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}_n^\star\mathcal{J} \mid \mathscr{D})\big) \\ &\quad + 2d_K\big(\mathcal{L}(\mathbb{Z}_n^\star\mathcal{J} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\mathcal{J})\big). \end{aligned} \tag{S36}$$

Proof. We first prove (S33) and (S34), and then prove (S35) and (S36).

Proofs of (S33) and (S34). We first observe that

$$\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\left(\mathbb{Z}\mathcal{I} \le t\right) - \mathrm{pr}\left(\mathbb{Z}\mathcal{J} \le t\right)\right| \le \sup_{t\in\mathbb{R}}\mathrm{pr}\{A(t) \cap B(t)\},$$

where

$$A(t) = \left\{\max_{g\in\mathcal{J}}\mathbb{Z}g \le t\right\} \qquad \text{and} \qquad B(t) = \left\{\max_{g\in\mathcal{J}^c}\mathbb{Z}g \ge t\right\},$$

with $\mathcal{J}^c$ denotes the complement of $\mathcal{J}$ relative to $\mathcal{I}$. It is further observed that

$$\sup_{t\in\mathbb{R}}\mathrm{pr}\big\{A(t) \cap B(t)\big\} \le \mathrm{pr}\{A(t_2)\} + \mathrm{pr}\{B(t_1)\}$$

for any pair (t_1, t_2) of real numbers satisfying $t_1 \le t_2$. In a similar manner, we have

$$\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\left(\mathbb{G}_n\mathcal{J} \le t\right) - \mathrm{pr}\left(\mathbb{G}_n\mathcal{I} \le t\right)\right| \le \sup_{t\in\mathbb{R}}\mathrm{pr}\{A'(t) \cap B'(t)\} \le \mathrm{pr}\{A'(t_2)\} + \mathrm{pr}\{B'(t_1)\}$$

$$\leq \text{pr}\{B'(t_1)\} + \text{pr}\{A(t_2)\} + \sup_{t\in\mathbb{R}} \big|\text{pr}(\mathbb{G}_n\mathcal{J} \leq t) - \text{pr}(\mathbb{Z}\mathcal{J} \leq t)\big|,$$

where

$$A'(t) = \left\{\max_{g\in\mathcal{J}} \mathbb{G}_n g \leq t\right\} \qquad \text{and} \qquad B'(t) = \left\{\max_{g\in\mathcal{J}^c} \mathbb{G}_n g \geq t\right\}.$$

Terms similar to $\text{pr}\{A(t_2)\}$ and $\text{pr}\{B(t_1)\}$ appear in Lopes et al. (2020); Lin et al. (2023) and are handled with additional conditions on the variance decay and correlation structure. However, we do not have such structures on $\mathbb{Z}g$ whose properties are determined by the function class $\mathcal{J}$ and the probability measures of X and Y. Our strategy to overcome the above challenge is to consider a smaller class $\mathcal{J}^\circ \subset \mathcal{J}$ of functions in view of the inequality $\text{pr}\{A(t_2)\} = \text{pr}\big(\mathbb{Z}\mathcal{J} \leq t_2\big) \leq \text{pr}\big(\mathbb{Z}\mathcal{J}^\circ \leq t_2\big)$. The key is to find a suitable class $\mathcal{J}^\circ$ so that

(i) the cardinality $|\mathcal{J}^\circ|$ is sufficiently large; otherwise, the bound by $\text{pr}\big(\mathbb{Z}\mathcal{J}^\circ \leq t_2\big)$ would be too conservative.

(ii) the correlations among the induced variables $\mathbb{Z}g$ for $g \in \mathcal{J}^\circ$ are not too strong; this will then allow us to apply Theorem 2.1 in Lopes & Yao (2022).

In Proposition S1 we show that such a class $\mathcal{J}^\circ$ exists.

Without loss of generality, we assume $S = [0,1]$ in Assumption 3. Take a constant $\varsigma_n \in (0, 1/4)$, to be chosen later, such that $\varsigma_n = r\epsilon_1$ for some potentially diverging integer $r \geq 2$, where ϵ_1 is as defined in Lemma S1. Let $\mathcal{F}^\circ_{\varsigma_n}$ be the class of functions asserted by Proposition S1 with $\varsigma = \varsigma_n$ and $\mathcal{J}^\circ = \mathcal{F}^\circ_{\varsigma_n} \circ v_\circ$, where $v_\circ$ is defined in Assumption 3. Then, for any $g, h \in \mathcal{J}^\circ$, $\text{cor}\{h(X), g(X)\} \leq \rho_0 < 1$ and $\text{cor}\{h(Y), g(Y)\} \leq \rho_0 < 1$. We claim that the class $\mathcal{J}^\circ$ satisfies the following properties.

- The cardinality $K = K_n = |\mathcal{J}^\circ| \geq c2^{1/(8\varsigma_n)}$.
- $\sup_{g\in\mathcal{J}^\circ} \text{var}(\mathbb{Z}g) \asymp \varsigma_n^2$ and $\inf_{g\in\mathcal{J}^\circ} \text{var}(\mathbb{Z}g) \asymp \varsigma_n^2$.
- $\text{cor}(\mathbb{Z}g, \mathbb{Z}h) \leq \rho_0 < 1$ for any two distinct $g, h \in \mathcal{J}^\circ$.

The first and second properties are directly obtained from (c) and (d) in Proposition S1 under Assumption 3. It suffices to prove the last property. The goal is to show that for some $\rho_0 < 1$ and any $g, h \in \mathcal{J}^\circ$,

$$\begin{aligned}\text{cor}(\mathbb{Z}h, \mathbb{Z}g) &= \frac{n^{-1}[n_2\text{cov}\{h(X), g(X)\} + n_1\text{cov}\{h(Y), g(Y)\}]}{(n^{-1}[n_2\text{var}\{h(X)\} + n_1\text{var}\{h(Y)\}])^{1/2}(n^{-1}[n_2\text{var}\{g(X)\} + n_1\text{var}\{g(Y)\}])^{1/2}} \\ &\leq \rho_0.\end{aligned}$$

To see this, we notice that

$$\begin{aligned}&n^{-1}[n_2\text{cov}\{h(X), g(X)\} + n_1\text{cov}\{h(Y), g(Y)\}] \\ &\leq n^{-1}n_2\rho_0[\text{var}\{h(X)\}\text{var}\{g(X)\}]^{1/2} + n^{-1}n_1\rho_0[\text{var}\{h(Y)\}\text{var}\{g(Y)\}]^{1/2} \\ &= \rho_0 n^{-1}\big(n_2^2\text{var}\{h(X)\}\text{var}\{g(X)\} + n_1^2\text{var}\{h(Y)\}\text{var}\{g(Y)\} +\end{aligned}$$

$$
\begin{aligned}
&\quad 2n_1n_2[\mathrm{var}\{h(X)\}\mathrm{var}\{g(X)\}\mathrm{var}\{h(Y)\}\mathrm{var}\{g(Y)\}]^{1/2}\big)^{1/2} \\
&\le \rho_0 n^{-1}[n_2^2\mathrm{var}\{h(X)\}\mathrm{var}\{g(X)\} + n_1^2\mathrm{var}\{h(Y)\}\mathrm{var}\{g(Y)\}+ \\
&\quad n_1n_2\mathrm{var}\{h(X)\}\mathrm{var}\{g(Y)\} + n_1n_2\mathrm{var}\{g(X)\}\mathrm{var}\{h(Y)\}]^{1/2} \\
&= \rho_0(n^{-1}[n_2\mathrm{var}\{h(X)\} + n_1\mathrm{var}\{h(Y)\}])^{1/2}(n^{-1}[n_2\mathrm{var}\{g(X)\} + n_1\mathrm{var}\{g(Y)\}])^{1/2},
\end{aligned}
$$

where the second inequality holds because $2ab \le a^2 + b^2$.

Take

$$
t_1 = C_1^\star\big[(\tau_n\kappa_n)^{1/2} + \tau_n\{s\log(pn)\}^{1/2}\big],
$$

$$
t_2 = C_2^\star\varsigma_n\delta_0\{2(1-\rho_0)\log K\}^{1/2},
$$

for some constants $C_1^\star, C_2^\star > 0$ and $\delta_0 \in (0,1)$. Letting $\varsigma_n \asymp (\log n)^{-1}$, the inequality $t_1 \le t_2$ can be established due to the definition K and the conditions on τ_n. Now we are ready to bound the terms of interest.

Bound $\mathrm{pr}\{A(t_2)\}$**.** We have $\varsigma_n \ge C\tau_n$ for some positive constant C such that $\mathcal{J}^\circ \subset \mathcal{J}$. According to Lemma S17, for some $\delta_0 \in (0,1)$,

$$
\begin{aligned}
\mathrm{pr}\{A(t_2)\} &= \mathrm{pr}\bigg(\max_{g\in\mathcal{J}} \mathbb{Z}g \le t_2\bigg) \le \mathrm{pr}\bigg(\max_{g\in\mathcal{J}^\circ} \mathbb{Z}g \le t_2\bigg) \\
&\le \mathrm{pr}\bigg[\max_{g\in\mathcal{J}^\circ} \mathbb{Z}g/\sigma_g \le \delta_0\{2(1-\rho_0)\log K\}^{1/2}\bigg] \\
&\le cK^{\frac{-(1-\rho_0)(1-\delta_0)^2}{\rho_0}}(\log K)^{\frac{1-\rho_0(2-\delta_0)-\delta_0}{2\rho_0}} \\
&\le c2^{\frac{-(1-\rho_0)(1-\delta_0)^2}{8\varsigma_n\rho_0}}(\varsigma_n^{-1})^{\frac{1-\rho_0(2-\delta_0)-\delta_0}{2\rho_0}} \le cn^{-1},
\end{aligned}
$$

where $\sigma_g^2 = \mathrm{var}(\mathbb{Z}g)$.

Bound $\mathrm{pr}\{B(t_1)\}$**.** Similar to the derivations that leads to (S23), we have

$$
E(\mathbb{Z}\mathcal{J}^c) \le c\tau_n\{s\log(pn)\}^{1/2} + c(\tau_n\kappa_n)^{1/2},
$$

and according to the Borell–Sudakov–Tsirel'son inequality,

$$
\begin{aligned}
\mathrm{pr}\{B(t_1)\} &= \mathrm{pr}\left(\mathbb{Z}\mathcal{J}^c \ge t_1\right) \\
&\le \mathrm{pr}\big\{\mathbb{Z}\mathcal{J}^c \ge E(\mathbb{Z}\mathcal{J}^c) + \tau_n(2\log n)^{1/2}\big\} \le n^{-1},
\end{aligned}
$$

where the first inequality follows from the definition of t_1 and $\log(1/\tau_n) \le c\log(pn)$.

Bound $\mathrm{pr}\{B'(t_1)\}$**.** Note that

$$
\begin{aligned}
E(\|\mathbb{G}_n\|_{\mathcal{J}^c}) &= \left(\frac{n_1n_2}{n_1+n_2}\right)^{1/2} E\bigg(\sup_{g\in\mathcal{J}^c}\bigg|\frac{1}{n_1}\sum_{i=1}^{n_1}\big[g(X_i) - E\{g(X)\}\big] - \frac{1}{n_2}\sum_{j=1}^{n_2}\big[g(Y_j) - E\{g(Y)\}\big]\bigg|\bigg) \\
&\le E(\|\mathbb{X}_{n_1}\|_{\mathcal{J}^c}) + E(\|\mathbb{Y}_{n_2}\|_{\mathcal{J}^c}).
\end{aligned}
$$

It suffices to establish the bound for $\mathbb{X}_{n_1}$, as similar arguments apply to $\mathbb{Y}_{n_2}$. By replacing $\mathcal{G}_n$ with $\mathcal{J}^c$ in the proof of Lemma S6 and using the same notation, we obtain

$$E(\|\mathbb{X}_{n_1}\|_{\mathcal{J}^c}) \leq c\|G\|_{P,2}E\{J(\sigma_{n,\mathcal{J}^c}/\|G\|_{P,2})\} + c\kappa_n n^{-1/2},$$

where $\sigma^2_{n,\mathcal{J}^c} = \sup_{g\in\mathcal{J}^c} n_1^{-1}\sum_{i=1}^{n_1} g^2(X_i)$. Together with the arguments in the proof of Theorem 5.2 in Chernozhukov et al. (2014), we conclude that

$$E(\|\mathbb{X}_{n_1}\|_{\mathcal{J}^c}) \leq c\|G\|_{P,2}J(\tau_n/\|G\|_{P,2}) + c\frac{J^2(\tau_n/\|G\|_{P,2})\|\Gamma\|_{P,2}}{(\tau_n/\|G\|_{P,2})^2 n^{1/2}} + c\kappa_n n^{-1/2}.$$

By setting $G \equiv \kappa_n$ and performing a further simple calculation, we have

$$E(\|\mathbb{X}_{n_1}\|_{\mathcal{J}^c}) \leq c\tau_n\{s\log(pn)\}^{1/2} + c(\kappa_n\tau_n)^{1/2} + cn^{-1/2}\{\kappa_n^2\tau_n^{-1} + s\kappa_n\log(pn)\}.$$

We apply the Talagrand concentration inequality in Lemma S20 to conclude that, for any $\delta > 0$,

$$\mathrm{pr}\{B'(t_1)\} \leq \mathrm{pr}\left\{\mathbb{G}_n\mathcal{J}^c \geq (1+\delta)E(\|\mathbb{G}_n\|_{\mathcal{J}^c}) + c\tau_n(\log n)^{1/2} + \left(c + \frac{c}{\delta}\right)\frac{\kappa_n(\log n)}{n^{1/2}}\right\} \leq cn^{-1},$$

where the first inequality follows from the definition of t_1 and the conditions $n^{-1/3}\kappa_n \leq c\tau_n$ and $n^{-1/2}\kappa_n\{s\log(pn)\}^{1/2} \leq c\tau_n$. Combining all the above arguments completes the proofs of (S33) and (S34).

Proofs of (S35) and (S36). Note that

$$\begin{aligned}
&\sup_{t\in\mathbb{R}}\left|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{I} \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\mathcal{I} \leq t)\right| \\
\leq &\sup_{t\in\mathbb{R}}|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{I} \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J} \leq t \mid \mathscr{D})| + \sup_{t\in\mathbb{R}}|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J} \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\mathcal{J} \leq t)| + \\
&\sup_{t\in\mathbb{R}}|\mathrm{pr}(\mathbb{Z}\mathcal{I} \leq t) - \mathrm{pr}(\mathbb{Z}\mathcal{J} \leq t)| \\
\leq &\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J} \leq t_2 \mid \mathscr{D}) + \mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}^c \geq t_1 \mid \mathscr{D}) + \mathrm{pr}(\mathbb{Z}\mathcal{J} \leq t_2) + \mathrm{pr}(\mathbb{Z}\mathcal{J}^c \geq t_1) + \\
&\sup_{t\in\mathbb{R}}|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J} \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\mathcal{J} \leq t)| \\
\leq &2\mathrm{pr}(\mathbb{Z}\mathcal{J} \leq t_2) + \mathrm{pr}(\mathbb{Z}\mathcal{J}^c \geq t_1) + \mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}^c \geq t_1 \mid \mathscr{D}) + 2\sup_{t\in\mathbb{R}}|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J} \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\mathcal{J} \leq t)|.
\end{aligned}$$

The first two terms have been controlled above. The third term is addressed below. Thus, we conclude that with probability at least $1 - cn^{-1}$,

$$\sup_{t\in\mathbb{R}}\left|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{I} \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\mathcal{I} \leq t)\right| \leq cn^{-1} + 2\sup_{t\in\mathbb{R}}|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J} \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\mathcal{J} \leq t)|.$$

Also,

$$\begin{aligned}
&\sup_{t\in\mathbb{R}}\left|\mathrm{pr}(\mathbb{V}_n^\star\mathcal{I} \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}_n^\star\mathcal{I} \leq t \mid \mathscr{D})\right| \\
\leq &\sup_{t\in\mathbb{R}}\left|\mathrm{pr}(\mathbb{V}_n^\star\mathcal{J} \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J} \leq t \mid \mathscr{D})\right| + \sup_{t\in\mathbb{R}}\left|\mathrm{pr}(\mathbb{V}_n^\star\mathcal{I} \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{V}_n^\star\mathcal{J} \leq t \mid \mathscr{D})\right| +
\end{aligned}$$

$$
\begin{aligned}
&\sup_{t\in\mathbb{R}}\big|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{I}\le t\mid\mathscr{D})-\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}\le t\mid\mathscr{D})\big|\\
&\le \mathrm{pr}(\mathbb{V}_n^\star\mathcal{J}\le t_2)+\mathrm{pr}(\mathbb{V}_n^\star\mathcal{J}^c\ge t_1)+\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}\le t_2)+\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}^c\ge t_1)+\\
&\quad\sup_{t\in\mathbb{R}}\big|\mathrm{pr}(\mathbb{V}_n^\star\mathcal{J}\le t\mid\mathscr{D})-\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}\le t\mid\mathscr{D})\big|\\
&\le 2\mathrm{pr}(\mathbb{Z}\mathcal{J}\le t_2)+\mathrm{pr}(\mathbb{V}_n^\star\mathcal{J}^c\ge t_1)+\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}^c\ge t_1)+2\sup_{t\in\mathbb{R}}\big|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}\le t\mid\mathscr{D})-\mathrm{pr}(\mathbb{Z}\mathcal{J}\le t)\big|+\\
&\quad 2\sup_{t\in\mathbb{R}}\big|\mathrm{pr}(\mathbb{V}_n^\star\mathcal{J}\le t\mid\mathscr{D})-\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}\le t\mid\mathscr{D})\big|.
\end{aligned}
$$

The first term has been controlled above. The second and third terms are addressed below. Thus, we conclude that with probability at least $1-cn^{-1}$,

$$
\begin{aligned}
\sup_{t\in\mathbb{R}}\big|\mathrm{pr}(\mathbb{V}_n^\star\mathcal{I}\le t\mid\mathscr{D})-\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{I}\le t\mid\mathscr{D})\big|\le cn^{-1}&+2\sup_{t\in\mathbb{R}}\big|\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}\le t\mid\mathscr{D})-\mathrm{pr}(\mathbb{Z}\mathcal{J}\le t)\big|+\\
&2\sup_{t\in\mathbb{R}}\big|\mathrm{pr}(\mathbb{V}_n^\star\mathcal{J}\le t\mid\mathscr{D})-\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}\le t\mid\mathscr{D})\big|.
\end{aligned}
$$

Bound $\mathrm{pr}(\mathbb{Z}_n^\star\mathcal{J}^c\ge t_1\mid\mathscr{D})$. We apply the Borell–Sudakov–Tsirel'son inequality to deduce that

$$
\mathrm{pr}\{\mathbb{Z}_n^\star\mathcal{J}^c>E(\mathbb{Z}_n^\star\mathcal{J}^c\mid\mathscr{D})+\sigma_{n,\mathcal{J}^c}(2\log n)^{1/2}\mid\mathscr{D}\}\le 2n^{-1},
$$

where

$$
\begin{aligned}
\sigma^2_{n,\mathcal{J}^c}&=\sup_{g\in\mathcal{J}^c}\frac{n_2}{n_1+n_2}\frac{1}{n_1}\sum_{i=1}^{n_1}[g(X_i)-\hat{E}_{n_1}\{g(X)\}]^2+\frac{n_1}{n_1+n_2}\frac{1}{n_2}\sum_{i=1}^{n_2}[g(Y_i)-\hat{E}_{n_2}\{g(Y)\}]^2\\
&\le\sup_{g\in\mathcal{J}^c}\frac{n_2}{n_1+n_2}\frac{1}{n_1}\sum_{i=1}^{n_1}[g(X_i)-E\{g(X)\}]^2+\frac{n_1}{n_1+n_2}\frac{1}{n_2}\sum_{i=1}^{n_2}[g(Y_i)-E\{g(Y)\}]^2\\
&\le\tau_n^2+\sup_{g\in\mathcal{J}^c}n_1^{-1}\sum_{i=1}^{n_1}\big([g(X_i)-E\{g(X)\}]^2-E[g(X_i)-E\{g(X)\}]^2\big)+\\
&\quad\sup_{g\in\mathcal{J}^c}n_2^{-1}\sum_{i=1}^{n_2}\big([g(Y_i)-E\{g(Y)\}]^2-E[g(Y_i)-E\{g(Y)\}]^2\big).
\end{aligned}
$$

It suffices to bound the second term, similar arguments apply to the third term. Applying Lemma S20 with $t=\log n$ yields

$$
\begin{aligned}
&\mathrm{pr}\Bigg[\sup_{g\in\mathcal{J}^c}n_1^{-1/2}\sum_{i=1}^{n_1}\big([g(X_i)-E\{g(X)\}]^2-E[g(X_i)-E\{g(X)\}]^2\big)\\
&\quad\ge(1+\delta)E\Bigg\{\sup_{g\in\mathcal{J}^c}n_1^{-1/2}\sum_{i=1}^{n_1}\big([g(X_i)-E\{g(X)\}]^2-E[g(X_i)-E\{g(X)\}]^2\big)\Bigg\}+\\
&\quad c\breve{\sigma}(\log n)^{1/2}+(c+c^2\delta^{-1})\frac{\kappa_n^2\log n}{n^{1/2}}\Bigg]\le n^{-1},
\end{aligned}
$$

where $\breve{\sigma}^2 = \sup_{g\in\mathcal{J}^c} n_1^{-1}\sum_{i=1}^{n_1} E\big([g(X_i)-E\{g(X)\}]^2 - E[g(X_i)-E\{g(X)\}]^2\big)^2 \le c\tau_n^2\kappa_n^2$ by Assumption 2. By similar arguments in Lemma S6, we deduce

$$
\begin{aligned}
&E\bigg\{\sup_{g\in\mathcal{J}^c} n_1^{-1/2}\sum_{i=1}^{n_1}\big([g(X_i)-E\{g(X)\}]^2 - E[g(X_i)-E\{g(X)\}]^2\big)\bigg\}\\
&\le c\bigg\{J(\tau_n/\kappa_n)\kappa_n^2 + \frac{J^2(\tau_n/\kappa_n)\kappa_n^2}{n^{1/2}(\tau_n/\kappa_n)^2}\bigg\}\\
&\le c[\tau_n\kappa_n\{s\log(pn)\}^{1/2} + (\tau_n\kappa_n^3)^{1/2} + n^{-1/2}\{\kappa_n^2 s\log(pn) + \kappa_n^3\tau_n^{-1}\}].
\end{aligned}
$$

Finally, with probability at least $1-cn^{-1}$, we have $\sigma^2_{n,\mathcal{J}^c}\le c\tau_n^2$, under the conditions $n^{-1/2}\kappa_n\{s\log(pn)\}^{1/2}\le c\tau_n$ and $n^{-1/3}\kappa_n\le c\tau_n$.

Next, we bound $E(\mathbb{Z}_n^\star\mathcal{J}^c \mid \mathscr{D})$. By the Dudley's maximal inequality, we have with probability at least $1-cn^{-1}$,

$$
\begin{aligned}
&E\bigg\{\sup_{g\in\mathcal{J}^c} n^{-1/2}\bigg(\sum_{i=1}^{n_1}\xi_i(n_2/n_1)^{1/2}[g(X_i)-\hat{E}_{n_1}\{g(X)\}]-\\
&\sum_{i=1}^{n_2}\xi_{n_1+i}(n_1/n_2)^{1/2}[g(Y_i)-\hat{E}_{n_2}\{g(Y)\}]\bigg) \mid \mathscr{D}\bigg\}\\
&\le c\int_0^{\sigma_{n,\mathcal{J}^c}}\sup_{m\in\mathfrak{M}_n}\{\log N(\delta,\mathcal{G}_n,e_m)\}^{1/2}d\delta\\
&\le c\int_0^{\tau_n}\sup_{m\in\mathfrak{M}_n}\{\log N(\delta,\mathcal{G}_n,e_m)\}^{1/2}d\delta\\
&\le c\tau_n\{s\log(pn)\}^{1/2} + c(\kappa_n\tau_n)^{1/2}.
\end{aligned}
$$

where $\mathfrak{M}_n$ is defined in Lemma S5. Combing all pieces together yields that with probability at least $1-cn^{-1}$,

$$
\mathrm{pr}\big(\mathbb{Z}_n^\star\mathcal{J}^c \ge t_1 \mid \mathscr{D}\big) \le cn^{-1}.
$$

Bound $\mathrm{pr}(\mathbb{V}_n^\star\mathcal{J}^c\ge t_1\mid\mathscr{D})$**.** Applying Lemma S20 together with $t=\log n$ yields

$$
\mathrm{pr}\Big\{\mathbb{V}_n^\star\mathcal{J}^c \ge (1+\delta)E(\mathbb{V}_n^\star\mathcal{J}^c\mid\mathscr{D}) + c\sigma_{n,\mathcal{J}^c}(\log n)^{1/2} + (c+c^2\delta^{-1})\frac{\kappa_n\log n}{n^{1/2}} \mid \mathscr{D}\Big\} \le n^{-1}.
$$

From the above, we already obtain that with probability at least $1-cn^{-1}$, $\sigma_{n,\mathcal{J}^c}\le c\tau_n$. It suffices to bound $E(\mathbb{V}_n^\star\mathcal{J}^c\mid\mathscr{D})$. Applying similar arguments in Lemma S6, we obtain with probability at least $1-cn^{-1}$,

$$
\begin{aligned}
&E(\mathbb{V}_n^\star\mathcal{J}^c\mid\mathscr{D})\\
&\le E\bigg\{\sup_{g\in\mathcal{J}^c}\bigg(\bigg|n_1^{-1/2}\sum_{i=1}^{n_1}[g(X_i^\star)-\hat{E}_{n_1}\{g(X)\}]\bigg| \mid \mathscr{D}\bigg)\bigg\}+
\end{aligned}
$$

$$
\begin{aligned}
&E\Big\{\sup_{g\in\mathcal{J}^c}\Big(\Big|n_2^{-1/2}\sum_{i=1}^{n_2}[g(Y_i^\star)-\hat{E}_{n_2}\{g(Y)\}]\Big| \mid \mathscr{D}\Big)\Big\}\\
&\le cJ(\tau_n/\kappa_n)\kappa_n + c\frac{J^2(\tau_n/\kappa_n)\kappa_n}{(\tau_n/\kappa_n)^2 n^{1/2}}\\
&\le c\tau_n\{s\log(pn)\}^{1/2} + c(\kappa_n\tau_n)^{1/2} + cn^{-1/2}\{\kappa_n^2\tau_n^{-1} + s\kappa_n\log(pn)\}\\
&\le c\tau_n\{s\log(pn)\}^{1/2} + c(\kappa_n\tau_n)^{1/2},
\end{aligned}
$$

where the last inequality follows from the conditions $n^{-1/2}\kappa_n\{s\log(pn)\}^{1/2} \le c\tau_n$ and $n^{-1/3}\kappa_n \le c\tau_n$. Consequently, with probability at least $1-cn^{-1}$, we conclude $\mathrm{pr}(\mathbb{V}_n^\star\mathcal{J}^c \ge t_1 \mid \mathscr{D}) \le cn^{-1}$. □

PROPOSITION S1. *Suppose Z is a real random variable such that $\mathrm{pr}(Z\in A)\ge C_0\mathrm{Leb}(A)$ for a fixed constant $C_0>0$ for all Borel sets $A\subset[0,1]$. Then, for some positive constants C_1, C_2, C_3 and $\rho<1$, for any $\varsigma\in(0,1/2]$, there exists $\mathcal{F}_\varsigma^\circ\subset\mathcal{F}$ with the following properties:*

(a) $\sup_{f\in\mathcal{F}_\varsigma^\circ}\sup_{z\in\mathbb{R}}|f(z)|\le\varsigma$.
(b) Each function $f\in\mathcal{F}_\varsigma^\circ$ is linear on the interval $[(j-1)\varsigma/2, j\varsigma/2]$ for each $j=1,\dots,2k$ with slope -1 or 1, and $f(z)=0$ for $z\notin[0,k\varsigma]$, where $k=\lfloor 1/\varsigma\rfloor$.
(c) The cardinality $|\mathcal{F}_\varsigma^\circ|\ge C_1 2^{1/(8\varsigma)}$.
(d) $C_2\varsigma^2\le\sup_{f\in\mathcal{F}_\varsigma^\circ}\mathrm{var}\{f(Z)\}\le C_3\varsigma^2$.
(e) $\mathrm{cor}\{f(Z),g(Z)\}\le\rho$ *for any two distinct $f,g\in\mathcal{F}_\varsigma^\circ$.*

Proof. Define $\varphi:\mathbb{R}\to\mathbb{R}$ by

$$
\varphi(x)=\begin{cases} x & \text{if } x\in[0,\varsigma/2],\\ \varsigma - x & \text{if } x\in[\varsigma/2,\varsigma],\\ 0 & \text{otherwise.}\end{cases}
$$

Consider the set of (basis) functions $\varphi_1,\dots,\varphi_k$ obtained by translating φ by ς, i.e., $\varphi_j(z)=\varphi\big(z-(j-1)\varsigma\big)$ for $z\in\mathbb{R}$. Take $\mathcal{H}_k$ to be a subset of $\{-1,1\}^k$ such that $d_H(\omega,\omega')\ge k/8$ for any $\omega,\omega'\in\mathcal{H}_k$, where d_H denotes the Hamming distance on $\mathcal{H}_k$. Let $\mathcal{F}_\varsigma^\circ=\{f_\omega(x)=\sum_{j=1}^k\omega_j\varphi_j(x),\omega\in\mathcal{H}_k\}$. We immediately see that $\mathcal{F}_\varsigma^\circ$ has the properties (a) and (b). According to the Varshamov–Gilbert bound (Tsybakov, 2009, Lemma 2.9), $\mathcal{F}_\varsigma^\circ$ satisfies (c). Below we show that it also has the properties (d) and (e).

Proof of Property (d). Define $I_j=[(j-1)\varsigma, j\varsigma]$ for $j=1,\dots,k$. Let J be the (random) index j such that $Z\in I_j$. We first observe

$$
\begin{aligned}
\mathrm{var}\{f_\omega(Z)\} &\ge E\big[\mathrm{var}\{f_\omega(Z)|J\}\big]\\
&=\sum_{j=1}^k \mathrm{pr}(J=j)\mathrm{var}\{f_\omega(Z)|J=j\}
\end{aligned}
$$

$$
\begin{aligned}
&= \sum_{j=1}^{k} \mathrm{pr}(J=j) \int_{I_j} [\varphi_j(z) - E\{\varphi_j(Z)|Z \in I_j\}]^2 \mathrm{d}F_{Z|j}(z) \\
&\geq C_0 \sum_{j=1}^{k} \mathrm{Leb}(I_j) \int_{I_j} [\varphi_j(z) - E\{\varphi_j(Z)|U \in I_j\}]^2 \frac{1}{\mathrm{Leb}(I_j)} \mathrm{d}z \\
&\geq C_0 \sum_{j=1}^{k} \mathrm{Leb}(I_j) \mathrm{var}_{U\sim \mathrm{Unif}(I_j)}\{\varphi_j(U)\} \\
&= C_0 \mathrm{Leb}([0, k\varsigma]) \frac{\varsigma^2}{12} \geq C_2 \varsigma^2,
\end{aligned}
$$

where $F_{Z|j}$ denotes the conditional distribution of Z given $J = j$ and $\mathrm{var}_{U\sim \mathrm{Unif}(I_j)}\{\varphi_j(U)\}$ stands for the variance of $\varphi_j(U)$ where U has a uniform distribution over I_j. The third inequality holds since the expected value minimizes the mean squared error, i.e., $u = EU$ minimizes $E(U - u)^2$. Then the property (d) follows from

$$
\mathrm{var}\big\{f_\omega(Z)\big\} \leq \sum_{j=1}^{k} \int_{I_j} \varphi_j^2(z) \mathrm{d}F_Z \leq \varsigma^2,
$$

where F_Z denotes the distribution of Z.

Proof of Property (e). First, we show that for some constant $c > 0$ not depending on ς,

$$
s(t) := \mathrm{var}\big\{f_\omega(Z) - t f_{\omega'}(Z)\big\} \geq c\varsigma^2 \tag{S37}
$$

for any number t and any two distinct $\omega, \omega' \in \mathcal{H}_k$. For this, define the set $\mathcal{D} = \{j : \omega_j \neq \omega'_j\}$; note that $|\mathcal{D}| \geq k/8$ according to the construction of $\mathcal{H}_k$. Let J be the (random) index j such that $Z \in I_j$. Observe that

$$
\begin{aligned}
s(t) &= \mathrm{var}\big\{f_\omega(Z) - t f_{\omega'}(Z)\big\} \\
&= \mathrm{var}\left\{ \sum_{j\in\mathcal{D}} (1+t)\omega_j\varphi_j(Z) + \sum_{j\notin\mathcal{D}} (1-t)\omega_j\varphi_j(Z) \right\} \\
&\geq E\left[\mathrm{var}\left\{ \sum_{j\in\mathcal{D}} (1+t)\omega_j\varphi_j(Z) + \sum_{j\notin\mathcal{D}} (1-t)\omega_j\varphi_j(Z) \middle| J \right\} \right] \\
&= \sum_{j\in\mathcal{D}} (1+t)^2 \mathrm{pr}(Z \in I_j) \mathrm{var}\left\{\varphi_j(Z)|Z \in I_j\right\} \\
&\quad + \sum_{j\notin\mathcal{D}} (1-t)^2 \mathrm{pr}(Z \in I_j) \mathrm{var}\left\{\varphi_j(Z)|U \in I_j\right\} \\
&\geq \sum_{j\in\mathcal{D}} C_0(1+t)^2 \mathrm{Leb}(I_j) \frac{\varsigma^2}{12} + \sum_{j\notin\mathcal{D}} C_0(1-t)^2 \mathrm{Leb}(I_j) \frac{\varsigma^2}{12} \\
&= \frac{C_0(k\varsigma)\varsigma^2}{12} t^2 - \frac{\delta C_0 \varsigma^2}{6} t + \frac{C_0(k\varsigma)\varsigma^2}{12},
\end{aligned}
$$

where $\delta = \sum_{j \notin \mathcal{D}} \mathrm{Leb}(I_j) - \sum_{j \in \mathcal{D}} \mathrm{Leb}(I_j) = k\varsigma - 2\sum_{j \in \mathcal{D}} \mathrm{Leb}(I_j)$. Since $d_{\mathbb{H}}(\omega, \omega') \geq k/8$, we have $\sum_{j \in \mathcal{D}} \mathrm{Leb}(I_j) \geq k\varsigma/8$ and thus $\delta \leq k\varsigma - k\varsigma/4 = 3k\varsigma/4$ since $1/2 \leq k\varsigma \leq 1$. This yields

$$\begin{aligned} s(t) &\geq \frac{C_0(k\varsigma)\varsigma^2}{12}t^2 - \frac{\delta C_0 \varsigma^2}{6}t + \frac{C_0(k\varsigma)\varsigma^2}{12} \\ &= \frac{C_0(k\varsigma)\varsigma^2}{12}[t^2 - 2\delta t/(k\varsigma) + \{\delta/(k\varsigma)\}^2 + 1 - \{\delta/(k\varsigma)\}^2] \\ &\geq \frac{C_0\varsigma^2}{24}(1 - 9/16) \geq c\varsigma^2 \end{aligned}$$

for some constant $c > 0$ not depending on t and ω. This proves (S37).

Now with (S37) and $t = \mathrm{cov}\{f_\omega(Z), f_{\omega'}(Z)\}/\mathrm{var}\{f_{\omega'}(Z)\}$, we deduce that

$$\begin{aligned} s(t) &= \mathrm{var}\{f_{\omega'}(Z)\}t^2 - 2\mathrm{cov}\{f_\omega(Z), f_{\omega'}(Z)\}t + \mathrm{var}\{f_\omega(Z)\} \\ &= \mathrm{var}\{f_\omega(Z)\} - \frac{\mathrm{cov}^2\{f_\omega(Z), f_{\omega'}(Z)\}}{\mathrm{var}\{f_{\omega'}(Z)\}} \geq c\varsigma^2 \geq c' \cdot \mathrm{var}\{f_\omega(Z)\}, \end{aligned}$$

where the last inequality is due to Part (d) and $c' > 0$ is a constant not depending on t and ω. This immediately implies

$$\mathrm{cor}^2\{f_\omega(Z), f_{\omega'}(Z)\} < 1 - c'$$

and establishes Part (e). □

LEMMA S3. *Let* $\mathcal{F}_L = \{f : [-L, L] \to [-L, L] \mid f$ *is 1-Lipschitz and* $f(0) = 0\}$ *with* $L > 0$. *Then for some universal constant* $c > 0$*, we have*

$$\log N(\epsilon, \mathcal{F}_L, \|\cdot\|_\infty) \leq c\frac{L}{\epsilon}, \textit{ for all } \epsilon > 0.$$

Proof. Define a new class $\mathcal{R}$ of functions $r(x) : [-1, 1] \to [-1, 1]$ with $r(x) = f(Lx)/L$. It is clear that $r(x)$ is 1-Lipschitz. Then $\log N(\epsilon, \mathcal{R}, \|\cdot\|_\infty) \leq c/\epsilon$, which is established in Example 5.10 of Wainwright (2019). Consequently, $\log N(\epsilon, \mathcal{F}_L, \|\cdot\|_\infty) \leq \log N(\epsilon/L, \mathcal{R}, \|\cdot\|_\infty) \leq cL/\epsilon$. □

LEMMA S4. *Let* $\mathcal{V}_{0,s} = \{v \in \mathbb{R}^p : \|v\|_2 = 1, \|v\|_0 \leq s\}$. *Then for any* $\epsilon > 0$,

$$\log N(\epsilon, \mathcal{V}_{0,s}, \|\cdot\|_2) \leq s\log(ep/s) + s\log(1 + 2/\epsilon).$$

$$\log N(\epsilon, \mathcal{V}_{0,s}, \|\cdot\|_1) \leq s\log(ep/s) + s\log(1 + 2s^{1/2}/\epsilon).$$

Proof. Given the locations of s non-zero elements, the ϵ-covering number of the unit sphere in $\mathbb{R}^s$ under the Euclidian metric is upper bounded by $(1 + 2/\epsilon)^s$, according to the volume ratio argument in Wainwright (2019, Lemma 5.7). Moreover, the number of ways to choose

s dimensions out of p dimensions is $\binom{p}{s}$, which is upper bounded by $(ep/s)^s$ by the Stirling's approximation. Combining these together establishes the first claim.

Let $\mathcal{B}_{1,s} = \{v \in \mathbb{R}^s : \|v\|_1 \le 1\}$ and $\mathcal{B}_{2,s} = \{v \in \mathbb{R}^s : \|v\|_2 \le 1\}$. Using Lemma 5.7 in Wainwright (2019), we deduce

$$
\begin{aligned}
N(\epsilon, \mathcal{V}_{0,s}, \|\cdot\|_1) &\le \binom{p}{s} N(\epsilon, \mathcal{B}_{s,2}, \|\cdot\|_1) \\
&\le \left(\frac{ep}{s}\right)^s \frac{\mathrm{vol}(\mathcal{B}_{s,2} + \epsilon/2\mathcal{B}_{1,s})}{\mathrm{vol}(\epsilon/2\mathcal{B}_{1,s})} \\
&\le \left(\frac{ep}{s}\right)^s \frac{\mathrm{vol}((s^{1/2} + \epsilon/2)\mathcal{B}_{1,s})}{\mathrm{vol}(\epsilon/2\mathcal{B}_{1,s})} \\
&\le \left(\frac{ep}{s}\right)^s \left(1 + \frac{2s^{1/2}}{\epsilon}\right)^s,
\end{aligned}
$$

where the third inequality is due to the fact that $\mathcal{B}_{2,s} \subset s^{1/2}\mathcal{B}_{1,s}$.

LEMMA S5. *For any* $\epsilon > 0$,

(1) $\sup_{m \in \mathfrak{M}} \log N(\epsilon, \mathcal{G}_n, e_m) \le s\log(ep/s) + s\log(1 + 8\nu/\epsilon) + c\kappa_n/\epsilon$, *where* $\mathfrak{M}$ *denotes the collection of probability measures on* $\mathbb{R}^p$ *satisfying Assumption 1.*

(2) $\sup_{m \in \mathfrak{M}_n} \log N(\epsilon, \mathcal{G}_n, e_m) \le c\kappa_n/\epsilon + s\log(ep/s) + s\log(1 + 4\kappa_n/\epsilon)$, *where* $\mathfrak{M}_n$ *denotes the collection of probability measures supported on* $[-\kappa_n/s^{1/2}, \kappa_n/s^{1/2}]^p$.

Proof. Below we establish the two claims for one m from the corresponding family of probability measures; the uniform bound follows from the fact that all constants involved below does not depend on m.

(1). Let $\mathcal{E}_{\mathcal{F}_n,\epsilon_1} = \{f_1, \dots, f_r\}$ be a minimal ϵ_1-net of $\mathcal{F}_n$ under the metric $\|\cdot\|_\infty$, and $\mathcal{E}_{\mathcal{V}_{0,s},\epsilon_2} = \{v_1, \dots, v_l\}$ be a minimal ϵ_2-net of $\mathcal{V}_{0,s}$ under the metric $\|\cdot\|_2$, with ϵ_1 and ϵ_2 to be determined later. By the construction of $\mathcal{G}_n$, we have, for any $g = f \circ v \in \mathcal{G}_n$, there exists $h = \check{f} \circ \check{v}$ where $\check{f} \in \mathcal{E}_{\mathcal{F}_n,\epsilon_1}$ and $\check{v} \in \mathcal{E}_{\mathcal{V}_{0,s},\epsilon_2}$ such that

$$
\begin{aligned}
\int |g(x) - h(x)|^2 dm(x) &= \int |f(v^\top x) - \check{f}(\check{v}^\top x)|^2 dm(x) \\
&= \int |f(v^\top x) - \check{f}(v^\top x) + \check{f}(v^\top x) - \check{f}(\check{v}^\top x)|^2 dm(x) \\
&\le 2\int |f(v^\top x) - \check{f}(v^\top x)|^2 dm(x) + 2\int |\check{f}(v^\top x) - \check{f}(\check{v}^\top x)|^2 dm(x) \\
&\le 2\int |f(v^\top x) - \check{f}(v^\top x)|^2 dm(x) + 2\int |v^\top x - \check{v}^\top x|^2 dm(x) \\
&\le 2\|f - \check{f}\|_\infty^2 + 2\|v - \check{v}\|_2^2 \int \left\{\frac{(v - \check{v})^\top x}{\|v - \check{v}\|_2}\right\}^2 dm(x) \le 2\epsilon_1^2 + 8\nu^2\epsilon_2^2,
\end{aligned}
$$

where the last inequality holds because $\sup_{v^\top v=1} \int (v^\top x)^2 dm(x) \le 4\nu^2$ by Assumption 1.

Taking $\epsilon_1 = \epsilon/2$ and $\epsilon_2 = \epsilon/(4\nu)$, we then see that $\{f \circ v : f \in \mathcal{E}_{\mathcal{F}_n,\epsilon_1}, v \in \mathcal{E}_{\mathcal{V}_{0,s},\epsilon_2}\}$ form an ϵ-net of $\mathcal{G}_n$ under e_P. From Lemma S4, we have $\log l = \log N(\epsilon/(4\nu), \mathcal{V}_{0,s}, \|\cdot\|_2) \le s\log(ep/s) + s\log(1+8\nu/\epsilon)$. Moreover, from Lemma S3, we obtain $\log r = \log N(\epsilon/2, \mathcal{F}_n, \|\cdot\|_\infty) \le c\kappa_n/\epsilon$. Thus, we conclude that

$$\log N(\epsilon, \mathcal{G}_n, e_m) \le s\log(ep/s) + s\log(1+8\nu/\epsilon) + c\kappa_n/\epsilon\,.$$

(2). Let $\mathcal{E}_{\mathcal{F}_n,\epsilon_1}$ be as in (1) and re-define $\mathcal{E}_{\mathcal{V}_{0,s},\epsilon_2} = \{v_1, \ldots, v_l\}$ as a minimal ϵ_2-net of $\mathcal{V}_{0,s}$ under the metric $\|\cdot\|_1$. Then for any $g = f \circ v \in \mathcal{G}_n$, there exists $h = \check{f} \circ \check{v}$ where $\check{f} \in \mathcal{E}_{\mathcal{F}_n,\epsilon_1}$ and $\check{v} \in \mathcal{E}_{\mathcal{V}_{0,s},\epsilon_2}$ such that

$$\begin{aligned}\int |g(x) - h(x)|^2 dm(x) &\le 2\int |f(v^\top x) - \check{f}(v^\top x)|^2 dm(x) + 2\int |v^\top x - \check{v}^\top x|^2 dm(x)\\ &\le 2\|f - \check{f}\|_\infty^2 + 2\int |v^\top x - \check{v}^\top x|^2 dm(x).\end{aligned}$$

To bound the second term, observe that for $v_1, v_2 \in \mathbb{R}^p$,

$$\int (v_1^\top x - v_2^\top x)^2 dm(x) = \int (\|v_1 - v_2\|_1 \|x\|_\infty)^2 dm(x) \le \|v_1 - v_2\|_1^2 \kappa_n^2/s,$$

where the last inequality is due to that m is supported on $[-\kappa_n/s^{1/2}, \kappa_n/s^{1/2}]^p$. Consequently,

$$\int |g(x) - h(x)|^2 dm(x) \le 2\|f - \check{f}\|_\infty^2 + 2\kappa_n^2/s\|v - \check{v}\|_1^2.$$

Taking $\epsilon_1 = \epsilon/2$ and $\epsilon_2 = \epsilon s^{1/2}/(2\kappa_n)$, we see that $\{f \circ v : f \in \mathcal{E}_{\mathcal{F}_n,\epsilon_1}, v \in \mathcal{E}_{\mathcal{V}_{0,s},\epsilon_2}\}$ is an ϵ-net of $\mathcal{G}_n$ under e_m. According to Lemma S3 and Lemma S4,

$$\begin{aligned}\log N(\epsilon, \mathcal{G}_n, e_m) &\le \log N(\epsilon/2, \mathcal{F}_n, \|\cdot\|_\infty) + \log N(\epsilon s^{1/2}/(2\kappa_n), \mathcal{V}_{0,s}, \|\cdot\|_1)\\ &\le c\kappa_n/\epsilon + s\log(ep/s) + s\log(1+4\kappa_n/\epsilon),\end{aligned}$$

which completes the proof. □

LEMMA S6. *For $\mathbb{X}_{n_1}$ and $\mathbb{Y}_{n_2}$ defined in (S5), under Assumptions 1 and 2,*

$$\max\{E(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n}), E(\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_n})\} \le c\kappa_n^{1/2} + c\{s\log(pn)\}^{1/2} + cn^{-1/2}\{s\kappa_n\log(pn) + \kappa_n^2\} \le cB_n.$$

Proof. It is sufficient to establish the bound for $\mathbb{X}_{n_1}$; similar arguments apply to $\mathbb{Y}_{n_2}$.

Let $\xi_1, \ldots, \xi_{n_1}$ be i.i.d. Rademacher random variables independent of $X_1, \ldots, X_{n_1}$. The symmetrization technique gives

$$\begin{aligned}E(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n}) &\le 2E\bigg\{\bigg\|n_1^{-1/2}\sum_{i=1}^{n_1}\xi_i g(X_i)\bigg\|_{\mathcal{G}_n}\bigg\}\\ &\le cE\bigg\{\bigg\|n_1^{-1/2}\sum_{i=1}^{n_1}\xi_i g(X_i)\bigg\|_{\mathcal{G}_n} 1_{\Omega_n}\bigg\} + 2n_1^{1/2}\kappa_n \mathrm{pr}(\Omega_n^c)\end{aligned}$$

$$\leq cE\biggl\{\biggl\|n_1^{-1/2}\sum_{i=1}^{n_1}\xi_i g(X_i)\biggr\|_{\mathcal{G}_n} 1_{\Omega_n}\biggr\} + c\kappa_n n^{-1/2},$$

where the event Ω_n is defined in (S3), and $1_{\Omega_n} = 1$ when Ω_n occurs and $1_{\Omega_n} = 0$ otherwise. Let $\sigma_n^2 = \sup_{g\in\mathcal{G}_n} n_1^{-1}\sum_{i=1}^{n_1} g^2(X_i)$. According to Dudley's maximal inequality,

$$\begin{aligned}
&E\biggl\{\biggl\|n_1^{-1/2}\sum_{i=1}^{n_1}\xi_i g(X_i)\biggr\|_{\mathcal{G}_n} 1_{\Omega_n}\biggr\}\\
&= E\biggl[1_{\Omega_n} E\biggl\{\biggl\|n_1^{-1/2}\sum_{i=1}^{n_1}\xi_i g(X_i)\biggr\|_{\mathcal{G}_n} \mid X_1,\ldots,X_{n_1}\biggr\}\biggr]\\
&\leq cE\biggl[1_{\Omega_n}\int_0^{\sigma_n}\{1+\log N(\delta,\mathcal{G}_n,e_{P_n})\}^{1/2}d\delta\biggr]\\
&\leq cE\biggl[1_{\Omega_n}\|G\|_{P,2}\int_0^{\sigma_n/\|G\|_{P,2}}\{1+\log N(\delta\|G\|_{P,2},\mathcal{G}_n,e_{P_n})\}^{1/2}d\delta\biggr]\\
&\leq c\|G\|_{P,2}E\{J(\sigma_n/\|G\|_{P,2})\},
\end{aligned}$$

where G is the envelope function of $\mathcal{G}_n$ and

$$J(\varepsilon) = \int_0^{\varepsilon}\varphi(\delta)d\delta = \int_0^{\varepsilon}\biggl\{1 + cs\log(ep/s) + cs\log\biggl(1+\frac{4\kappa_n}{\delta\|G\|_{P,2}}\biggr) + \frac{c\kappa_n}{\delta\|G\|_{P,2}}\biggr\}^{1/2}d\delta.$$

In the above, the last inequality is due to Lemma S5(2).

We claim that J has the following two properties.

- $J(\varepsilon)$ is concave since its second derivative is negative.
- $J(\varepsilon)/\varepsilon$ is non-increasing. To see this, it is clear that $\varphi(\delta)$ is non-increasing with δ. Then $J(\varepsilon)/\varepsilon = \int_0^1\varphi(\delta\varepsilon)d\delta$, which clearly indicates that $J(\varepsilon)/\varepsilon$ is non-increasing.
- $J(1)\geq 1$. This is because $\varphi(\delta)\geq 1$.

These properties enable us to apply the argument in the proof of Theorem 5.2 in Chernozhukov et al. (2014) and conclude that

$$E\biggl\{\biggl\|n_1^{-1/2}\sum_{i=1}^{n_1}\xi_i g(X_i)\biggr\|_{\mathcal{G}_n} 1_{\Omega_n}\biggr\} \leq cJ(\vartheta)\|G\|_{P,2} + c\frac{J^2(\vartheta)\|\Gamma\|_{P,2}}{\vartheta^2 n^{1/2}},$$

where $\Gamma = \max_{i=1,\ldots,n_1} G(X_i)$, $\vartheta = \sigma/\|G\|_{P,2}$. Taking $G\equiv\kappa_n$ leads to the desired result. □

LEMMA S7. *Let $U_1,\ldots,U_n$ be independent centered random vectors supported on $[-b_n,b_n]^r$. Denote*

$$S_n = n^{-1/2}\sum_{i=1}^{n} U_i.$$

Let $\tilde S$ be the Gaussian analog of S_n such that $\tilde S \sim N(0, n^{-1}\sum_{i=1}^n E(U_iU_i^\top))$ and $n^{-1}\sum_{i=1}^n E(U_{ij}^2) \ge \underline{\sigma}_n^2$ for all $j = 1, \dots, r$. Define

$$\rho = \sup_{\varrho\in[0,1], t\in\mathbb{R}} \left| \mathrm{pr}\left\{ \max_{1\le j\le r} (\varrho^{1/2} S_{n,j} + (1-\varrho)^{1/2}\tilde S_j) \le t \right\} - \mathrm{pr}\left(\max_{1\le j\le r} \tilde S' \le t \right) \right|,$$

where $\tilde S'$ is an independent copy of $\tilde S$ and also independent of everything else. If $n^{-1/2} b_n \phi \log r \le c_\circ$ for some sufficiently small positive constant $c_\circ$, then

$$\rho \le c\Bigg[\frac{\phi(\log r)^{3/2}}{\underline{\sigma}_n} E \max_{1\le j,k\le r} \left| \frac{1}{n}\sum_{i=1}^n \{U_{ij}U_{ik} - E(U_{ij}U_{ik})\} \right| + \sup_{t\in\mathbb{R}} \mathrm{pr}\left(|\max_{1\le j\le r} \tilde S_j - t| \le \phi^{-1} \right)$$
$$+ \frac{\phi^4(\log r)^3}{n} \max_{1\le j\le r} \left\{ \frac{1}{n}\sum_{i=1}^n E(U_{ij}^4) \right\} \left\{ \rho + \sup_{t\in\mathbb{R}} \mathrm{pr}\left(|\max_{1\le j\le r} \tilde S_j - t| \le \phi^{-1} \right) \right\} \Bigg].$$

Proof. The proof below combines various ideas and techniques from Chernozhukov et al. (2017a, 2022) and Fang & Koike (2021).

Pick any $t \in \mathbb{R}$. Let $\beta = \phi \log r$, where ϕ is a nonrandom number to be determined later. Define the function

$$F_\beta^t(w) = \beta^{-1} \log\left[\sum_{j=1}^r \exp\{\beta(w_j - t)\} \right], \; w \in \mathbb{R}^r.$$

The function $F_\beta^t(w)$ has the following property,

$$0 \le F_\beta^t(w) - \max_{j=1,\dots,r} (w_j - t) \le \beta^{-1}\log r = \phi^{-1}, \quad \text{for all } w \in \mathbb{R}^r.$$

Fix any five-times continuously differentiable and non-increasing function $h_0 : \mathbb{R} \to \mathbb{R}$ such that (i) $h_0(t) \ge 0$ for all $t \in \mathbb{R}$, (ii) $h_0(t) = 0$ for all $t \ge 1$, and (iii) $h_0(t) = 1$ for all $t \le 0$. There exists a constant $C_h > 0$ such that

$$\sup_{t\in\mathbb{R}} |h_0^{(1)}(t)| \vee |h_0^{(2)}(t)| \vee |h_0^{(3)}(t)| \vee |h_0^{(4)}(t)| \le C_h,$$

where $h_0^{(k)}(t)$ denotes the kth derivative of the function $h_0(t)$. Let $h(t) = h_0(\phi t)$, and define the function $m^t : \mathbb{R}^r \to \mathbb{R}$ by

$$m^t(w) = h\big(F_\beta^t(w)\big).$$

The function m^t has the following properties established in Chernozhukov et al. (2022). For all $j, k, l, r = 1, \dots, r$, there exist functions $M_{jk}^t : \mathbb{R}^r \to \mathbb{R}$, $M_{jkl}^t : \mathbb{R}^r \to \mathbb{R}$ and $M_{jklr}^t : \mathbb{R}^r \to \mathbb{R}$ such that (i) for all $w \in \mathbb{R}^r$, we have

$$|\partial_{jk} m^t(w)| \le M_{jk}^t(w), \;\; |\partial_{jkl} m^t(w)| \le M_{jkl}^t(w), \;\; |\partial_{jklr} m^t(w)| \le M_{jklr}^t(w),$$

(ii) for all $w_1 \in \mathbb{R}^r$ and $w_2 \in \mathbb{R}^r$ such that $\beta\|w_2\|_\infty \le 1$, we have

$$M^t_{jklr}(w_1 + w_2) \le cM^t_{jklr}(w_1), \tag{S38}$$

and (iii) for all $w \in \mathbb{R}^r$,

$$\sum_{j,k=1}^{r} M^t_{jk}(w) \le c\phi^2 \log r, \quad \sum_{j,k,l=1}^{r} M^t_{jkl}(w) \le c\phi^3 \log^2 r, \quad \sum_{j,k,l,r=1}^{r} M^t_{jklr}(w) \le c\phi^4 \log^3 r. \tag{S39}$$

Note that for any $\varrho \in [0,1]$ and $t \in \mathbb{R}$,

$$\begin{aligned}
&\mathrm{pr}\left[\max_{1\le j\le r}\{\varrho^{1/2}S_{n,j} + (1-\varrho)^{1/2}\tilde{S}_j\} \le t\right] \\
&= \mathrm{pr}\left[\max_{1\le j\le r}\{\varrho^{1/2}S_{n,j} + (1-\varrho)^{1/2}\tilde{S}_j - t - 1/\phi\} \le -1/\phi\right] \\
&\le \mathrm{pr}\big\{F_\beta^{t+1/\phi}\big(\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\big) \le 0\big\} \le E\big\{m^{t+1/\phi}\big(\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\big)\big\} \\
&= E\big\{m^{t+1/\phi}\big(\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\big)\big\} - E\{m^{t+1/\phi}(\tilde{S}')\} + E\{m^{t+1/\phi}(\tilde{S}')\} \\
&\le \mathrm{pr}\left(\max_{1\le j\le r}\tilde{S}'_j \le t + 2\phi^{-1}\right) + E\big\{m^{t+1/\phi}\big(\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\big)\big\} - E\{m^{t+1/\phi}(\tilde{S}')\} \\
&\le \mathrm{pr}\left(\max_{1\le j\le r}\tilde{S}'_j \le t\right) + \mathrm{pr}\left(\left|\max_{1\le j\le r}\tilde{S}'_j - t\right| \le 2\phi^{-1}\right) + \\
&\quad E\big\{m^{t+1/\phi}\big(\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\big)\big\} - E\{m^{t+1/\phi}(\tilde{S}')\}.
\end{aligned}$$

Similarly,

$$\begin{aligned}
&\mathrm{pr}\left(\max_{1\le j\le r}\tilde{S}'_j \le t\right) \le \mathrm{pr}\left(\max_{1\le j\le r}\tilde{S}'_j \le t - 2\phi^{-1}\right) + \mathrm{pr}\left(\left|\max_{1\le j\le r}\tilde{S}'_j - t\right| \le 2\phi^{-1}\right) \\
&\le E\{m^{t-1/\phi}(\tilde{S}')\} + \mathrm{pr}\left(\left|\max_{1\le j\le r}\tilde{S}'_j - t\right| \le 2\phi^{-1}\right) \\
&\le \mathrm{pr}\left[\max_{1\le j\le r}\{\varrho^{1/2}S_{n,j} + (1-\varrho)^{1/2}\tilde{S}_j\} \le t\right] + E\{m^{t-1/\phi}(\tilde{S}')\} - E\big\{m^{t-1/\phi}\big(\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\big)\big\} \\
&\quad + \mathrm{pr}\left(\left|\max_{1\le j\le r}\tilde{S}'_j - t\right| \le 2\phi^{-1}\right).
\end{aligned}$$

Hence,

$$\begin{aligned}
&\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\left[\max_{1\le j\le r}\{\varrho^{1/2}S_{n,j} + (1-\varrho)^{1/2}\tilde{S}_j\} \le t\right] - \mathrm{pr}\left(\max_{1\le j\le r}\tilde{S}'_j \le t\right)\right| \\
&\le \sup_{t\in\mathbb{R}}\left|E\big\{m^t\big(\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\big)\big\} - E\{m^t(\tilde{S}')\}\right| + \sup_{t\in\mathbb{R}}\mathrm{pr}\left(\left|\max_{1\le j\le r}\tilde{S}_j - t\right| \le 2\phi^{-1}\right).
\end{aligned} \tag{S40}$$

Comparison bounds. In this part, we will bound $\sup_{t\in\mathbb{R}}|E\{m^t(\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S})\} - E\{m^t(\tilde{S}')\}|$. For notational convenience, we write $m(w) = m^t(w)$.

We first observe that

$$
\begin{aligned}
& E\{m\big(\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\big)\}-E\{m(\tilde{S}')\} \\
&= E\bigg\{\int_0^1 \frac{dm\big(u^{1/2}\{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)}{du}du\bigg\} \\
&= E\bigg\{\int_0^1 \langle\frac{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}}{2u^{1/2}},\partial m\big(u^{1/2}\{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\rangle \\
&\quad -\langle\frac{\tilde{S}'}{2(1-u)^{1/2}},\partial m\big(u^{1/2}\{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\rangle du\bigg\} \\
&= \int_0^1 E\bigg\{\langle\frac{\varrho^{1/2}S_n}{2u^{1/2}},\partial m\big(u^{1/2}\{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\rangle\bigg\}du+ \\
&\quad \frac{1-\varrho}{2}\int_0^1 E\bigg\{\sum_{j,k=1}^r \Sigma_{jk}\partial_{jk}m\big(u^{1/2}\{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\bigg\}du- \\
&\quad \frac{1}{2}\int_0^1 E\bigg\{\sum_{j,k=1}^r \Sigma_{jk}\partial_{jk}m\big(u^{1/2}\{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\bigg\}du \\
&= \int_0^1 E\bigg\{\langle\frac{\varrho^{1/2}S_n}{2u^{1/2}},\partial m\big(u^{1/2}\{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\rangle\bigg\}du \\
&\quad -\frac{\varrho}{2}\int_0^1 E\bigg\{\sum_{j,k=1}^r \Sigma_{jk}\partial_{jk}m\big(u^{1/2}\{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\bigg\}du,
\end{aligned}
$$

where $\Sigma=E(\tilde{S}\tilde{S}^\top)$, and the third equality is due to Stein's identity.

Next we employ the exchangeable pair approach in Stein's method for multivariate normal approximation by (Chatterjee & Meckes, 2008; Reinert & Röllin, 2009) along with a symmetry argument in Fang & Koike (2021) and Chernozhukov et al. (2023). Let $(U_i')_{i=1}^n$ be an independent copy of $(U_i)_{i=1}^n$. Let I be a random index uniformly chosen from $\{1,\dots,n\}$ and independent of $(U_i)_{i=1}^n$ and $(U_i')_{i=1}^n$. In addition, define $\Delta_i=(U_i'-U_i)/n^{1/2}$ and $S_n'=S_n+\Delta_I$. Then, it is easy to verify that (S_n,S_n') has the same distribution as (S_n',S_n) (exchangeability) and

$$
E(S_n'-S_n|S_n')=\frac{S_n'}{n},\quad E(S_n'-S_n|S_n)=-\frac{S_n}{n}.
$$

Denote $D=S_n'-S_n$, we have

$$
\begin{aligned}
& E\bigg\{\bigg\langle\frac{\varrho^{1/2}S_n}{2u^{1/2}},\partial m\big(u^{1/2}\{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\bigg\rangle\bigg\} \\
&= E\bigg\{\bigg\langle\frac{\varrho^{1/2}S_n'}{2u^{1/2}},\partial m\big(u^{1/2}(\varrho^{1/2}S_n'+(1-\varrho)^{1/2}\tilde{S})+(1-u)^{1/2}\tilde{S}'\big)\bigg\rangle\bigg\} \\
&= E\bigg\{\bigg\langle\frac{\varrho^{1/2}nD}{2u^{1/2}},\partial m\big(u^{1/2}(\varrho^{1/2}S_n'+(1-\varrho)^{1/2}\tilde{S})+(1-u)^{1/2}\tilde{S}'\big)\bigg\rangle\bigg\}
\end{aligned}
$$

$$
\begin{aligned}
&= E\Big\{\Big\langle \frac{\varrho^{1/2} nD}{2u^{1/2}}, \partial m\big(u^{1/2}\{\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)\Big\rangle\Big\} \\
&\quad + \frac{\varrho}{2}\sum_{j,k=1}^{r} E\big\{nD_jD_k\partial_{jk}m\big(u^{1/2}\{\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)\big\} \\
&\quad + \frac{\varrho^{3/2}nu^{1/2}}{2}\sum_{j,k,l=1}^{r}\int_0^1 E\big\{\delta D_jD_kD_l\partial_{jkl}m\big(u^{1/2}[\varrho^{1/2}\{S_n + (1-\delta)D\} + \\
&\quad (1-\varrho)^{1/2}\tilde{S}] + (1-u)^{1/2}\tilde{S}'\big)\big\}d\delta \\
&= E\Big\{\Big\langle \frac{-\varrho^{1/2}S_n}{2u^{1/2}}, \partial m\big(u^{1/2}\{\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)\Big\rangle\Big\} \\
&\quad + \frac{\varrho}{2}\sum_{j,k=1}^{r} E\big\{nD_jD_k\partial_{jk}m\big(u^{1/2}\{\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)\big\} \\
&\quad + \frac{\varrho^{3/2}nu^{1/2}}{2}\sum_{j,k,l=1}^{r}\int_0^1 E\big\{\delta D_jD_kD_l\partial_{jkl}m\big(u^{1/2}[\varrho^{1/2}\{S_n + (1-\delta)D\} + \\
&\quad (1-\varrho)^{1/2}\tilde{S}] + (1-u)^{1/2}\tilde{S}'\big)\big\}d\delta.
\end{aligned}
$$

Thus,

$$
\begin{aligned}
&E\Big\{\Big\langle \frac{\varrho^{1/2}S_n}{2u^{1/2}}, \partial m\big(u^{1/2}S_n + (1-u)^{1/2}\tilde{S}\big)\Big\rangle\Big\} \\
&= \frac{\varrho}{4}\sum_{j,k=1}^{r} E\big\{nD_jD_k\partial_{jk}m\big(u^{1/2}\{\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)\big\} + \frac{\varrho^{3/2}nu^{1/2}}{4}\sum_{j,k,l=1}^{r} \\
&\int_0^1 E\big\{\delta D_jD_kD_l\partial_{jkl}m\big(u^{1/2}[\varrho^{1/2}\{S_n + (1-\delta)D\} + (1-\varrho)^{1/2}\tilde{S}] + (1-u)^{1/2}\tilde{S}'\big)\big\}d\delta.
\end{aligned}
$$

Consequently,

$$
E\big\{m\big(\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\big)\big\} - E\{m(\tilde{S}')\} = \int_0^1 \{R_1(u) + R_2(u)\}du, \tag{S41}
$$

where

$$
\begin{aligned}
R_1(u) &= \frac{\varrho}{2}\sum_{j,k=1}^{r} E\big\{(nD_jD_k/2 - \Sigma_{jk})\partial_{jk}m\big(u^{1/2}\{\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)\big\}, \\
R_2(u) &= \frac{\varrho^{3/2}nu^{1/2}}{4}\sum_{j,k,l=1}^{r}\int_0^1 E\big\{\delta D_jD_kD_l\partial_{jkl}m\big(u^{1/2}[\varrho^{1/2}\{S_n + (1-\delta)D\} + (1-\varrho)^{1/2}\tilde{S}] + \\
&\quad (1-u)^{1/2}\tilde{S}'\big)\big\}d\delta.
\end{aligned}
$$

We begin with bounding $R_1(u)$. Note that

$$
E(nDD^\top | U_{i=1}^n) = \frac{1}{n}\sum_{i=1}^{n} E(U_iU_i^\top) + \frac{1}{n}\sum_{i=1}^{n} U_iU_i^\top = \Sigma + \frac{1}{n}\sum_{i=1}^{n} U_iU_i^\top.
$$

Therefore,

$$
\begin{aligned}
\int_0^1 |R_1(u)|du &= \int_0^1 \left| \frac{\varrho}{4n} \sum_{i=1}^n \sum_{j,k=1}^r E\big\{(U_{ij}U_{ik} - \Sigma_{jk})\partial_{jk} m\big(u^{1/2}\{\varrho^{1/2}S_n + \right. \\
&\qquad \left. (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)\big\} \right| du \\
&= \int_0^1 \left| \frac{\varrho}{4n} \sum_{i=1}^n \sum_{j,k=1}^r E\big[(U_{ij}U_{ik} - \Sigma_{jk}) E\big\{\chi \partial_{jk} m\big(u^{1/2}\{\varrho^{1/2}S_n + \right. \\
&\qquad \left. (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big) \mid U_{i=1}^n\big\}\big] \right| du \\
&\le c \int_0^1 \frac{\varrho\phi^2 \log r}{4} E\left\{ \left\| n^{-1} \sum_{i=1}^n U_i U_i^\top - \Sigma \right\|_\infty E(\chi \mid U_{i=1}^n) \right\} du,
\end{aligned}
$$

where the last inequality is due to (S39), and

$$
\chi = \begin{cases} 1 & \text{if } -\phi^{-1} \le \max_{1\le j\le r}[u^{1/2}\{\varrho^{1/2}S_{n,j} + (1-\varrho)^{1/2}\tilde{S}_j\} + (1-u)^{1/2}\tilde{S}_j'] - t \le \phi^{-1} \\ 0 & \text{otherwise.} \end{cases}
$$

Observing that $u^{1/2}\{\varrho^{1/2}S_n + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'$ and $(\varrho u)^{1/2}S_n + (1-\varrho u)^{1/2}\tilde{S}$ have the same distribution and the same conditional distribution given $U_{i=1}^n$, we deduce

$$
\begin{aligned}
E(\chi \mid U_{i=1}^n) &= E\left\{ 1\big(-\phi^{-1} \le \max_{1\le j\le r}\{(\varrho u)^{1/2}S_{n,j} + (1-\varrho u)^{1/2}\tilde{S}_j\} - t \le \phi^{-1}\big) \mid U_{i=1}^n \right\} \\
&= \mathrm{pr}\left[\left| \max_{1\le j\le r} \left\{ \tilde{S}_j - \frac{t - (\varrho u)^{1/2}S_{n,j}}{(1-\varrho u)^{1/2}} \right\} \right| \le \frac{\phi^{-1}}{(1-\varrho u)^{1/2}} \mid U_{i=1}^n \right] \\
&\le \sup_{y\in\mathbb{R}^r} \mathrm{pr}\left\{ \left| \max_{1\le j\le r}(\tilde{S}_j - y_j) \right| \le \phi^{-1}/(1-\varrho u)^{1/2} \right\}.
\end{aligned}
$$

Hence,

$$
\begin{aligned}
\int_0^1 |R_1(u)|du &\le c \int_0^1 \varrho\phi^2(\log r) E\left(\left\| n^{-1}\sum_{i=1}^n U_i U_i^\top - \Sigma \right\|_\infty \right) \\
&\qquad \left[\sup_{y\in\mathbb{R}^r} \mathrm{pr}\left\{ \left| \max_{1\le j\le r}(\tilde{S}_j - y_j) \right| \le \phi^{-1}/(1-\varrho u)^{1/2} \right\} \right] du \\
&\le c \frac{\phi \log^{3/2}(r)}{\underline{\sigma}_n} E\left(\left\| n^{-1}\sum_{i=1}^n U_i U_i^\top - \Sigma \right\|_\infty \right) \\
&= c \frac{\phi \log^{3/2}(r)}{\underline{\sigma}_n} E\left[\max_{1\le j,k\le r} \left| \frac{1}{n}\sum_{i=1}^n \{U_{ij}U_{ik} - E(U_{ij}U_{ik})\} \right| \right], \qquad \text{(S42)}
\end{aligned}
$$

where the second inequality is partially based on Lemma S18.

Next, we bound $R_2(u)$. By exchangeability, we have

$$R_2(u) = \frac{\varrho^{3/2} n u^{1/2}}{4} \sum_{j,k,l=1}^{r} \int_0^1 E\{\delta D_j D_k D_l \partial_{jkl} m\big(u^{1/2}[\varrho^{1/2}\{S_n + (1-\delta)D\} + (1-\varrho)^{1/2}\tilde{S}] + (1-u)^{1/2}\tilde{S}'\big)\} d\delta$$
$$= \frac{\varrho^{3/2} n u^{1/2}}{4} \sum_{j,k,l=1}^{r} \int_0^1 -E\big\{\delta D_j D_k D_l \partial_{jkl} m\big(u^{1/2}[\varrho^{1/2}\{S_n' - (1-\delta)D\} + (1-\varrho)^{1/2}\tilde{S}] + (1-u)^{1/2}\tilde{S}'\big)\big\} d\delta$$
$$= \frac{\varrho^{3/2} n u^{1/2}}{4} \sum_{j,k,l=1}^{r} \int_0^1 -E\big\{\delta D_j D_k D_l \partial_{jkl} m\big(u^{1/2}\{\varrho^{1/2}(S_n + \delta D) + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)\big\} d\delta.$$

Hence, we obtain

$$R_2(u) = \frac{\varrho^{3/2} n u^{1/2}}{8} \sum_{j,k,l=1}^{r} \int_0^1 E\big\{\delta D_j D_k D_l \partial_{jkl} m\big(u^{1/2}[\varrho^{1/2}\{S_n + (1-\delta)D\} + (1-\varrho)^{1/2}\tilde{S}] + (1-u)^{1/2}\tilde{S}'\big)\big\} d\delta - \frac{\varrho^{3/2} n u^{1/2}}{8} \sum_{j,k,l=1}^{r} \int_0^1 E\big\{\delta D_j D_k D_l \partial_{jkl} m\big(u^{1/2}\{\varrho^{1/2}(S_n + \delta D) + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)\big\} d\delta$$
$$\le \frac{\varrho^2 n u}{8} \sum_{j,k,l,r=1}^{r} \int_0^1 \int_0^1 E\big\{|D_j D_k D_l D_r \partial_{jklr} m\big(u^{1/2}[\varrho^{1/2}\{S_n + \delta D + \omega(1-2\delta)D\} + (1-\varrho)^{1/2}\tilde{S}] + (1-u)^{1/2}\tilde{S}'\big)|\big\} d\omega d\delta$$
$$= \frac{\varrho^2 u}{8} \sum_{j,k,l,r=1}^{r} \sum_{i=1}^{n} \int_0^1 \int_0^1 E\big\{|\Delta_{ij}\Delta_{ik}\Delta_{il}\Delta_{ir} \partial_{jklr} m\big(u^{1/2}[\varrho^{1/2}\{S_n + \delta\Delta_i + \omega(1-2\delta)\Delta_i\} + (1-\varrho)^{1/2}\tilde{S}] + (1-u)^{1/2}\tilde{S}'\big)|\big\} d\omega d\delta$$
$$= \frac{\varrho^2 u}{8} \sum_{j,k,l,r=1}^{r} \sum_{i=1}^{n} \int_0^1 \int_0^1 E\big\{|\Delta_{ij}\Delta_{ik}\Delta_{il}\Delta_{ir} \partial_{jklr} m\big(u^{1/2}\{\varrho^{1/2}(S_{n,-i} + \Xi_i) + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)|\big\} d\omega d\delta,$$

where $\Xi_i = n^{-1/2}U_i + \delta\Delta_i + \omega(1-2\delta)\Delta_i$ and $S_{n,-i} = S_n - n^{-1/2}U_i$. Note that $|\Xi_i| \le c b_n n^{-1/2}$ and $n^{-1/2} b_n \phi \log r \le c_\circ$ for some sufficiently small positive constant $c_\circ$. This enables us to apply (S38) and deduce

$$|R_2(u)| \le c\varrho^2 u \sum_{j,k,l,r=1}^{r} \sum_{i=1}^{n} \int_0^1 \int_0^1 E\big\{|\Delta_{ij}\Delta_{ik}\Delta_{il}\Delta_{ir} \partial_{jklr} m\big(u^{1/2}\{\varrho^{1/2} S_{n,-i} + (1-\varrho)^{1/2}\tilde{S}\} + (1-u)^{1/2}\tilde{S}'\big)|\big\} d\omega d\delta$$

$$= c\sum_{j,k,l,r=1}^{r}\sum_{i=1}^{n}\int_0^1\int_0^1 E\big(|\Delta_{ij}\Delta_{ik}\Delta_{il}\Delta_{ir}|\big)\cdot E\{\big|\partial_{jklr}m\big(u^{1/2}\{\varrho^{1/2}S_n+$$
$$(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\big|\}d\omega d\delta$$
$$\le c\int_0^1\int_0^1 \max_{1\le j,k,l,r\le r}\sum_{i=1}^{n}E\big(|\Delta_{ij}\Delta_{ik}\Delta_{il}\Delta_{ir}|\big)\sum_{j,k,l,r=1}^{r}E\{\big|\partial_{jklr}m\big(u^{1/2}\{\varrho^{1/2}S_n+$$
$$(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\big|\}d\omega d\delta$$
$$\le \max_{1\le j\le r}\frac{c}{n^2}\sum_{i=1}^{n}E(U_{ij}^4)\int_0^1\int_0^1\sum_{j,k,l,r=1}^{r}E\{\chi\big|\partial_{jklr}m\big(u^{1/2}\{\varrho^{1/2}S_n+$$
$$(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'\big)\big|\}d\omega d\delta,$$

where the first equality is due to independence of Δ_i and $S_{n,-i}$. Observing that $u^{1/2}\{\varrho^{1/2}S_n+(1-\varrho)^{1/2}\tilde{S}\}+(1-u)^{1/2}\tilde{S}'$ and $(u\varrho)^{1/2}S_n+(1-u\varrho)^{1/2}\tilde{S}$ have the same distribution, by the definition of ρ, we deduce

$$E(\chi)\le 2\rho+\sup_{t\in\mathbb{R}}\mathrm{pr}\bigg(\bigg|\max_{1\le j\le r}\tilde{S}-t\bigg|\le\phi^{-1}\bigg).$$

Hence,

$$\int_0^1|R_2(u)|du\le c\frac{\phi^4\log^3 r}{n^2}\max_{1\le j\le r}\bigg\{\sum_{i=1}^{n}E(U_{ij}^4)\bigg\}\bigg\{\rho+\sup_{t\in\mathbb{R}}\mathrm{pr}\bigg(\bigg|\max_{1\le j\le r}\tilde{S}-t\bigg|\le\phi^{-1}\bigg)\bigg\}. \tag{S43}$$

Combining (S40), (S41), (S42) and (S43) completes the proof. □

LEMMA S8. *Let $\mathcal{J}=\{g_1,\ldots,g_r\}$ be defined in (S10) and r is the number of elements in $\mathcal{J}$. For sequences a_n and b_n such that $\max\{|a_n|,|b_n|\}\le c_\star$ for a constant $c_\star<\infty$, for $j=1,\ldots,r$ and $i=1,\ldots,n$, define*

$$U_{ij}=\begin{cases}a_n g_j(X_i) & \text{if } 1\le i\le n_1\\ b_n g_j(Y_{i-n_1}) & \text{if } n_1+1\le i\le n.\end{cases}$$

Then, under Assumptions 1 and 2,

$$E\bigg\{\max_{1\le j,k\le r}\bigg|\frac{1}{n}\sum_{i=1}^{n}\big(\{U_{ij}-E(U_{ij})\}\{U_{ik}-E(U_{ik})\}-E[\{U_{ij}-E(U_{ij})\}\{U_{ik}-E(U_{ik})\}]\big)\bigg|\bigg\}$$
$$\le cn^{-1/2}B_n, \tag{S44}$$

and if $n^{-1/2}B_n \leq c$, with probability at least $1 - cn^{-1}$,

$$\max_{1\leq j,k\leq r}\left|\frac{1}{n}\left\{\sum_{i=1}^{n_1}(U_{ij}-\bar{U}_{1j})(U_{ik}-\bar{U}_{1k})+\sum_{i=n_1+1}^{n}(U_{ij}-\bar{U}_{2j})(U_{ik}-\bar{U}_{2k})\right\}-\right.$$
$$\left.\frac{1}{n}\sum_{i=1}^{n}E[\{U_{ij}-E(U_{ij})\}\{U_{ik}-E(U_{ik})\}]\right|\leq cn^{-1/2}B_n, \tag{S45}$$

where B_n is defined in (S13), $\bar{U}_{1j} = n_1^{-1}\sum_{i=1}^{n_1} U_{ij}$ and $\bar{U}_{2j} = n_2^{-1}\sum_{i=n_1+1}^{n} U_{ij}$.

Proof. We first observe that

$$E\left\{\max_{1\leq j,k\leq r}\left|\frac{1}{n^{1/2}}\sum_{i=1}^{n}\left(\{U_{ij}-E(U_{ij})\}\{U_{ik}-E(U_{ik})\}-E[\{U_{ij}-E(U_{ij})\}\{U_{ik}-E(U_{ik})\}]\right)\right|\right\}$$
$$\leq E\left[\max_{1\leq j,k\leq r}\left|\frac{1}{n^{1/2}}\sum_{i=1}^{n}\{U_{ij}U_{ik}-E(U_{ij}U_{ik})\}\right|\right]+2E\left[\max_{1\leq j,k\leq r}\left|\frac{1}{n^{1/2}}\sum_{i=1}^{n}\{U_{ik}-E(U_{ik})\}E(U_{ij})\right|\right]$$
$$\leq cE(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n\times\mathcal{G}_n}+\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_n\times\mathcal{G}_n}+\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n}+\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_n}), \tag{S46}$$

where $\mathbb{X}_{n_1}$ and $\mathbb{Y}_{n_2}$ are defined in (S5), and $\mathcal{G}_n \times \mathcal{G}_n = \{f \times g : f, g \in \mathcal{G}_n\}$ with $(f \times g)(x) = f(x)g(x)$. Combining (S46)–(S49) below establishes (S44). Moreover,

$$\max_{1\leq j,k\leq r}\left|\frac{1}{n^{1/2}}\left\{\sum_{i=1}^{n_1}(U_{ij}-\bar{U}_{1j})(U_{ik}-\bar{U}_{1k})+\sum_{i=n_1+1}^{n}(U_{ij}-\bar{U}_{2j})(U_{ik}-\bar{U}_{2k})\right\}-\right.$$
$$\left.\frac{1}{n^{1/2}}\sum_{i=1}^{n}E[\{U_{ij}-E(U_{ij})\}\{U_{ik}-E(U_{ik})\}]\right|$$
$$\leq \max_{1\leq j,k\leq r}\left|\frac{1}{n^{1/2}}\sum_{i=1}^{n}\{U_{ij}U_{ik}-E(U_{ij}U_{ik})\}\right|+\max_{1\leq j,k\leq r}\frac{1}{n^{1/2}}\left|\sum_{i=1}^{n_1}\{\bar{U}_{1j}\bar{U}_{1k}-E(U_{ij})E(U_{ik})\}+\right.$$
$$\left.\sum_{i=n_1+1}^{n}\{\bar{U}_{2j}\bar{U}_{2k}-E(U_{ij})E(U_{ik})\}\right|.$$

The first term of the above is bounded by $c(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n\times\mathcal{G}_n}+\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_n\times\mathcal{G}_n})$. For the second term, we further observe

$$\max_{1\leq j,k\leq r}\frac{1}{n^{1/2}}\left|\sum_{i=1}^{n_1}\{\bar{U}_{1j}\bar{U}_{1k}-E(U_{ij})E(U_{ik})\}+\sum_{i=n_1+1}^{n}\{\bar{U}_{2j}\bar{U}_{2k}-E(U_{ij})E(U_{ik})\}\right|$$
$$\leq \max_{1\leq j,k\leq r}\frac{1}{n^{1/2}}\left|\bar{U}_{1j}\sum_{i=1}^{n_1}\{U_{ik}-E(U_{ik})\}+\sum_{i=1}^{n_1}[E(U_{ik})\{\bar{U}_{1j}-E(U_{ij})\}]+\bar{U}_{2j}\sum_{i=n_1+1}^{n}\{U_{ik}-E(U_{ik})\}+\right.$$
$$\left.\sum_{i=n_1+1}^{n}[E(U_{ik})\{\bar{U}_{2j}-E(U_{ij})\}]\right|$$
$$\leq c\left\{\max_{1\leq j\leq r}|\bar{U}_{1j}|+\max_{1\leq j\leq r}|\bar{U}_{2j}|+\max_{1\leq i\leq n,1\leq k\leq r}|E(U_{ik})|\right\}\left(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n}+\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_n}\right)$$

$$\le c\left\{\frac{1}{n^{1/2}}\left(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n}+\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_n}\right)+\max_{1\le i\le n,1\le k\le r}|E(U_{ik})|\right\}\left(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n}+\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_n}\right).$$

Then (S45) follows from (S47), (S49) below, $n^{-1/2}B_n \le c$ and $\max_{1\le i\le n,1\le k\le r}|E(U_{ik})| \le c$ due to Assumption 1.

Next, we bound $\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n}$, $\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n\times\mathcal{G}_n}$ and their expectations. Applying Lemma S20 with $t=\log n$, we have for $\delta>0$,

$$\mathrm{pr}\left\{\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n}\ge(1+\delta)E(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n})+c\sigma(\log n)^{1/2}+(c+c^2\delta^{-1})\frac{\kappa_n\log n}{n_1^{1/2}}\right\}\le n^{-1},$$

where $\sigma^2=\sup_{g\in\mathcal{G}_n}Eg^2(X)\le c$. Meanwhile, applying Lemma S6,

$$E(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n})\le c[\{s\log(pn)\}^{1/2}+\kappa_n^{1/2}+n^{-1/2}\{s\kappa_n\log(pn)+\kappa_n^2\}]\le cB_n.$$

Since $\kappa_n\ge c\log n$, we conclude that, with probability at least $1-cn^{-1}$,

$$\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n}\le cB_n. \tag{S47}$$

Again, applying Lemma S20 with $t=\log n$, we have for $\delta>0$,

$$\mathrm{pr}\left\{\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n\times\mathcal{G}_n}\ge(1+\delta)E(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n\times\mathcal{G}_n})+c\breve{\sigma}(\log n)^{1/2}+(c+c^2\delta^{-1})\frac{\kappa_n^2(\log n)}{n_1^{1/2}}\right\}\le n^{-1},$$

where $\breve{\sigma}^2=\sup_{g\in\mathcal{G}_n\times\mathcal{G}_n}E\{g^2(X)\}\le c$ due to Assumption 1. Applying similar arguments in Lemma S6 together with the fact that $N(\epsilon,\mathcal{G}_n\times\mathcal{G}_n,e_{P_n})\le N^2(\epsilon/(2\kappa_n),\mathcal{G}_n,e_{P_n})$,

$$E(\|\mathbb{X}_{n_1}\|_{\mathcal{G}_n\times\mathcal{G}_n})\le c[\{s\log(pn)\}^{1/2}+\kappa_n+n^{-1/2}\{s\kappa_n^2\log(pn)+\kappa_n^4\}]\le cB_n.$$

Therefore, with probability at least $1-cn^{-1}$,

$$\|\mathbb{X}_{n_2}\|_{\mathcal{G}_n\times\mathcal{G}_n}\le cB_n. \tag{S48}$$

Similar arguments lead to

$$\|\mathbb{Y}_{n_2}\|_{\mathcal{G}_n}\le cB_n \quad\text{and}\quad \|\mathbb{Y}_{n_2}\|_{\mathcal{G}_n\times\mathcal{G}_n}\le cB_n \tag{S49}$$

with probability at least $1-cn^{-1}$. □

S.4. Additional Discussions

S.4.1. *Testing with simultaneous confidence intervals*

Rejecting the global null hypothesis when $T_n>\hat{q}_{T^\star}(1-\alpha)$ is equivalent to rejecting it when $0\notin SCI(f,v)$ for some $f\in Lip_1$ and $v\in\mathcal{V}_{0,s}$. To see this, note that

$$T_n>\hat{q}_{T^\star}(1-\alpha)$$

$$\iff \left(\frac{n_1 n_2}{n_1+n_2}\right)^{1/2}\left[\sup_{v\in\mathcal{V}_{0,s}}\sup_{f\in Lip_1}\left\{\frac{1}{n_1}\sum_{i=1}^{n_1} f(v^\top X_i)-\frac{1}{n_2}\sum_{j=1}^{n_2} f(v^\top Y_j)\right\}\right] > \hat{q}_{T^\star}(1-\alpha)$$

$$\iff \frac{1}{n_1}\sum_{i=1}^{n_1} f(v^\top X_i)-\frac{1}{n_2}\sum_{j=1}^{n_2} f(v^\top Y_j) > \frac{\hat{q}_{T^\star}(1-\alpha)}{\{(n_1n_2)/(n_1+n_2)\}^{1/2}}$$

for some $f\in Lip_1$ and $v\in\mathcal{V}_{0,s}$

$$\iff 0\notin SCI(f,v) \text{ for some } f\in Lip_1 \text{ and } v\in\mathcal{V}_{0,s}.$$

Therefore, we can alternatively and equivalently conduct the test by utilizing SCIs. Specifically, we reject the null hypothesis if $0\notin SCI(f,v)$ for some $f\in Lip_1$ and $v\in\mathcal{V}_{0,s}$.

Moreover, the SCIs allow us to detect directions along which the two distributions are significantly different; we refer to these as significant directions. Let $q_{1-\alpha}$ denote the $1-\alpha$ quantile of $\mathbb{G}_n\mathcal{G}$. Based on the Gaussian and bootstrap approximations in Theorems 1 and 2, we can consistently estimate this quantile $q_{1-\alpha}$ by $\hat{q}_{T^\star}(1-\alpha)$. Consequently, by Proposition S2, using these SCIs which are asymptotically valid under the null and alternative hypotheses, we can identify/estimate significant directions once the null hypothesis is rejected without performing additional tests, while asymptotically controlling the family-wise error rate.

PROPOSITION S2. *Under Assumptions 1–3, we have*

$$\mathrm{pr}\left\{W_1(v^\top X, v^\top Y)\in\left[\hat{W}_1(v^\top X, v^\top Y)-\frac{q_{1-\alpha}}{r_n^{1/2}},\ +\infty\right),\ \forall v\in\mathcal{V}_{0,s}\right\}\geq 1-\alpha,$$

where $r_n = n_1n_2/(n_1+n_2)$. Moreover, the family-wise error rate is bounded by α, that is,

$$\mathrm{FWER}=\mathrm{pr}\left(\bigcup_{v\in\mathcal{L}}\{r_n^{1/2}\hat{W}_1(v^\top X, v^\top Y)>q_{1-\alpha}\}\right)\leq\alpha,$$

where $\mathcal{L}=\{v\in\mathcal{V}_{0,s}: v^\top X \overset{d}{=} v^\top Y\}$.

Proof. Note that

$$\begin{aligned}
&\mathrm{pr}\left\{W_1(v^\top X, v^\top Y)\in\left[\hat{W}_1(v^\top X, v^\top Y)-\frac{q_{1-\alpha}}{r_n^{1/2}},\ +\infty\right),\ \forall v\in\mathcal{V}_{0,s}\right\}\\
&=\mathrm{pr}\Bigg(\sup_{f\in Lip_1}\left\{n_1^{-1}\sum_{i=1}^{n_1} f(v^\top X_i)-n_2^{-1}\sum_{j=1}^{n_2} f(v^\top Y_j)\right\}-\\
&\qquad\sup_{f\in Lip_1}\left[E\{f(v^\top X)\}-E\{f(v^\top Y)\}\right]\leq\frac{q_{1-\alpha}}{r_n^{1/2}},\ \forall v\in\mathcal{V}_{0,s}\Bigg)\\
&\geq\mathrm{pr}\Bigg\{\sup_{f\in Lip_1}\Bigg(n_1^{-1}\sum_{i=1}^{n_1}[f(v^\top X_i)-E\{f(v^\top X)\}]-\\
&\qquad n_2^{-1}\sum_{j=1}^{n_2}[f(v^\top Y_j)-E\{f(v^\top Y)\}]\Bigg)\leq\frac{q_{1-\alpha}}{r_n^{1/2}},\ \forall v\in\mathcal{V}_{0,s}\Bigg\}
\end{aligned}$$

$$
\begin{aligned}
&= \mathrm{pr}\Bigg\{ \sup_{v\in\mathcal{V}_{0,s}} \sup_{f\in Lip_1} \Bigg(n_1^{-1}\sum_{i=1}^{n_1}[f(v^\top X_i) - E\{f(v^\top X)\}] - \\
&\qquad n_2^{-1}\sum_{j=1}^{n_2}[f(v^\top Y_j) - E\{f(v^\top Y)\}]\Bigg) \le \frac{q_{1-\alpha}}{r_n^{1/2}} \Bigg\} \\
&= \mathrm{pr}(\mathbb{G}_n\mathcal{G} \le q_{1-\alpha}) = 1-\alpha.
\end{aligned}
$$

To bound the family-wise error rate, noting that $\mathcal{L} \subset \mathcal{V}_{0,s}$, we deduce that

$$
\begin{aligned}
\mathrm{FWER} &= \mathrm{pr}\Bigg(\bigcup_{v\in\mathcal{L}} \{ r_n^{1/2}\hat{W}_1(v^\top X, v^\top Y) > q_{1-\alpha}\} \Bigg) \\
&= \mathrm{pr}\Bigg(\bigcup_{v\in\mathcal{L}} \Bigg\{ \sup_{f\in Lip_1} \Bigg(n_1^{-1}\sum_{i=1}^{n_1}[f(v^\top X_i) - E\{f(v^\top X)\}] - \\
&\qquad n_2^{-1}\sum_{j=1}^{n_2}[f(v^\top Y_j) - E\{f(v^\top Y)\}]\Bigg) > \frac{q_{1-\alpha}}{r_n^{1/2}} \Bigg\}\Bigg) \\
&\le \mathrm{pr}\Bigg(\sup_{v\in\mathcal{V}_{0,s}, f\in Lip_1} \Bigg(\frac{1}{n_1}\sum_{i=1}^{n_1}[f(v^\top X_i) - E\{f(v^\top X)\}] - \\
&\qquad \frac{1}{n_2}\sum_{j=1}^{n_2}[f(v^\top Y_j) - E\{f(v^\top Y)\}]\Bigg) > \frac{q_{1-\alpha}}{r_n^{1/2}} \Bigg) \\
&\le \mathrm{pr}\left(\mathbb{G}_n\mathcal{G} > q_{1-\alpha}\right) \\
&\le \alpha.
\end{aligned}
$$

This completes the proof. $\square$

A key feature of our approach enabling automatic detection of significant marginals is that our bootstrapping procedure approximates the distribution of

$$
\mathbb{G}_n\mathcal{G} = r_n^{1/2} \sup_{f\in Lip_1, v\in\mathcal{V}_{0,s}} \Bigg(\frac{1}{n_1}\sum_{i=1}^{n_1}[f(v^\top X_i) - E\{f(v^\top X)\}] - \frac{1}{n_2}\sum_{j=1}^{n_2}[f(v^\top Y_j) - E\{f(v^\top Y)\}]\Bigg),
$$

irrespective of whether the null or alternative hypothesis is true, where $r_n = n_1 n_2/(n_1+n_2)$. This allows us to construct the SCIs, thus enabling the identification of significant directions without additional tests.

This desirable feature is absent in other bootstrapping or projection-based tests (Zhou et al., 2017), which therefore do not guarantee automatic detection of significant marginals. Additionally, it is unclear to us whether permutation-based methods (Biswas & Ghosh, 2014; Zhu & Shao, 2021; Wang et al., 2021) can achieve this desirable property or not, as discussed in Section S.4.3.

S.4.2. Interpretation of the optimal $\hat{v}$

When the null hypothesis is rejected, the nonzero coordinates of $\hat{v}$ collectively provide an estimate for the subspace on which the two distributions differ. This estimate is, of course, subject to sampling variability, as $\hat{v}$ depends on the data. Notably, we have accounted for the variability of $\hat{v}$ when constructing the one-sided confidence interval for $W_1(\hat{v}^\top X, \hat{v}^\top Y)$, and thus do not require additional adjustment/correction for this variability. To see this, first observe that

$$W_1(\hat{v}^\top X, \hat{v}^\top Y) = \sup_{f \in Lip_1} [E\{f(\hat{v}^\top X)\} - E\{f(\hat{v}^\top Y)\}],$$

$$\begin{aligned}\widehat{W}_1(\hat{v}^\top X, \hat{v}^\top Y) &= \sup_{f \in Lip_1} \left\{ \frac{1}{n_1} \sum_{i=1}^{n_1} f(\hat{v}^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(\hat{v}^\top Y_j) \right\} \\ &= \sup_{v \in \mathcal{V}_{0,s}} \sup_{f \in Lip_1} \left\{ \frac{1}{n_1} \sum_{i=1}^{n_1} f(v^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(v^\top Y_j) \right\},\end{aligned}$$

and

$$\begin{aligned}&\sup_{f \in Lip_1} \left\{ \frac{1}{n_1} \sum_{i=1}^{n_1} f(\hat{v}^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(\hat{v}^\top Y_j) \right\} - \sup_{f \in Lip_1} [E\{f(\hat{v}^\top X)\} - E\{f(\hat{v}^\top Y)\}] \\ &\leq \sup_{f \in Lip_1} \left(\frac{1}{n_1} \sum_{i=1}^{n_1} f(\hat{v}^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(\hat{v}^\top Y_j) - [E\{f(\hat{v}^\top X)\} - E\{f(\hat{v}^\top Y)\}] \right) \\ &\leq \sup_{v \in \mathcal{V}_{0,s}} \sup_{f \in Lip_1} \left(\frac{1}{n_1} \sum_{i=1}^{n_1} f(v^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(v^\top Y_j) - [E\{f(v^\top X)\} - E\{f(v^\top Y)\}] \right),\end{aligned}$$

where the last inequality is due to $\hat{v} \in \mathcal{V}_{0,s}$. Thus,

$$\begin{aligned}&\mathrm{pr}\left\{ \widehat{W}_1(\hat{v}^\top X, \hat{v}^\top Y) - W_1(\hat{v}^\top X, \hat{v}^\top Y) \leq \frac{\hat{q}_{T^\star}(1-\alpha)}{\{(n_1 n_2)/(n_1+n_2)\}^{1/2}} \right\} \\ &= \mathrm{pr}\left\{ \sup_{f \in Lip_1} \left[\frac{1}{n_1} \sum_{i=1}^{n_1} f(\hat{v}^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(\hat{v}^\top Y_j) \right] - \sup_{f \in Lip_1} [E\{f(\hat{v}^\top X)\} - E\{f(\hat{v}^\top Y)\}] \leq \right. \\ &\qquad \left. \frac{\hat{q}_{T^\star}(1-\alpha)}{\{(n_1 n_2)/(n_1+n_2)\}^{1/2}} \right\} \\ &\geq \mathrm{pr}\left\{ \sup_{v \in \mathcal{V}_{0,s}} \sup_{f \in Lip_1} \left(\frac{1}{n_1} \sum_{i=1}^{n_1} f(v^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(v^\top Y_j) - [E\{f(v^\top X)\} - E\{f(v^\top Y)\}] \right) \leq \right. \\ &\qquad \left. \frac{\hat{q}_{T^\star}(1-\alpha)}{\{(n_1 n_2)/(n_1+n_2)\}^{1/2}} \right\} \\ &= \mathrm{pr}\{\mathbb{G}_n \mathcal{G} \leq \hat{q}_{T^\star}(1-\alpha)\} \geq 1 - \alpha - o(1),\end{aligned}$$

since $\hat{q}_{T^\star}(1-\alpha) \to q_{T^\star}(1-\alpha)$ and $\mathrm{pr}\{\mathbb{G}_n\mathcal{G} \le q_{T^\star}(1-\alpha)\} \ge 1-\alpha-o(1)$ according to Theorems 1 and 2. This implies that

$$\mathrm{pr}\left\{W_1(\hat{v}^\top X, \hat{v}^\top Y) \in \left[\widehat{W}_1(\hat{v}^\top X, \hat{v}^\top Y) - \frac{\hat{q}_{T^\star}(1-\alpha)}{\{(n_1n_2)/(n_1+n_2)\}^{1/2}},\ +\infty\right)\right\} \ge 1-\alpha-o(1).$$

Consequently, if $T_n > \hat{q}_{T^\star}(1-\alpha)$, i.e., $\widehat{W}_1(\hat{v}^\top X, \hat{v}^\top Y) > \hat{q}_{T^\star}(1-\alpha)/\{(n_1n_2)/(n_1+n_2)\}^{1/2}$, we conclude that the two distributions are significantly different (asymptotically at the level α) along the optimal direction $\hat{v}$ and within the subspace spanned by the nonzero coordinates of $\hat{v}$.

S.4.3. Permutation versus bootstrapping

While permutation is a popular alternative method to estimate the test threshold, we opt for bootstrapping due to the following two considerations.

- Firstly, from the methodological aspect, bootstrapping enables us to construct simultaneous confidence intervals (SCI) for $E\{f(v^\top X)\} - E\{f(v^\top Y)\}$ for $f \in Lip_1$ and $v \in \mathcal{V}_{0,s}$. Based on these SCIs which are valid under the null and alternative hypotheses, we can identify significant directions once the null hypothesis is rejected without performing additional tests, while controlling the family-wise error rate.

  For permutation, it is not clear to us how to achieve this practically desirable feature. Consider a projection-based metric $\Psi = \sup_{v\in\mathcal{V}} d(v^\top X, v^\top Y)$ and its sample version $\hat{\Psi}$ as a test statistic, where d denotes a univariate probability metric such as the Wasserstein distance. The permutation method would approximate the null distribution of $\hat{\Psi}$ by pooling all observations of X and Y together, randomly dividing the pooled observations into two groups with n_1 and n_2 observations, and then calculating the permutation statistic $\breve{\Psi}$ of $\hat{\Psi}$ based on these groups. Intuitively, under the alternative hypothesis, the distribution of $\breve{\Psi}$ would mimic that of the sample version $\hat{\Phi}$ of $\Phi = \sup_{v\in\mathcal{V}} d(v^\top U, v^\top V)$, where U and V are i.i.d. drawn from a mixture distribution of X and Y (the mixture weights are the limits of $n_1/(n_1+n_2)$ and $n_2/(n_1+n_2)$). However, it is unclear whether $\breve{\Psi}$ and $\hat{\Phi}$ share the same asymptotic distribution with $\sup_{v\in\mathcal{V}}(\hat{\Psi}_v - \Psi_v)$ under the alternative hypothesis since these statistics are in quite different functional forms of the data, where $\Psi_v = d(v^\top X, v^\top Y)$ and $\hat{\Psi}_v$ is its sample version, while a shared asymptotic distribution is desirable for accurately identifying significant directions while controlling the family-wise error rate.
- Secondly, from the theoretical aspect, by utilizing bootstrapping, we can leverage techniques of high-dimensional CLTs and empirical process theories to analyse the theoretical properties (e.g., validity, consistency and power) of the proposed test. In contrast, a theoretical investigation on permutation for the proposed test statistic in high dimensions requires a different set of techniques. Nevertheless, such theoretical investigation is of independent interest and could be a topic for future research, since the

permutation method has its own advantages, such as requiring less/weaker distributional assumptions.

S.4.4. The max-sliced Wasserstein distance versus other distances on distributions

While there are various alternative distances and divergence metrics on distributions, we opt for the max-sliced Wasserstein distance, due to the following considerations.

- *Max-sliced Wasserstein distance versus average-sliced Wasserstein distance*. Compared with the average-sliced Wasserstein distance and other average-based metrics, the max-sliced can better exploit the sparsity of differences between high-dimensional distributions, given the prevalence of the sparsity principle in high-dimensional contexts. Specifically, in the context of two-sample distributional tests, the sparsity principle suggests that a few directions are sufficient to detect the essential distributional differences between two samples. Numerical results in Section S.2 demonstrate the advantage of the max-sliced Wasserstein distance against sparse alternatives.
- *Wasserstein distance versus MMD and energy distance*. Recent endeavors by Yan & Zhang (2023); Gao & Shao (2023) have revealed some limitations of the methods of MMD and the energy distance in detecting the differences of high-order moments of two high-dimensional samples. In contrast, the Wasserstein distance and its variants do not have these limitations, as illustrated by the numeric studies in Section S.2.4.
- *Wasserstein distance versus other probability metrics/divergences*. Compared to traditional metrics such as the KL divergence, Wasserstein distances offer advantageous geometric properties, including robustness to support mismatches, such as when the distributions of two samples are supported on different regions.

While highlighting the various advantages of the max-sliced Wasserstein distance, it is important to acknowledge that other distances and divergences also offer distinct advantages in certain contexts. For example, MMD-based methods demonstrate higher power against dense alternatives. In summary, we consider the max-sliced Wasserstein distance in this article, primarily for its following desirable features in high dimensions: 1) exploiting the sparsity principle, 2) detecting differences in high-order moments, and 3) allowing unaligned supports of distributions.

S.4.5. The higher criticism test

The higher criticism (HC) test is designed for settings involving a finite number of hypothesis tests, with a focus on distinguishing between scenarios where no signal is present and those where there are some signals. It is particularly effective for detecting weak, rare signals, as demonstrated in Donoho & Jin (2004); Hall & Jin (2010).

In the context of two-sample testing, the HC test could theoretically be used to detect whether the marginal distributions of two samples differ in certain coordinates. This could be done by performing a univariate two-sample homogeneity test (e.g., the univariate MMD test) on each marginal distribution and then combining the resulting p-values. However, since the univariate two-sample tests may be dependent, combining p-values in this way appears nontrivial. Additionally, this approach lacks power if the two samples share marginal distributions across individual coordinates but differ in their joint distribution, making it less suitable for testing the global null homogeneity hypothesis $X \overset{d}{=} Y$.

The HC test also fundamentally differs from our proposed method and the specific two-sample homogeneity testing problem we address in several ways. First, the HC test is primarily used for signal detection (e.g., testing for mean differences or regression coefficients in linear models) rather than for detecting distributional differences between two high-dimensional samples. Second, the HC test is typically applied in contexts with a finite number of hypothesis tests. By contrast, our problem involves an infinite number of hypothesis tests: $H_{0,v} : v^\top X \overset{d}{=} v^\top Y$ versus $H_{1,v} : v^\top X \overset{d}{\neq} v^\top Y$, where $v \in \mathcal{V}_{0,s}$. These key differences highlight that the HC test and our method are suited to distinct applications: while the HC test targets sparse and rare signals from a finite set of hypotheses, our approach is specifically designed to test high-dimensional distributional homogeneity between two samples.

S.5. RESULTS UNDER ℓ_1 SPARSITY

Define $\mathcal{V}_{1,\lambda} = \{v : \|v\|_2 = 1, \|v\|_1 \leq \lambda\}$. Consider the following empirical ℓ_1-constrained max-sliced Wasserstein distance

$$\acute{D}_n = \sup_{v\in\mathcal{V}_{1,\lambda}} \sup_{f\in Lip_1} \left\{ \frac{1}{n_1}\sum_{i=1}^{n_1} f(v^\top X_i) - \frac{1}{n_2}\sum_{j=1}^{n_2} f(v^\top Y_j) \right\},$$

and the test statistic

$$\acute{T}_n = \left(\frac{n_1 n_2}{n_1+n_2}\right)^{1/2} \acute{D}_n.$$

Analogously, define $\acute{D}^\star$ and $\acute{T}^\star$ by replacing $\mathcal{V}_{0,s}$ with $\mathcal{V}_{1,\lambda}$. For a significance level α, we reject the null hypothesis if $\acute{T}_n > \hat{q}_{\acute{T}^\star}(1-\alpha)$, where $\hat{q}_{\acute{T}^\star}(\cdot)$ is the empirical quantile function of $\acute{T}^{\star,1}, \ldots, \acute{T}^{\star,B}$. The algorithms in Section S.1 can be readily applied in this context. Moreover, we can also alternatively and equivalently conduct the test via constructing the approximate $1-\alpha$ one-sided SCI for $E\{f(v^\top X)\} - E\{f(v^\top Y)\}$ for $f \in Lip_1$ and $v \in \mathcal{V}_{1,\lambda}$.

The automatic detection of significant directions and marginals is also achievable with the ℓ_1 sparsity constrained test. Specifically, we can analogously construct a one-sided simultaneous confidence interval for $W_1(v^\top X, v^\top Y)$ for each $v \in \mathcal{V}_{1,\lambda}$,

$$SCI(W_1(v^\top X, v^\top Y)) = \left[\hat{W}_1(v^\top X, v^\top Y) - \frac{\hat{q}_{\acute{T}^\star}(1-\alpha)}{\{(n_1 n_2)/(n_1+n_2)\}^{1/2}},\ \ +\infty\right).$$

Based on these SCIs, given a projection direction $v \in \mathcal{V}_{1,\lambda}$, one can conclude that $v^\top X$ and $v^\top Y$ have significantly different distributions if $\hat{W}_1(v^\top X, v^\top Y) > \hat{q}_{\acute{T}^\star}(1-\alpha)/\{(n_1 n_2/(n_1+n_2))\}^{1/2}$. Taking v to be the canonical unit vectors $e_1, \ldots, e_p$ leads to detection of significant marginals, where e_k is the p-dimensional vector having 1 in the kth coordinate and 0 in others.

S.5.1. Main theoretical results under ℓ_1 sparsity

Let $\acute{\kappa}_n = 2\lambda\nu\log(pn)$. Recall that $\mathcal{F} = \{f : \mathbb{R} \to \mathbb{R} \mid f \text{ is 1-Lipschitz and } f(0) = 0\}$. Denote

$$\acute{\mathcal{F}}_n = \{f \in \mathcal{F} : f(0) = 0, f(x) = f(-\acute{\kappa}_n) \text{ if } x < -\acute{\kappa}_n, f(x) = f(\acute{\kappa}_n) \text{ if } x > \acute{\kappa}_n\},$$

and

$$\acute{\mathcal{G}} = \{f \circ v : f \in \mathcal{F}, v \in \mathcal{V}_{1,\lambda}\},$$

$$\acute{\mathcal{G}}_n = \{f \circ v : f \in \acute{\mathcal{F}}_n, v \in \mathcal{V}_{1,\lambda}\}.$$

We continue to use the notations $\mathbb{G}_n$ and $\mathbb{Z}$ for brevity. Under the null hypothesis, $\acute{T}_n = \mathbb{G}_n\acute{\mathcal{G}}$. We first adapt Assumption 3 to the current setting, as follows.

Assumption S1. There exists one projection $\acute{v}_\circ \in \mathcal{V}_{1,\lambda}$, a constant $C_\circ > 0$ and a measurable subset $S \subset \mathbb{R}$ containing an open interval, such that $\mathrm{pr}(\acute{v}_\circ^\top X \in A \cap S) \geq C_\circ \mathrm{Leb}(A \cap S)$ for all Borel sets $A \subset \mathbb{R}$ or $\mathrm{pr}(\acute{v}_\circ^\top Y \in A \cap S) \geq C_\circ \mathrm{Leb}(A \cap S)$ for all Borel sets $A \subset \mathbb{R}$, where $\mathrm{Leb}(\cdot)$ denotes the Lebesgue measure.

All previous theoretical results under ℓ_0 sparsity have their counterparts under ℓ_1 sparsity. Let $\acute{\tau}_n \asymp (\log n)^{-1}(p^{1/2} + \acute{\kappa}_n)^{-1}$.

LEMMA S9. *Under Assumption 1, with probability at least* $1 - 2/n$, $\mathbb{G}_n\acute{\mathcal{G}} = \mathbb{G}_n\acute{\mathcal{G}}_n$.

THEOREM S3 (GAUSSIAN APPROXIMATION). *Assume* $n^{-1/3} \lesssim \epsilon \lesssim p^{-1}\log^{-2}(pn)$ *and* $p^5/\lambda^4 \lesssim n\{\log(pn)\}^{-10}$. *Under Assumptions 1–2 and S1, there is an absolute constant* $c > 0$ *such that*

$$d_K\big(\mathcal{L}(\mathbb{G}_n\acute{\mathcal{G}}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{G}}_n)\big) \leq c\frac{\lambda\log^{3/2}(pn)}{\acute{\tau}_n^2}\left(\epsilon^{1/2}\acute{\kappa}_n + n^{-1/4}\epsilon^{-3/4}\right).$$

Taking $\epsilon \asymp (n\acute{\kappa}_n^4)^{-1/5}$ *yields*

$$d_K\big(\mathcal{L}(\mathbb{G}_n\acute{\mathcal{G}}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{G}}_n)\big) \leq c\lambda^{8/5}pn^{-1/10}\{\log(pn)\}^7.$$

THEOREM S4 (BOOTSTRAP APPROXIMATION). *Assume* $n^{-1/3} \lesssim \epsilon \lesssim p^{-1}\log^{-2}(pn)$ *and* $p^5/\lambda^4 \lesssim n\{\log(pn)\}^{-10}$. *Under Assumptions 1–2 and S1, for an absolute constant* $c > 0$, *with* *probability at least* $1 - cn^{-1}$,

$$d_K\big(\mathcal{L}(\acute{T}^\star \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{G}}_n)\big) \le c\frac{\lambda \log^{3/2}(pn)}{\acute{\tau}_n^2}(n^{-1/4}\epsilon^{-3/4} + \epsilon^{1/2}\acute{\kappa}_n),$$

and taking $\epsilon \asymp (n\acute{\kappa}_n^4)^{-1/5}$ *yields*

$$d_K\big(\mathcal{L}(\acute{T}^\star \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{G}}_n)\big) \le c\lambda^{8/5}pn^{-1/10}\{\log(pn)\}^7.$$

The above theorems jointly imply that $d_K\big(\mathcal{L}(\mathbb{G}_n\acute{\mathcal{G}}), \mathcal{L}(\acute{T}^\star \mid \mathscr{D})\big) \le c\lambda^{8/5}pn^{-1/10}\{\log(pn)\}^7$ with probability at least $1 - cn^{-1}$, where the dimension p is allowed to grow at a polynomial order of the sample size n. Theorem S5 below is a direct corollary of Theorems S3 and S4. It implies that the size of the proposed test is asymptotically controlled at the nominal level.

THEOREM S5. *Suppose Assumptions 1–2 and S1 hold. If* $p^5/\lambda^4 \lesssim n\{\log(pn)\}^{-10}$, *then under the null hypothesis, we have*

$$\big|\mathrm{pr}\{\acute{T}_n > q_{\acute{T}^\star}(1-\alpha)\} - \alpha\big| \le c\lambda^{8/5}pn^{-1/10}\{\log(pn)\}^7,$$

for some universal constant $c > 0$, *where* $q_{\acute{T}^\star}(\cdot)$ *is the quantile function of* $\acute{T}^\star$.

Remark S.2. It is noteworthy that the Lipschitz class is not VC-type; the metric entropy of the Lipschitz class $\mathcal{F}_n$ under consideration is of the order $1/\epsilon$, rather than $\log(1/\epsilon)$ that is usually encountered in the literature. Consequently, techniques and results for VC-type classes, such as Corollary 2.2 of Chernozhukov et al. (2014) and the theories of Chernozhukov et al. (2016), cannot be directly employed. This often poses challenges in the theoretical analysis and contributes to a relatively slow rate in n. For instance, in a bound from Theorem 3 of Sadhu et al. (2022) for bootstrapping the one-sample smoothed Wasserstein distance, the rate in n is at best $n^{-1/12}$.

Remark S.3. The empirical studies in Section S.5.2, showcasing the superior performance of the proposed test with ℓ_1 sparsity even when $p = 500$ and $n_1 = n_2 = 250$, suggest that dependence of the convergence rates on both p and n may be improved. The linear dependence on p in Theorems S3-S4 is introduced by the constant $\acute{\tau}_n$ for applying the anti-concentration inequality. We conjecture that, when $X \sim P$ and $Y \sim Q$ have a nonzero max-sliced Wasserstein distance $D(X,Y)$, the pair $(\bar{v}, \bar{f})$ that achieves $D(X,Y)$ in (2) may give rise to sufficiently large $\mathrm{var}\{\bar{f}(\bar{v}^\top X)\}$ or $\mathrm{var}\{\bar{f}(\bar{v}^\top Y)\}$. If so, it may be possible to eliminate the need of $\acute{\tau}_n$ and also the linear dependence on p, potentially allowing p to grow at a rate much faster than n. Investigation along this direction, highly challenging in nature, is beyond the scope of the paper and left for future studies.

The theorem below investigates the power of the proposed test against local alternatives.

THEOREM S6 (CONSISTENCY). *Suppose Assumptions 1–2 hold and* $n^{-1}p^2\log^3(pn) \lesssim 1$, $n^{-1}\lambda^6\log^7(pn) \lesssim 1$. *If there is some sufficiently large constant* $c_a > 0$ *such that*

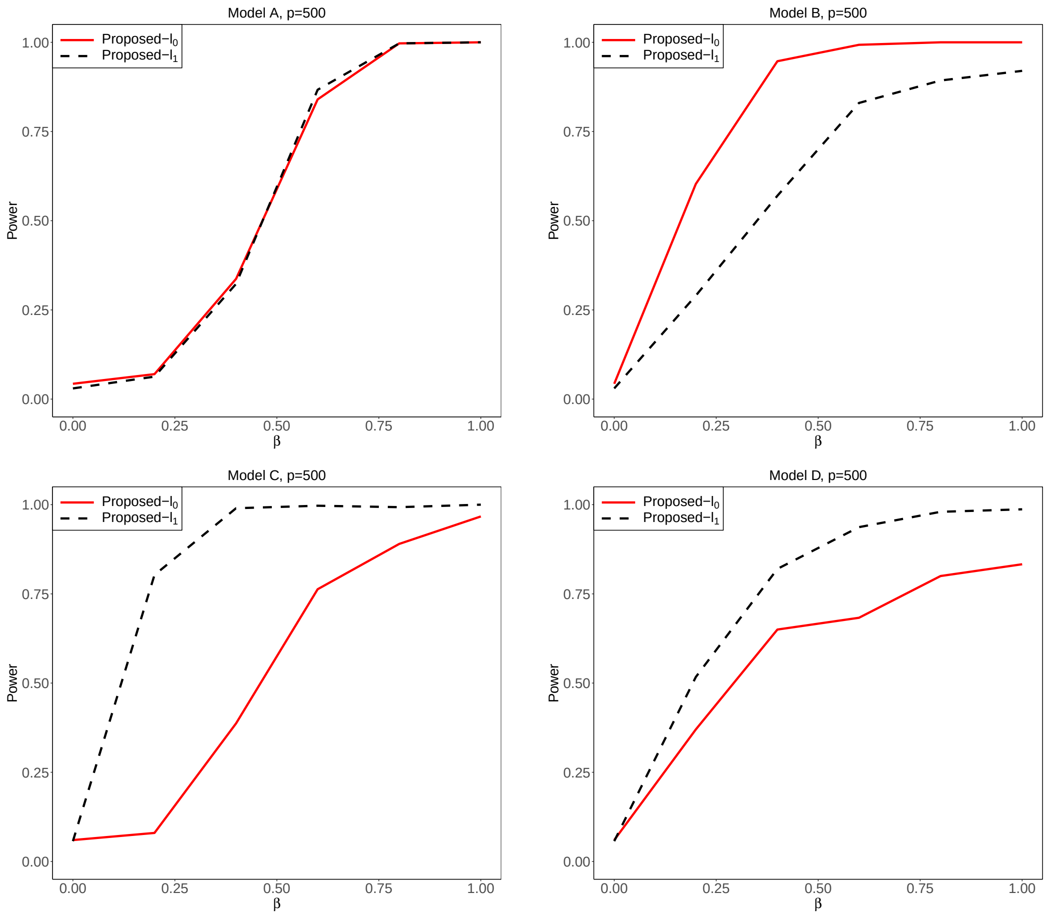

Fig. S5. Empirical size ($\beta = 0$) and power ($\beta > 0$) of the proposed method with ℓ_0 sparsity (red, solid) and the proposed method with ℓ_1 sparsity (black, dashed) for different models with $n_1 = n_2 = 250$.

$\sup_{g\in\acute{\mathcal{G}}}[E\{g(X)\} - E\{g(Y)\}] \geq c_a n^{-1/2}\lambda\log^{3/2}(pn)$, *then the null hypothesis will be rejected with probability tending to 1.*

Remark S.4. Note that if we replace the sub-exponential tail condition with the sub-Gaussian condition, that is, $\sup_{v^\top v=1}\|v^\top X\|_{\psi_2} \leq \nu$ and $\sup_{v^\top v=1}\|v^\top Y\|_{\psi_2} \leq \nu$, then the requirement on the dimension p in Theorem S6 can be relaxed to $p \lesssim n$.

S.5.2. Numerical results under ℓ_1 sparsity

The results across different signal levels under Models A-D are displayed in Figure S5. The proposed method using ℓ_1 sparsity and ℓ_0 sparsity achieves comparable performance under Model A. For Model B, ℓ_0 sparsity leads to higher power compared to ℓ_1 sparsity. In contrast, ℓ_1 sparsity outperforms ℓ_0 sparsity under more challenging scenarios, i.e., Models C and D, probably due to its ability to incorporate a broader range of potential directions.

To illustrate the effect of dimensionality, we investigate the power under $p = 60, 200, 500$, with a fixed signal level for each model. Specifically, we set $\mu_j = 0.64j^{-3}$ for Model A and $\Sigma'_{jj} = 1 + 6.4j^{-3}$ for Model B. Moreover, we set $s_0 = 5$ and $s_0 = 60$ for Models C and D, respectively. As shown in Table S6, the proposed method with ℓ_1 sparsity yields more robust performance as dimensionality increases, in comparison to ℓ_0 sparsity.

Table S6. *The power of the proposed method using ℓ_1 and ℓ_0 sparsity under different dimensions with $n_1 = n_2 = 250$*

	Model A			Model B		
	$p = 60$	$p = 200$	$p = 500$	$p = 60$	$p = 200$	$p = 500$
Proposed$-\ \ell_1$	1.000	0.997	0.997	0.997	0.967	0.893
Proposed$-\ \ell_0$	1.000	1.000	0.997	1.000	1.000	1.000
	Model C			Model D		
	$p = 60$	$p = 200$	$p = 500$	$p = 60$	$p = 200$	$p = 500$
Proposed$-\ \ell_1$	1.000	0.997	0.997	1.000	0.993	0.697
Proposed$-\ \ell_0$	0.980	0.823	0.630	0.973	0.943	0.523

We further explore the impact of the sparsity parameter λ on the performance of the proposed method with ℓ_1 sparsity. We take 10 equally spaced values on a logarithmic scale within $[0, \log(500)/2]$. From Figure S6, the type I errors are well controlled across all considered values of λ, showing the promising size performance of this approach. Regarding power performance, according to Figure S7, the parameter λ should be chosen appropriately to achieve higher power. The numerical results of the proposed method with ℓ_1 sparsity under dense models and high-order moment discrepancies show a pattern similar to that of ℓ_0 sparsity, which are not presented for space economy.

In summary, for the proposed method with ℓ_1 sparsity, although the relationship between p and n is relatively restrictive as outlined in Theorems S3-S6, the numerical results demonstrate its efficacy in high-dimensional two-sample homogeneity tests.

S.5.3. Preliminaries under ℓ_1 sparsity

We continue to use the notations $\mathbb{X}_{n_1}, \mathbb{Y}_{n_2}, \mathbb{V}_n, \mathbb{V}_n^\star, \mathbb{Z}_n^\star$ defined in Section S.3.1 for processes, but indexed by the functional class $\acute{\mathcal{G}}_n$. For $\epsilon > 0$, let

$$\acute{\mathcal{I}} = \{g_1, \ldots, g_{\acute{N}}\} \text{ be an } \epsilon\|\acute{G}\|_{(P+Q)/2,2}\text{-net of } \acute{\mathcal{G}}_n \text{ under the metric } e_{(P+Q)/2}, \tag{S50}$$

which is given in Lemma S13, where $\acute{G}$ is the envelope function of $\acute{\mathcal{G}}_n$. Define

$$\acute{\mathcal{G}}_\epsilon = \{f - g : f, g \in \acute{\mathcal{G}}_n, e_{(P+Q)/2}(f, g) \le \epsilon\|\acute{G}\|_{(P+Q)/2,2}\}.$$

Let $\acute{\mathcal{I}} = \acute{\mathcal{J}} \cup \acute{\mathcal{J}}^c$, where

$$\acute{\mathcal{J}} = \{g \in \acute{\mathcal{I}} : E\{(\mathbb{Z}g)^2\} \ge \acute{\tau}_n^2\}, \tag{S51}$$

where $\acute{\tau}_n \asymp (\log n)^{-1} \min\{p^{-1/2}, \acute{\kappa}_n^{-1}\}$. Denote

$$\acute{r} = |\acute{\mathcal{J}}| \le \acute{N}. \tag{S52}$$

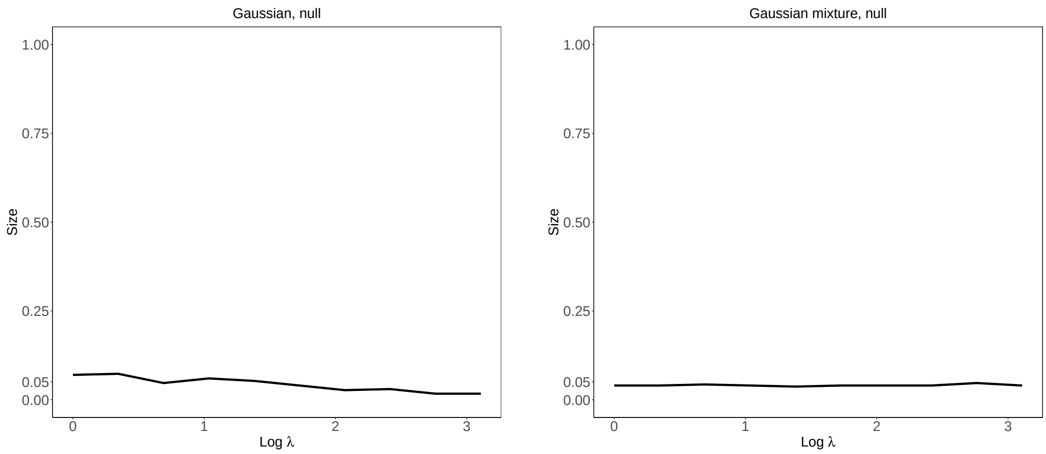

Fig. S6. The size performance of the proposed method with different ℓ_1 sparsity parameters λ under $n_1 = n_2 = 250, p = 500$. The Gaussian, null model corresponds to $\beta = 0$ in Model A or B, while the Gaussian mixture, null model corresponds to $\beta = 0$ in Model C or D.

Let

$$\acute{A}_n \asymp \acute{\kappa}_n^{1/2} + \lambda\{\log(2p/\lambda^2)\}^{3/2}, \tag{S53}$$
$$\acute{B}_n \asymp \lambda\acute{\kappa}_n \log^{3/2}(pn) + \acute{\kappa}_n + n^{-1/2}\lambda^2\acute{\kappa}_n^4 \log^3(pn). \tag{S54}$$

S.5.4. Proofs of main results under ℓ_1 sparsity

Proof of Theorem S3. Recall that $\acute{\kappa}_n = 2\lambda\nu\log(pn)$. With the arguments in the proof of Theorem 1, we establish the claim

$$d_K(\mathcal{L}(\mathbb{G}_n\acute{\mathcal{G}}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{G}}_n)) \le c\left[\frac{\acute{\eta}}{\acute{\tau}_n^2}\{\acute{A}_n + 1 \vee \log^{1/2}(\acute{\tau}_n/\acute{\eta})\} + \frac{\acute{A}_n + \acute{B}_n\acute{\tau}_n}{n^{1/4}\acute{\tau}_n^2}\log^{3/4}(\acute{r}) + \frac{1}{n}\right],$$

with the conditions

$$\acute{\eta} \ge c_5[\epsilon^{1/2}\acute{\kappa}_n + \epsilon\acute{\kappa}_n\{p\log(1/\epsilon)\}^{1/2} + \epsilon\acute{\kappa}_n(\log n)^{1/2} + n^{-1/2}\acute{\kappa}_n\{p\log(1/\epsilon) + \epsilon^{-1} + \log n\}], \tag{S55}$$
$$\acute{\tau}_n \le c_5^{-1}(\log n)^{-1}\min\{p^{-1/2}, \acute{\kappa}_n^{-1}\}, \tag{S56}$$
$$\acute{\tau}_n \ge c_5[n^{-1/3}\acute{\kappa}_n + n^{-1/2}\acute{\kappa}_n\{p\log(\acute{\kappa}_n/\acute{\tau}_n)\}^{1/2}], \tag{S57}$$
$$\log\acute{r} \le c_5^{-1}n\acute{\kappa}_n^{-4},$$
$$\epsilon\acute{\kappa}_n \le c_5^{-1}(\log n)^{-1}, \tag{S58}$$

for some sufficiently large constant $c_5 > 0$. If we take $\acute{\tau}_n = c(\log n)^{-1}(p^{1/2} + \acute{\kappa}_n)^{1/2}$ and $\acute{\eta} = c\epsilon^{1/2}\acute{\kappa}_n$, and choose ϵ so that $cn^{-1/3} \le \epsilon \le cp^{-1}\log^{-2}(pn)$, then under the assumption $p^5/\lambda^4 \le$

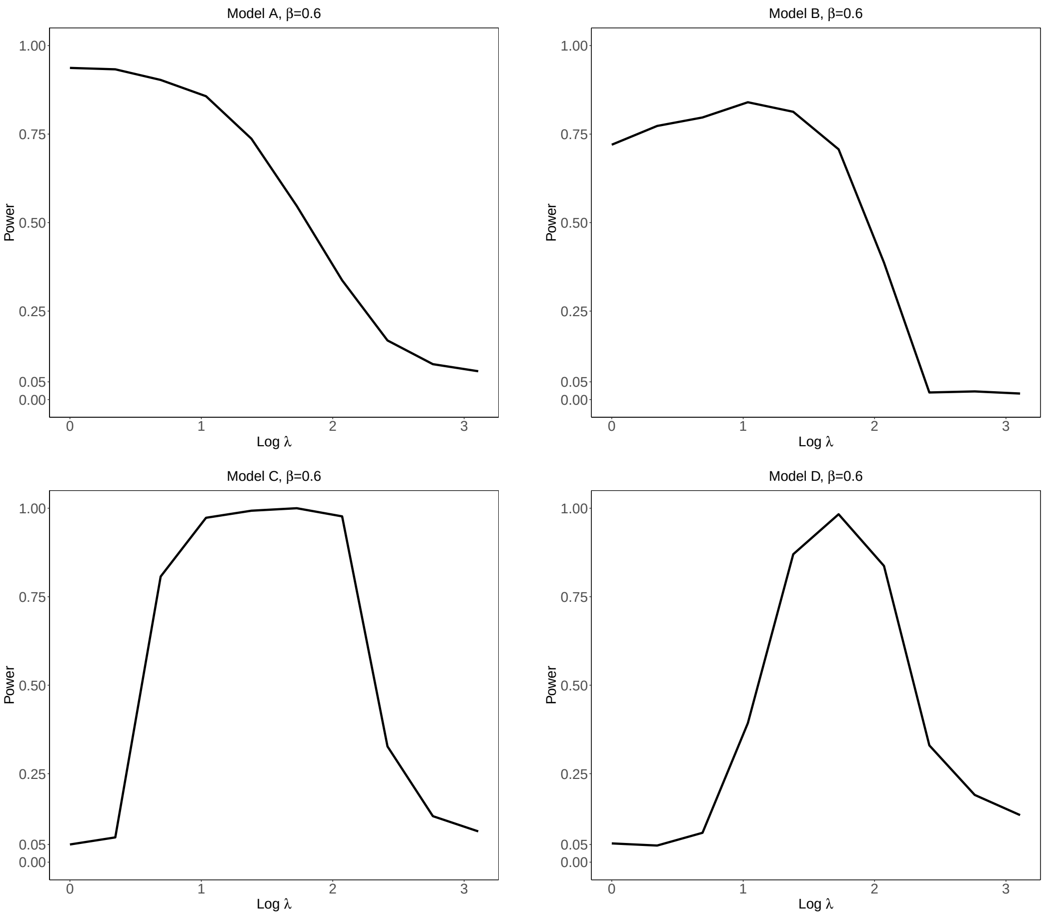

Fig. S7. The power performance of the proposed method with different ℓ_1 sparsity parameters λ under $n_1 = n_2 = 250, p = 500$.

$cn \log^{-10}(pn)$, all the above conditions can be satisfied. In this case, the rate becomes

$$d_K(\mathcal{L}(\mathbb{G}_n\acute{\mathcal{G}}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{G}}_n)) \leq c\frac{\lambda \log^{3/2}(pn)}{\acute{\tau}_n^2}(\epsilon^{1/2}\acute{\kappa}_n + n^{-1/4}\epsilon^{-3/4}).$$

Taking $\epsilon \asymp (n\acute{\kappa}_n^4)^{-1/5}$ yields the rate

$$d_K(\mathcal{L}(\mathbb{G}_n\acute{\mathcal{G}}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{G}}_n)) \leq cn^{-1/10}p\lambda^{8/5}\{\log(pn)\}^{61/10} \leq cn^{-1/10}p\lambda^{8/5}\log^7(pn).$$

Proof of Theorem S4. First, by using the arguments in Theorem 2 and applying Theorem S7 and Theorem S8, we obtain that with probability at least $1 - cn^{-1}$,

$$d_K\big(\mathcal{L}(\acute{T}^\star \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{G}}_n)\big) \leq c\frac{\acute{A}_n + \acute{B}_n\acute{\tau}_n + (\acute{A}_n\acute{B}_n\acute{\tau}_n)^{1/2}}{n^{1/4}\acute{\tau}_n^2}(\log \acute{r})^{3/4} + c\frac{\eta}{\acute{\tau}_n^2}\big[\acute{A}_n + 1 \vee \{\log(\acute{\tau}_n/\acute{\eta})\}^{1/2}\big],$$

under the conditions (S55), (S56), (S57), (S58), and

$$n^{-1}\acute{\kappa}_n^4(\log \acute{r} + \log n) \lesssim 1,\ n^{-1/2}\acute{B}_n \lesssim \acute{\tau}_n^2,\ n^{-1/2}\epsilon^{-3/2} \lesssim 1,\ n^{-1/2}\epsilon^{-1}\{p\log(1/\epsilon)\}^{1/2} \lesssim 1.$$

If we take $\acute{\eta} \asymp \epsilon^{1/2}\acute{\kappa}_n$ and $\acute{\tau}_n \asymp (\log n)^{-1}\min\{p^{-1/2}, \acute{\kappa}_n^{-1}\}$, and choose ϵ so that $cn^{-1/3} \leq \epsilon \leq cp^{-1}\log^{-2}(pn)$, then under the assumption $p^5/\lambda^4 + p^2\lambda^4 \leq cn\log^{-13}(pn)$, all the above conditions can be satisfied. In this case, the rate becomes

$$\sup_{t\in\mathbb{R}} |\mathrm{pr}(\acute{T}^\star \leq t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\acute{\mathcal{G}}_n \leq t)| \leq c\frac{\lambda\log^{3/2}(pn)}{\acute{\tau}_n^2}(n^{-1/4}\epsilon^{-3/4} + \epsilon^{1/2}\acute{\kappa}_n).$$

Taking $\epsilon \asymp (n\acute{\kappa}_n^4)^{-1/5}$ yields the rate

$$d_K\big(\mathcal{L}(\acute{T}^\star \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{G}}_n)\big) \leq c\lambda^{8/5}\{\log(pn)\}^{61/10}pn^{-1/10} \leq c\lambda^{8/5}\{\log(pn)\}^{7}pn^{-1/10}.$$

THEOREM S7. *Suppose Assumptions 1–2 and S1 hold. If $\acute{\eta}$ satisfies (S55), $\acute{\tau}_n$ satisfies (S56) and (S57), and*

$$c_5\epsilon\acute{\kappa}_n(\log n) \leq 1, \quad c_5 n^{-1/3} \leq \epsilon, \quad c_5\epsilon^{-1}\{p\log(1/\epsilon)\}^{1/2} \leq n^{1/2},$$

then we have with probability at least $1 - cn^{-1}$,

$$d_K\big(\mathcal{L}(\mathbb{Z}_n^\star\acute{\mathcal{G}}_n \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{G}}_n)\big) \leq c\frac{\acute{\eta}}{\acute{\tau}_n^2}\big[\acute{A}_n + 1 \vee \{\log(\acute{\tau}_n/\acute{\eta})\}^{1/2}\big] + c\frac{(\acute{A}_n\acute{B}_n)^{1/2}(\log\acute{r})^{3/4}}{n^{1/4}\acute{\tau}_n^{3/2}},$$

where $\acute{A}_n$ and $\acute{B}_n$ are respectively defined in (S53) and (S54), and $\acute{r}$ is defined in (S52).

Proof. The proof is straightforward by using the arguments in Theorem S1, hence it is omitted. □

THEOREM S8. *Under Assumptions 1–2, S1 and the conditions in Theorem S7, if*

$$c_5\acute{\kappa}_n^4(\log\acute{r} + \log n) \leq n \quad \textit{and} \quad c_5 n^{-1/2}\acute{B}_n \leq \acute{\tau}_n^2,$$

then we have with probability at least $1 - cn^{-1}$,

$$d_K\big(\mathcal{L}(\mathbb{V}_n^\star\acute{\mathcal{G}}_n \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}_n^\star\acute{\mathcal{G}}_n \mid \mathscr{D})\big) \leq c\frac{(\log\acute{r})^{3/4}\big(\acute{A}_n + \acute{B}_n\acute{\tau}_n + (\acute{A}_n\acute{B}_n\acute{\tau}_n)^{1/2}\big)}{n^{1/4}\acute{\tau}_n^2} + c\frac{\acute{\eta}}{\acute{\tau}_n^2}\big[\acute{A}_n + 1 \vee \{\log(\acute{\tau}_n/\acute{\eta})\}^{1/2}\big],$$

where $\acute{A}_n$ and $\acute{B}_n$ are defined in (S53) and (S54), and $\acute{r}$ is defined in (S52).

Proof. We omit the details, since the derivation is similar to the proof of Theorem S2. □

Proof of Theorem S6. To prove the consistency, we need to show that $\mathrm{pr}\big\{\acute{T}_n > q_{\acute{T}^\star}(1-\alpha)\big\} \to 1$ under the alternative for any $\alpha \in (0,1)$. Recall that $\acute{T}^\star \overset{d}{=} \sup_{g\in\acute{\mathcal{G}}}\mathbb{V}_n^\star g$. Lemma S9 implies that with probability at least $1 - cn^{-1}$, $\sup_{g\in\acute{\mathcal{G}}}\mathbb{V}_n^\star g = \sup_{g\in\acute{\mathcal{G}}_n}\mathbb{V}_n^\star g$.

Based on the proof of Theorem 4, it suffices to bound $q_{\acute{T}^\star}(1-\alpha)$ for $\alpha \in (0,1)$. By the Talagrand concentration inequality,

$$\mathrm{pr}\left\{ \sup_{g\in\acute{\mathcal{G}}_n} |\mathbb{V}_n^\star g| \geq cE\left(\sup_{g\in\acute{\mathcal{G}}_n} |\mathbb{V}_n^\star g| \mid \mathscr{D} \right) + c\acute{\sigma}_n \log^{1/2} n + cn^{-1/2}\acute{\kappa}_n \log n \mid \mathscr{D} \right\} \leq n^{-1},$$

where

$$\begin{aligned}\acute{\sigma}_n^2 &= \sup_{g\in\acute{\mathcal{G}}_n} \left(\frac{n_2}{n_1+n_2} n_1^{-1} \sum_{i=1}^{n_1} [g(X_i) - \hat{E}_{n_1}\{g(X)\}]^2 + \frac{n_1}{n_1+n_2} n_2^{-1} \sum_{i=1}^{n_2} [g(Y_i) - \hat{E}_{n_2}\{g(Y)\}]^2 \right) \\ &\leq c \sup_{g\in\acute{\mathcal{G}}_n} [E\{g^2(X)\} + E\{g^2(Y)\}] + cn^{-1/2}\|\mathbb{X}_{n_1}\|_{\acute{\mathcal{G}}_n^2} + cn^{-1/2}\|\mathbb{Y}_{n_2}\|_{\acute{\mathcal{G}}_n^2}.\end{aligned}$$

Together with the Talagrand concentration inequality and the arguments in Lemma S6, we conclude that with probability at least $1 - cn^{-1}$,

$$\acute{\sigma}_n^2 \leq c + cn^{-1/2}\acute{B}_n \leq c,$$

and

$$E\left(\sup_{g\in\acute{\mathcal{G}}_n} |\mathbb{V}_n^\star g| \mid \mathscr{D} \right) \leq c\lambda \log^{3/2}(pn) + cn^{-1/2}\lambda^3 \log^4(pn) \leq c\lambda \log^{3/2}(pn),$$

under the condition $n^{-1}\lambda^6 \log^7(pn) \lesssim 1$. Thus, we show that

$$\mathrm{pr}\{|\acute{T}^\star| \leq c\lambda \log^{3/2}(pn)\} \geq 1 - cn^{-1}.$$

It implies that $q_{\acute{T}^\star}(1-\alpha) \leq c\lambda \log^{3/2}(pn)$ for $\alpha \in (0,1)$.

Consequently, if $\sup_{g\in\acute{\mathcal{G}}}[E\{g(X)\} - E\{g(Y)\}] \geq c_a n^{-1/2}\lambda \log^{3/2}(pn)$ for a sufficiently large constant $c_a > 0$, the null hypothesis will be rejected with probability tending to 1. □

S.5.5. *Technical lemmas and proofs under ℓ_1 sparsity*

LEMMA S10. *Let $\mathcal{V}_{1,\lambda} = \{v \in \mathbb{R}^p : \|v\|_2 = 1, \|v\|_1 \leq \lambda\}$. Then*

$$\log N(\epsilon, \mathcal{V}_{1,\lambda}, \|\cdot\|_1) \leq p \log\left(1 + \frac{2\lambda}{\epsilon}\right),$$

$$\log N(\epsilon, \mathcal{V}_{1,\lambda}, \|\cdot\|_2) \leq c \min\left\{ p\log\left(1 + \frac{2\lambda}{\epsilon p^{1/2}}\right), \frac{\lambda^2}{\epsilon^2}\log\left(1 + \frac{2p\epsilon^2}{\lambda^2}\right)\right\}.$$

Proof. The first claim is a direct consequence of Lemma 5.7 in Wainwright (2019). Below we provide a proof for the second claim.

We start with proving that

$$\log N(\epsilon, \mathcal{V}_{1,\lambda}, \|\cdot\|_2) \leq cp \log\left(1 + \frac{2\lambda}{\epsilon p^{1/2}}\right).$$

Let $\mathcal{B}_1 = \{v \in \mathbb{R}^p : \|v\|_1 \le 1\}$ and $\mathcal{B}_2 = \{v \in \mathbb{R}^p : \|v\|_2 \le 1\}$. Also, $\mathrm{vol}(A)$ denotes the volume of $A \subset \mathbb{R}^p$, and $aA + bB = \{au + bv : u \in A, v \in B\}$ for $a, b \in \mathbb{R}$ and $A, B \subset \mathbb{R}^p$. Using Lemma 5.7 in Wainwright (2019) we deduce

$$\begin{aligned} N(\epsilon, \mathcal{V}_{1,\lambda}, \|\cdot\|_2) &\le N(\epsilon/\lambda, \mathcal{B}_1, \|\cdot\|_2) \\ &\le \frac{\mathrm{vol}(2\lambda/\epsilon \mathcal{B}_1 + \mathcal{B}_2)}{\mathrm{vol}(\mathcal{B}_2)} = \frac{\mathrm{vol}(\mathcal{B}_1 + \epsilon/(2\lambda)\mathcal{B}_2)}{\mathrm{vol}(\epsilon/(2\lambda)\mathcal{B}_2)} \\ &\le \frac{\mathrm{vol}((1 + \epsilon p^{1/2}/(2\lambda))\mathcal{B}_1)}{\mathrm{vol}(\epsilon/(2\lambda)\mathcal{B}_2)} \\ &\le \left\{\frac{1 + \epsilon p^{1/2}/(2\lambda)}{\epsilon/(2\lambda)}\right\}^p \left(\frac{c}{p^{1/2}}\right)^p, \end{aligned}$$

where the last two inequalities are due to the fact that $\mathcal{B}_2 \subset p^{1/2}\mathcal{B}_1$, $\mathrm{vol}(\mathcal{B}_1)^{1/p} \asymp 1/p$ and $\mathrm{vol}(\mathcal{B}_2)^{1/p} \asymp 1/p^{1/2}$.

Next, we use the Maurey's empirical method (as in Theorem 0.0.2 and Corollary 0.0.4 of Vershynin, 2018) to prove

$$\log N(\epsilon, \mathcal{V}_{1,\lambda}, \|\cdot\|_2) \le c\frac{\lambda^2}{\epsilon^2} \log\left(1 + \frac{2p\epsilon^2}{\lambda^2}\right).$$

It is sufficient to establish

$$\log N(\epsilon, \mathcal{B}_1, \|\cdot\|_2) \le c\frac{1}{\epsilon^2} \log\left(1 + 2p\epsilon^2\right),$$

and then the desired claim follows from $N(\epsilon, \mathcal{V}, \|\cdot\|_2) \le N(\epsilon/\lambda, \mathcal{B}_1, \|\cdot\|_2)$. Let e_j be the p-dimensional vector that takes value 1 in the j-th coordinate and 0 otherwise. For an arbitrary $v \in \mathcal{B}_1$, define a random vector R in $\mathbb{R}^p$ such that $\mathrm{pr}(R = 0) = 1 - \|v\|_1$ and

$$\mathrm{pr}\big\{R = \mathrm{sgn}(v_j)e_j\big\} = |v_j|, \qquad \text{for } j = 1, \dots, p.$$

Then $E(R) = \sum_{j=1}^p v_j e_j = v$. Let $R_1, \dots, R_k$ be i.i.d. copies of R and $\bar{R} = k^{-1}\sum_{i=1}^k R_i$. Then

$$E\big(\|\bar{R} - v\|_2^2\big) = k^{-1}E\big(\|R - v\|_2^2\big) \le k^{-1}.$$

Taking $k = 1/\epsilon^2$, we conclude that there exists a realization $\bar{r}$ of $\bar{R}$ such that $\|\bar{r} - v\|_2 \le \epsilon$. It remains to find the number of possible realizations $\bar{r}$ regardless of v.

Write $\bar{r} = k^{-1}(z_1, \dots, z_p)^\top$ and observe $\|\bar{r}\|_1 \le 1$. Consequently,

$$\sum_{j=1}^p |z_j| = \sum_{j=1}^p z_j^+ + z_j^- \le k, \text{ where } z_j^+, z_j^- \in \mathbb{Z} \text{ and } 0 \le z_j^+, z_j^- \le k, \tag{S59}$$

with $z_j^+ = \max(z_j, 0)$ and $z_j^- = \max(0, -z_j)$. Thus, the number of possible realizations is upper bounded by the number of solutions of the inequality (S59) which is

$$\binom{0+2p-1}{2p-1} + \binom{1+2p-1}{2p-1} + \cdots + \binom{k+2p-1}{2p-1} = \binom{k+2p}{k}.$$

With $k = 1/\epsilon^2$ we conclude

$$\log N(\epsilon, \mathcal{B}_1, \|\cdot\|_2) \leq \frac{c}{\epsilon^2} \log(1 + 2p\epsilon^2).$$

LEMMA S11. *For any $\epsilon > 0$,*

(1) $\sup_{m \in \mathfrak{M}} \log N(\epsilon, \acute{\mathcal{G}}_n, e_m) \leq c \min\left\{p \log\left(1 + \frac{2\lambda}{\epsilon p^{1/2}}\right), \frac{\lambda^2}{\epsilon^2} \log\left(1 + \frac{2p\epsilon^2}{\lambda^2}\right)\right\} + c\acute{\kappa}_n/\epsilon$, *where $\mathfrak{M}$ denotes the collection of probability measures on $\mathbb{R}^p$ satisfying Assumption 1.*

(2) if $n^{-1}p^2 \log^3(pn) \lesssim 1$, with probability at least $1 - cn^{-1}$,

$$\log N(\epsilon, \acute{\mathcal{G}}_n, e_m) \leq c\acute{\kappa}_n/\epsilon + c \min\left\{p \log\left(1 + \frac{2\lambda}{\epsilon p^{1/2}}\right), \frac{\lambda^2}{\epsilon^2} \log\left(1 + \frac{2p\epsilon^2}{\lambda^2}\right)\right\},$$

where m can be either P_n or Q_n.

Proof. (1). Below we establish the first claim for one m from the corresponding family of probability measures; the uniform bound follows from the fact that all constants involved below does not depend on m.

Let $\mathcal{E}_{\acute{\mathcal{F}}_n, \epsilon_1} = \{f_1, \ldots, f_r\}$ be a minimal ϵ_1-net of $\acute{\mathcal{F}}_n$ under the metric $\|\cdot\|_\infty$, and $\mathcal{E}_{\mathcal{V}_{1,\lambda}, \epsilon_2} = \{v_1, \ldots, v_l\}$ be a minimal ϵ_2-net of $\mathcal{V}_{1,\lambda}$ under the metric $\|\cdot\|_2$. Taking $\epsilon_1 = \epsilon/2$ and $\epsilon_2 = \epsilon/(4\nu)$, based on the proof of Lemma S5(1), we then see that $\{f \circ v : f \in \mathcal{E}_{\acute{\mathcal{F}}_n, \epsilon_1}, v \in \mathcal{E}_{\mathcal{V}_{1,\lambda}, \epsilon_2}\}$ form an ϵ-net of $\acute{\mathcal{G}}_n$ under e_P. From Lemma S10, we have $\log l = \log N(\epsilon/(4\nu), \mathcal{V}_{1,\lambda}, \|\cdot\|_2) \leq c \min\left\{p \log\left(1 + \frac{2\lambda}{\epsilon p^{1/2}}\right), \frac{\lambda^2}{\epsilon^2} \log\left(1 + \frac{2p\epsilon^2}{\lambda^2}\right)\right\}$. Moreover, from Lemma S3, we obtain $\log r = \log N(\epsilon/2, \acute{\mathcal{F}}_n, \|\cdot\|_\infty) \leq c\acute{\kappa}_n/\epsilon$. Thus, we conclude that

$$\log N(\epsilon, \acute{\mathcal{G}}_n, e_m) \leq c \min\left\{p \log\left(1 + \frac{2\lambda}{\epsilon p^{1/2}}\right), \frac{\lambda^2}{\epsilon^2} \log\left(1 + \frac{2p\epsilon^2}{\lambda^2}\right)\right\} + \acute{\kappa}_n/\epsilon\,.$$

(2). We provide the proof under $m = P_n$, and the proof for $m = Q_n$ is analogous. Let $\mathcal{E}_{\acute{\mathcal{F}}_n, \epsilon_1}$ and $\mathcal{E}_{\mathcal{V}_{1,\lambda}, \epsilon_2}$ be as in (1), with ϵ_1, ϵ_2 to be determined later. Then for any $g = f \circ v \in \acute{\mathcal{G}}_n$, there exists $h = \check{f} \circ \check{v}$ where $\check{f} \in \mathcal{E}_{\acute{\mathcal{F}}_n, \epsilon_1}$ and $\check{v} \in \mathcal{E}_{\mathcal{V}_{1,\lambda}, \epsilon_2}$ such that

$$\begin{aligned}
\int |g(x) - h(x)|^2 dP_n(x) &\leq 2 \int |f(v^\top x) - \check{f}(v^\top x)|^2 dP_n(x) + 2 \int |v^\top x - \check{v}^\top x|^2 dP_n(x) \\
&\leq 2\|f - \check{f}\|_\infty^2 + \frac{2}{n_1} \sum_{i=1}^{n_1} \frac{|(v - \check{v})^\top X_i|^2}{\|v - \check{v}\|_2^2} \|v - \check{v}\|_2^2 \\
&\leq 2\|f - \check{f}\|_\infty^2 + 2\|v - \check{v}\|_2^2 \sup_{v \in \mathcal{V}} n_1^{-1} \sum_{i=1}^{n_1} (v^\top X_i)^2, \qquad \text{(S60)}
\end{aligned}$$

□

where $\mathcal{V} = \{v \in \mathbb{R}^p : v^\top v = 1\}$.

Next, we aim to bound $\sup_{v\in\mathcal{V}} n_1{}^{-1}\sum_{i=1}^{n_1}(v^\top X_i)^2$ with a high probability. Denote $\zeta_n = 2p^{1/2}\nu\log(pn)$. Under the event Ω_n defined in (S3), we have $\max_{i=1,\ldots,n_1}\sup_{v\in\mathcal{V}}|v^\top X_i| \le \zeta_n$. Note that

$$\Big\{\sup_{v\in\mathcal{V}} n_1{}^{-1}\sum_{i=1}^{n_1}(v^\top X_i)^2\Big\}1_{\Omega_n} \le \sup_{v\in\mathcal{V}} n_1^{-1}\sum_{i=1}^{n_1}(v^\top X_i)^2 1_{\Omega_{n,i}},$$

where $\Omega_{n,i} = \{\|X_i\|_\infty \le 2\nu\log(pn)\}$. Applying Assumption 1 leads to

$$\begin{aligned}
&\sup_{v\in\mathcal{V}} n_1{}^{-1}\sum_{i=1}^{n_1}(v^\top X_i)^2 1_{\Omega_{n,i}}\\
&\le \sup_{v\in\mathcal{V}}\Big|n_1{}^{-1}\sum_{i=1}^{n_1}[(v^\top X_i)^2 1_{\Omega_{n,i}} - E\{(v^\top X_i)^2 1_{\Omega_{n,i}}\}]\Big| + \sup_{v\in\mathcal{V}} E\{(v^\top X)^2\}\\
&\le \sup_{v\in\mathcal{V}}\Big|n_1{}^{-1}\sum_{i=1}^{n_1}[(v^\top X_i)^2 1_{\Omega_{n,i}} - E\{(v^\top X_i)^2 1_{\Omega_{n,i}}\}]\Big| + 4\nu^2.
\end{aligned}$$

By the Talagrand's inequality in Lemma S20, we have

$$\begin{aligned}
&\mathrm{pr}\Bigg\{\sup_{v\in\mathcal{V}}\Big|n_1^{-1}\sum_{i=1}^{n_1}[(v^\top X_i)^2 1_{\Omega_{n,i}} - E\{(v^\top X_i)^2 1_{\Omega_{n,i}}\}]\Big|\\
&\quad\le cE\Bigg(\sup_{v\in\mathcal{V}}\Big|n_1^{-1}\sum_{i=1}^{n_1}[(v^\top X_i)^2 1_{\Omega_{n,i}} - E\{(v^\top X_i)^2 1_{\Omega_{n,i}}\}]\Big|\Bigg) + \frac{c\log^{1/2} n}{n_1^{1/2}} + c\frac{\zeta_n^2\log n}{n_1}\Bigg\}\\
&\ge 1 - 1/n.
\end{aligned}$$

For any $v_1, v_2 \in \mathcal{V}$, define the distance

$$\begin{aligned}
d(v_1, v_2) &= \frac{1}{n_1}\sum_{i=1}^{n_1}\{(v_1^\top X_i)^2 1_{\Omega_{n,i}} - (v_2^\top X_i)^2 1_{\Omega_{n,i}}\}^2\\
&\le \frac{1}{n_1}\sum_{i=1}^{n_1}(|v_1^\top X_i|1_{\Omega_{n,i}} + |v_2^\top X_i|1_{\Omega_{n,i}})^2\{|(v_1 - v_2)^\top X_i|1_{\Omega_{n,i}}\}^2\\
&\le 4\zeta_n^4\|v_1 - v_2\|_2^2.
\end{aligned}$$

Thus, we have $\log N(\epsilon, \mathcal{V}, d(\cdot,\cdot)) \le \log N(\epsilon/(2\zeta_n^2), \mathcal{V}, \|\cdot\|_2) \le p\log(1 + 4\zeta_n^2/\epsilon)$, according to Lemma 5.7 in Wainwright (2019).

Let $\xi_1, \ldots, \xi_{n_1}$ be i.i.d. Rademacher random variables independent of $X_1, \ldots, X_{n_1}$. By symmetrization,

$$\begin{aligned}
&E\Bigg(\sup_{v\in\mathcal{V}}\Big|n_1{}^{-1}\sum_{i=1}^{n_1}[(v^\top X_i)^2 1_{\Omega_{n,i}} - E\{(v^\top X_i)^2 1_{\Omega_{n,i}}\}]\Big|\Bigg)\\
&\le 2E\Bigg\{\sup_{v\in\mathcal{V}}\Big|n_1{}^{-1}\sum_{i=1}^{n_1}\xi_i(v^\top X_i)^2 1_{\Omega_{n,i}}\Big|\Bigg\}
\end{aligned}$$

$$\leq c n_1^{-1/2}\left\{\check{J}_1(1/\zeta_n^2)\zeta_n^2+\frac{\check{J}_1^2(1/\zeta_n^2)\zeta_n^2}{n_1^{1/2}(1/\zeta_n^2)^2}\right\}$$
$$\leq \frac{cp^{1/2}\log^{1/2}(1+4\zeta_n^2)}{n_1^{1/2}}+\frac{cp\log(1+4\zeta_n^2)\zeta_n^2}{n_1},$$

where

$$\check{J}_1(\varepsilon)=\int_0^{\varepsilon}\{p\log(1+4/\delta)\}^{1/2}d\delta,$$

and the last two inequalities use similar arguments in Lemma S6 and Lemma S23.

Denote

$$\check{\Omega}_n=\Omega_n\cap\left\{\sup_{v\in\mathcal{V}} n_1{}^{-1}\sum_{i=1}^{n_1}(v^\top X_i)^2\leq a_n\right\}, \tag{S61}$$

where $a_n \asymp 1+n^{-1/2}\{p\log(pn)\}^{1/2}+n^{-1}p^2\log^3(pn)$. We conclude $\mathrm{pr}(\check{\Omega}_n)\geq 1-cn^{-1}$.

Consequently, for (S60), we have on the event $\check{\Omega}_n$,

$$\int|g(x)-h(x)|^2dP_n(x)\leq 2\epsilon_1^2+2a_n\epsilon_2^2.$$

On the event $\check{\Omega}_n$, taking $\epsilon_1=\epsilon/2$ and $\epsilon_2=\epsilon/2a_n^{1/2}$, we see that $\{f\circ v: f\in\mathcal{E}_{\acute{\mathcal{F}}_n,\epsilon_1}, v\in\mathcal{E}_{\mathcal{V},\epsilon_2}\}$ is an ϵ-net of $\acute{\mathcal{G}}_n$ under e_{P_n}. According to Lemma S3 and Lemma S10, on the event $\check{\Omega}_n$,

$$\begin{aligned}\log N(\epsilon,\acute{\mathcal{G}}_n,e_{P_n})&\leq\log N(\epsilon/2,\acute{\mathcal{F}}_n,\|\cdot\|_\infty)+\log N(\epsilon/2a_n^{1/2},\mathcal{V}_{1,\lambda},\|\cdot\|_2)\\&\leq c\acute{\kappa}_n/\epsilon+c\min\left\{p\log\left(1+\frac{2\lambda}{\epsilon p^{1/2}}\right),\frac{\lambda^2}{\epsilon^2}\log\left(1+\frac{2p\epsilon^2}{\lambda^2}\right)\right\},\end{aligned}$$

due to $n^{-1}p^2\log^3(pn)\lesssim 1$.

LEMMA S12. *For $\mathbb{X}_{n_1}$ and $\mathbb{Y}_{n_2}$ defined in (S5), under Assumptions 1 and 2,*

$$\begin{aligned}\max\{E(\|\mathbb{X}_{n_1}\|_{\acute{\mathcal{G}}_n}),E(\|\mathbb{Y}_{n_2}\|_{\acute{\mathcal{G}}_n})\}\leq c\lambda\log^{3/2}(2p/\lambda^2)+c\acute{\kappa}_n^{1/2}+\\ cn^{-1/2}\{\lambda^2\acute{\kappa}_n\log^3(2p/\lambda^2)+\acute{\kappa}_n^2\}\leq c\acute{B}_n.\end{aligned}$$

Proof. Following the arguments in the proof of Lemma S6, we replace Ω_n with $\check{\Omega}_n$ defined in (S61) and use the covering number result in Lemma S10. Note that $\lambda/p^{1/2}\leq 1$,

$$\begin{aligned}&\int_0^{1/\acute{\kappa}_n}c\min\left\{p^{1/2}\log^{1/2}\left(1+\frac{2\lambda}{\delta\acute{\kappa}_np^{1/2}}\right),\frac{\lambda}{\delta\acute{\kappa}_n}\log^{1/2}\left(1+\frac{2p\delta^2\acute{\kappa}_n^2}{\lambda^2}\right)\right\}d\delta\\&\leq c\int_0^{\lambda/(\acute{\kappa}_np^{1/2})}p^{1/2}\log^{1/2}\left(1+\frac{2\lambda}{p^{1/2}\delta\acute{\kappa}_n}\right)d\delta+c\int_{\lambda/(\acute{\kappa}_np^{1/2})}^{1/\acute{\kappa}_n}\frac{\lambda}{\delta\acute{\kappa}_n}\log^{1/2}\left(1+\frac{2p\delta^2\acute{\kappa}_n^2}{\lambda^2}\right)d\delta\\&\leq c(\lambda/\acute{\kappa}_n)\log^{3/2}(1+2p/\lambda^2),\end{aligned}$$

due to Lemma S23 and

$$\int_{\lambda/(\acute{\kappa}_n p^{1/2})}^{1/\acute{\kappa}_n} \frac{\lambda}{\delta \acute{\kappa}_n} \log^{1/2}\left(1+\frac{2p\delta^2\acute{\kappa}_n^2}{\lambda^2}\right) d\delta$$
$$\leq c\frac{\lambda}{\acute{\kappa}_n}\left[\log^{3/2}\left(1+\frac{2p\delta^2\acute{\kappa}_n^2}{\lambda^2}\right)\right]\Bigg|_{\lambda/(\acute{\kappa}_n p^{1/2})}^{1/\acute{\kappa}_n} \leq c(\lambda/\acute{\kappa}_n)\log^{3/2}(1+2p/\lambda^2).$$

Other details are omitted for conciseness. □

LEMMA S13. *Let $L=\lfloor \acute{\kappa}_n/\epsilon_1\rfloor$ and consider the points in $[-\acute{\kappa}_n,\acute{\kappa}_n]$ given by*

$$x_j=(j-1)\epsilon_1,\ \ x_{-j}=(-j+1)\epsilon_1,\ \ j=1,\ldots,L.$$

Moreover, define the functions

$$\chi_+(x)=\begin{cases}0 & \text{for } x<0,\\ x & \text{for } x\in[0,1],\\ 1 & \text{otherwise.}\end{cases}\qquad \chi_-(x)=\begin{cases}0 & \text{for } x>0,\\ x & \text{for } x\in[-1,0],\\ -1 & \text{otherwise.}\end{cases}$$

For each $\beta=(\beta_{-L},\ldots,\beta_{-1},\beta_1,\ldots,\beta_L)^\top\in\{-1,1\}^{2L}$, it induces a function $f_\beta: [-\acute{\kappa}_n,\acute{\kappa}_n]\to[-\acute{\kappa}_n,\acute{\kappa}_n]$ via

$$f_\beta(x)=\sum_{j=1}^{L}\beta_j\epsilon_1\chi_+\left(\frac{x-x_j}{\epsilon_1}\right)+\sum_{j=1}^{L}\beta_{-j}\epsilon_1\chi_-\left(\frac{x-x_{-j}}{\epsilon_1}\right).$$

Define the collection of functions $\mathcal{E}_{\acute{\mathcal{F}}_n,\epsilon_1}=\{f_\beta,\beta\in\{-1,1\}^{2L}\}$. Let $\acute{\mathcal{I}}=\{f\circ v: f\in\mathcal{E}_{\acute{\mathcal{F}}_n,\epsilon_1}, v\in\mathcal{E}_{\mathcal{V}_{1,\lambda},\epsilon_2}\}$, where $\mathcal{E}_{\mathcal{V}_{1,\lambda},\epsilon_2}$ is a minimal ϵ_2-net of $\mathcal{V}_{1,\lambda}$ under the $\|\cdot\|_2$. Let $\epsilon_1=\epsilon\|\acute{G}\|_{(P+Q)/2,2}/2$, $\epsilon_2=\epsilon\|\acute{G}\|_{(P+Q)/2,2}/(4\nu)$. Then, we have $\acute{\mathcal{I}}$ is an $\epsilon\|\acute{G}\|_{(P+Q)/2,2}$-net of $\acute{\mathcal{G}}_n$ under the metric $e_{(P+Q)/2}$ and

$$\log\acute{N}=\log|\acute{\mathcal{I}}|\leq c\min\left\{p\log\left(1+\frac{2\lambda}{\epsilon\|\acute{G}\|_{(P+Q)/2,2}p^{1/2}}\right),\right.$$
$$\left.\frac{\lambda^2}{\epsilon^2\|\acute{G}\|^2_{(P+Q)/2,2}}\log\left(1+\frac{2p\epsilon^2\|\acute{G}\|^2_{(P+Q)/2,2}}{\lambda^2}\right)\right\}+\frac{c\acute{\kappa}_n}{\epsilon\|\acute{G}\|_{(P+Q)/2,2}}.$$

Proof. The proof is analogous to that of Lemma S1 and is therefore omitted. □

LEMMA S14. *Let $\acute{\mathcal{I}}$ and $\acute{\mathcal{J}}$ be defined (S8) and (S10). Suppose $\epsilon\acute{\kappa}_n(\log n)\leq c$, $n^{-1/3}\acute{\kappa}_n\leq c\acute{\tau}_n$, $n^{-1/2}\acute{\kappa}_n\{p\log(\acute{\kappa}_n/\acute{\tau}_n)\}^{1/2}\leq c\acute{\tau}_n$ and $\acute{\tau}_n\leq c(\log n)^{-1}\min\{p^{-1/2},\acute{\kappa}_n^{-1}\}$. Then,*

$$\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\left(\mathbb{Z}\acute{\mathcal{J}}\leq t\right)-\mathrm{pr}\left(\mathbb{Z}\acute{\mathcal{I}}\leq t\right)\right|\leq cn^{-1}, \tag{S62}$$

$$\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\left(\mathbb{G}_n\acute{\mathcal{I}}\leq t\right)-\mathrm{pr}\left(\mathbb{Z}\acute{\mathcal{I}}\leq t\right)\right|\leq cn^{-1}+2\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\left(\mathbb{G}_n\acute{\mathcal{J}}\leq t\right)-\mathrm{pr}\left(\mathbb{Z}\acute{\mathcal{J}}\leq t\right)\right|. \tag{S63}$$

In addition, with probability at least $1 - cn^{-1}$,

$$d_K\big(\mathcal{L}(\mathbb{Z}_n^\star \acute{\mathcal{I}} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{I}})\big) \le cn^{-1} + 2\sup_{t\in\mathbb{R}} \big|\mathrm{pr}(\mathbb{Z}_n^\star \acute{\mathcal{J}} \le t \mid \mathscr{D}) - \mathrm{pr}(\mathbb{Z}\acute{\mathcal{J}} \le t)\big|, \tag{S64}$$

$$\begin{aligned} d_K\big(\mathcal{L}(\mathbb{V}_n^\star \acute{\mathcal{I}} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}_n^\star \acute{\mathcal{I}} \mid \mathscr{D})\big) &\le cn^{-1} + 2d_K\big(\mathcal{L}(\mathbb{V}_n^\star \acute{\mathcal{J}} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}_n^\star \acute{\mathcal{J}} \mid \mathscr{D})\big) \\ &\quad + 2d_K\big(\mathcal{L}(\mathbb{Z}_n^\star \acute{\mathcal{J}} \mid \mathscr{D}), \mathcal{L}(\mathbb{Z}\acute{\mathcal{J}})\big). \end{aligned} \tag{S65}$$

Proof. Here we only highlight some important steps and the main quantities involved, the claims can be easily established by using arguments in the proof of Lemma S2.

We begin with proving (S62) and (S63). Note that

$$\sup_{t\in\mathbb{R}} \big|\mathrm{pr}\big(\mathbb{Z}\acute{\mathcal{J}} \le t\big) - \mathrm{pr}\big(\mathbb{Z}\acute{\mathcal{I}} \le t\big)\big| \le \mathrm{pr}\bigg(\max_{g\in\acute{\mathcal{J}}} \mathbb{Z}g \le \acute{t}_2\bigg) + \mathrm{pr}\bigg(\max_{g\in\acute{\mathcal{J}}^c} \mathbb{Z}g \ge \acute{t}_1\bigg),$$

$$\begin{aligned} \sup_{t\in\mathbb{R}} \big|\mathrm{pr}\big(\mathbb{G}_n\acute{\mathcal{I}} \le t\big) - \mathrm{pr}\big(\mathbb{G}_n\acute{\mathcal{J}} \le t\big)\big| &\le \mathrm{pr}\bigg(\max_{g\in\acute{\mathcal{J}}} \mathbb{Z}g \le \acute{t}_2\bigg) + \mathrm{pr}\bigg(\max_{g\in\acute{\mathcal{J}}^c} \mathbb{G}_n g \ge \acute{t}_1\bigg) \\ &\quad + \sup_{t\in\mathbb{R}} \big|\mathrm{pr}\big(\mathbb{G}_n\acute{\mathcal{J}} \le t\big) - \mathrm{pr}\big(\mathbb{Z}\acute{\mathcal{J}} \le t\big)\big|, \end{aligned}$$

for some real numbers satisfying $\acute{t}_1 \le \acute{t}_2$. Let $\mathcal{F}_{\varsigma_n}^\circ$ be the same as in the proof of Lemma S2, define $\acute{\mathcal{J}}^\circ = \mathcal{F}_{\varsigma_n}^\circ \circ \acute{v}^\circ$, where $\acute{v}^\circ$ is defined in Assumption S1. It is straightforward to verify that the class $\acute{\mathcal{J}}^\circ$ satisfies the following properties.

- The cardinality $\acute{K} = \acute{K}_n = |\acute{\mathcal{J}}^\circ| \ge c2^{1/(8\varsigma_n)}$.
- $\sup_{g\in\acute{\mathcal{J}}^\circ} \mathrm{var}(\mathbb{Z}g) \asymp \varsigma_n^2$ and $\inf_{g\in\acute{\mathcal{J}}^\circ} \mathrm{var}(\mathbb{Z}g) \asymp \varsigma_n^2$.
- $\mathrm{cor}(\mathbb{Z}g, \mathbb{Z}h) \le \rho_0 < 1$ for any two distinct $g, h \in \acute{\mathcal{J}}^\circ$. □

Take

$$\acute{t}_1 = \acute{C}_1\{(\acute{\tau}_n\acute{\kappa}_n)^{1/2} + \acute{\tau}_n(p\log n)^{1/2}\}, \quad \acute{t}_2 = \acute{C}_2\varsigma_n \log^{1/2}(\acute{K}),$$

for some constants $\acute{C}_1, \acute{C}_2 > 0$. Recall that $\varsigma_n \asymp (\log n)^{-1}$. The inequality $\acute{t}_1 \le \acute{t}_2$ can be established due to the construction of the classes and the condition on $\acute{\tau}_n$.

Applying Lemma S17 easily leads to

$$\mathrm{pr}\bigg(\max_{g\in\acute{\mathcal{J}}} \mathbb{Z}g \le \acute{t}_2\bigg) \le cn^{-1}.$$

According to the Borell–Sudakov–Tsirel'son inequality and the Dudley's maximal inequality,

$$\mathrm{pr}\bigg[\max_{g\in\acute{\mathcal{J}}^c} \mathbb{Z}g \ge \acute{C}_1\{(\acute{\tau}_n\acute{\kappa}_n)^{1/2} + \acute{\tau}_n(p\log n)^{1/2}\}\bigg] \le n^{-1}.$$

Note that

$$E(\|\mathbb{X}_{n_1}\|_{\acute{\mathcal{J}}^c}) \le c\acute{\tau}_n\{p\log(\acute{\kappa}_n/\acute{\tau}_n)\}^{1/2} + c(\acute{\kappa}_n\acute{\tau}_n)^{1/2} + cn^{-1/2}\{\acute{\kappa}_n^2\acute{\tau}_n^{-1} + p\acute{\kappa}_n\log(\acute{\kappa}_n/\acute{\tau}_n)\}.$$

The Talagrand concentration inequality implies

$$\mathrm{pr}\left(\max_{g\in\acute{\mathcal{J}}^c}\mathbb{G}_n g \geq \acute{t}_1\right) \leq cn^{-1},$$

since $n^{-1/3}\acute{\kappa}_n \leq c\acute{\tau}_n$ and $n^{-1/2}\acute{\kappa}_n\{p\log(\acute{\kappa}_n/\acute{\tau}_n)\}^{1/2} \leq c\acute{\tau}_n$. Combining all the arguments completes the proof of (S62) and (S63).

To prove (S64) and (S65), according to Lemma S2, it suffices to bound $\mathrm{pr}\big(\sup_{g\in\acute{\mathcal{J}}^c}\mathbb{Z}_n^\star g \geq \acute{t}_1 \mid \mathscr{D}\big)$ and $\mathrm{pr}\big(\sup_{g\in\acute{\mathcal{J}}^c}\mathbb{V}_n^\star g \geq \acute{t}_1 \mid \mathscr{D}\big)$.

Denote

$$\sigma^2_{n,\acute{\mathcal{J}}^c} = \sup_{g\in\acute{\mathcal{J}}^c}\frac{n_2}{n_1+n_2}\frac{1}{n_1}\sum_{i=1}^{n_1}[g(X_i)-\hat{E}_{n_1}\{g(X)\}]^2 + \frac{n_1}{n_1+n_2}\frac{1}{n_2}\sum_{i=1}^{n_2}[g(Y_i)-\hat{E}_{n_2}\{g(Y)\}]^2.$$

Applying similar arguments in the proof of Lemma S2, we obtain that with probability at least $1-cn^{-1}$,

$$\sigma^2_{n,\acute{\mathcal{J}}^c} \leq \acute{\tau}_n^2 + cn^{-1/2}[\acute{\tau}_n^{1/2}\acute{\kappa}_n^{3/2} + \acute{\tau}_n\acute{\kappa}_n\{p\log(\acute{\kappa}_n/\acute{\tau}_n)\}^{1/2}] + cn^{-1}\{\acute{\tau}_n^{-1}\acute{\kappa}_n^3 + \acute{\kappa}_n^2 p\log(\acute{\kappa}_n/\acute{\tau}_n)\}.$$

Under the conditions $n^{-1/2}\acute{\kappa}_n\{p\log(\acute{\kappa}_n/\acute{\tau}_n)\}^{1/2} \leq c\acute{\tau}_n$ and $n^{-1/3}\acute{\kappa}_n \leq c\acute{\tau}_n$, we have with probability at least $1-cn^{-1}$, $\sigma^2_{n,\acute{\mathcal{J}}^c} \leq c\acute{\tau}_n^2$. The Borell–Sudakov–Tsirel'son inequality and the Dudley's maximal inequality imply that with probability at least $1-cn^{-1}$,

$$\mathrm{pr}\big[\mathbb{Z}_n^\star\acute{\mathcal{J}}^c \geq \acute{C}_1\{(\acute{\tau}_n\acute{\kappa}_n)^{1/2} + \acute{\tau}_n(p\log n)^{1/2}\} \mid \mathscr{D}\big] \leq cn^{-1}.$$

Analogously, we show that with probability at least $1-cn^{-1}$,

$$\mathrm{pr}\big[\mathbb{V}_n^\star\acute{\mathcal{J}}^c \geq \acute{C}_1\{(\acute{\tau}_n\acute{\kappa}_n)^{1/2} + \acute{\tau}_n(p\log n)^{1/2}\} \mid \mathscr{D}\big] \leq cn^{-1}.$$

LEMMA S15. *Let $\acute{\mathcal{J}} = \{g_1,\ldots,g_{\acute{r}}\}$ be defined in (S10) and $\acute{r}$ is the number of elements in $\acute{\mathcal{J}}$. For sequences a_n and b_n such that $\max\{|a_n|,|b_n|\} \leq c_\star$ for a constant $c_\star < \infty$, for $j = 1,\ldots,\acute{r}$ and $i = 1,\ldots,n$, define*

$$U_{ij} = \begin{cases} a_n g_j(X_i) & \text{if } 1 \leq i \leq n_1 \\ b_n g_j(Y_{i-n_1}) & \text{if } n_1+1 \leq i \leq n. \end{cases}$$

Then, under Assumptions 1 and 2,

$$E\left\{\max_{1\leq j,k\leq\acute{r}}\left|\frac{1}{n}\sum_{i=1}^{n}\big(\{U_{ij}-E(U_{ij})\}\{U_{ik}-E(U_{ik})\} - E[\{U_{ij}-E(U_{ij})\}\{U_{ik}-E(U_{ik})\}]\big)\right|\right\} \leq cn^{-1/2}\acute{B}_n,$$

and with probability at least $1 - cn^{-1}$,

$$\max_{1\le j,k\le \acute{r}} \left| \frac{1}{n} \left\{ \sum_{i=1}^{n_1} (U_{ij} - \bar{U}_{1j})(U_{ik} - \bar{U}_{1k}) + \sum_{i=n_1+1}^{n} (U_{ij} - \bar{U}_{2j})(U_{ik} - \bar{U}_{2k}) \right\} - \frac{1}{n} \sum_{i=1}^{n} E[\{U_{ij} - E(U_{ij})\}\{U_{ik} - E(U_{ik})\}] \right| \le cn^{-1/2} \acute{B}_n,$$

where $\acute{B}_n$ *is defined in* (S54), $\bar{U}_{1j} = n_1^{-1} \sum_{i=1}^{n_1} U_{ij}$ *and* $\bar{U}_{2j} = n_2^{-1} \sum_{i=n_1+1}^{n} U_{ij}$.

Proof. The proof is similar to that of Lemma S8. Here, we only provide the proof of the following bound

$$E(\|\mathbb{X}_{n_1}\|_{\acute{\mathcal{G}}_n \times \acute{\mathcal{G}}_n}) \le c[\lambda \acute{\kappa}_n \log^{3/2}(pn) + \acute{\kappa}_n + n^{-1/2}\{\lambda^2 \acute{\kappa}_n^4 \log^3(pn) + \acute{\kappa}_n^4\}] \le c\acute{B}_n.$$

If $\acute{\kappa}_n \le p^{1/2}/\lambda$,

$$\begin{aligned}
&\int_0^{1/\acute{\kappa}_n^2} c \min\left\{ p^{1/2} \log^{1/2}\left(1 + \frac{2\lambda}{\delta \acute{\kappa}_n p^{1/2}}\right), \frac{\lambda}{\delta \acute{\kappa}_n} \log^{1/2}\left(1 + \frac{2p\delta^2 \acute{\kappa}_n^2}{\lambda^2}\right) \right\} d\delta \\
&\le c \int_0^{\lambda/(\acute{\kappa}_n p^{1/2})} p^{1/2} \log^{1/2}\left(1 + \frac{2\lambda}{p^{1/2}\delta \acute{\kappa}_n}\right) d\delta + c \int_{\lambda/(\acute{\kappa}_n p^{1/2})}^{1/\acute{\kappa}_n^2} \frac{\lambda}{\delta \acute{\kappa}_n} \log^{1/2}\left(1 + \frac{2p\delta^2 \acute{\kappa}_n^2}{\lambda^2}\right) d\delta \\
&\le c(\lambda/\acute{\kappa}_n) \log^{3/2}(pn).
\end{aligned}$$

If $\acute{\kappa}_n > p^{1/2}/\lambda$,

$$\begin{aligned}
&\int_0^{1/\acute{\kappa}_n^2} c \min\left\{ p^{1/2} \log^{1/2}\left(1 + \frac{2\lambda}{\delta \acute{\kappa}_n p^{1/2}}\right), \frac{\lambda}{\delta \acute{\kappa}_n} \log^{1/2}\left(1 + \frac{2p\delta^2 \acute{\kappa}_n^2}{\lambda^2}\right) \right\} d\delta \\
&\le c \int_0^{1/\acute{\kappa}_n^2} p^{1/2} \log^{1/2}\left(1 + \frac{2\lambda}{p^{1/2}\delta \acute{\kappa}_n}\right) d\delta \\
&\le c \frac{p^{1/2}}{\acute{\kappa}_n^2} \log^{1/2}(pn) \le c(\lambda/\acute{\kappa}_n) \log^{1/2}(pn).
\end{aligned}$$

Putting the above pieces together leads to

$$\int_0^{1/\acute{\kappa}_n^2} c \min\left\{ p^{1/2} \log^{1/2}\left(1 + \frac{2\lambda}{\delta \acute{\kappa}_n p^{1/2}}\right), \frac{\lambda}{\delta \acute{\kappa}_n} \log^{1/2}\left(1 + \frac{2p\delta^2 \acute{\kappa}_n^2}{\lambda^2}\right) \right\} d\delta \le c(\lambda/\acute{\kappa}_n) \log^{3/2}(pn).$$

By further applying the arguments in the proof of Lemma S6, we establish the claim. □

S.6. BACKGROUND RESULTS

LEMMA S16 (GAUSSIAN COMPARISON OF DISTRIBUTIONS). *Let* U *and* W *be random vectors in* $\mathbb{R}^r$ *with* $U \sim N(0, \Sigma^U)$ *and* $W \sim N(0, \Sigma^W)$. *Let* $\Delta = \max_{1\le j,k\le r} |\sigma_{jk}^U - \sigma_{jk}^W|$, $\bar{\sigma}^W = \max_{1\le j\le r} \sigma_{jj}^W$ *and* $\sigma_{jj}^W \ge b$ *where* σ_{jk}^U *and* σ_{jk}^W *denote the* (j,k)*-th elements of* Σ^U *and*

Σ^W, *respectively. Then*

$$\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\left(\max_{1\le j\le r} U_j \le t\right) - \mathrm{pr}\left(\max_{1\le j\le r} W_j \le t\right)\right|$$
$$\le c\frac{\Delta^{1/2}\log^{3/4} r}{b^{3/2}}\left[E\left(\max_{1\le j\le r} W_j\right) + \bar{\sigma}^W\left\{1 \vee \log^{1/2}\left(\frac{b}{\Delta\log^{3/2} r}\right)\right\}\right]^{1/2}.$$

Proof. Without loss of generality, we will assume that U and W are independent. Let $m^t : \mathbb{R}^r \to \mathbb{R}$ be defined as in the proof of Lemma S7 with a nonrandom number ϕ. Thus,

$$\sup_{t\in\mathbb{R}}\left|\mathrm{pr}\left(\max_{1\le j\le r} U_j \le t) - \mathrm{pr}(\max_{1\le j\le r} W_j \le t\right)\right|$$
$$\le \sup_{t\in\mathbb{R}}|E\{m^t(U)\} - E\{m^t(W)\}| + \sup_{t\in\mathbb{R}}\mathrm{pr}\left(\left|\max_{1\le j\le r} W_j - t\right| \le 2/\phi\right)$$
$$\le \sup_{t\in\mathbb{R}}|E\{m^t(U)\} - E\{m^t(W)\}| + \frac{c}{b^2\phi}\left[E\left(\max_{1\le j\le r} W_j\right) + \bar{\sigma}^W\{1\vee\log^{1/2}(b\phi)\}\right], \quad \text{(S66)}$$

where the second inequality is due to Lemma S19. It remains to bound the first term on the right-hand side. Note that

$$|E\{m^t(U)\} - E\{m^t(W)\}|$$
$$= \left|E\left\{\int_0^1 \frac{dm^t(\delta^{1/2}U + (1-\delta)^{1/2}W)}{d\delta}d\delta\right\}\right|$$
$$= \left|\int_0^1 \left[\frac{1}{2\delta^{1/2}}\sum_{j=1}^r E\{U_j\partial_j m^t(\delta^{1/2}U + (1-\delta)^{1/2}W)\} - \right.\right.$$
$$\left.\left.\frac{1}{2(1-\delta)^{1/2}}\sum_{j=1}^r E\{W_j\partial_j m^t(\delta^{1/2}U + (1-\delta)^{1/2}W)\}\right]d\delta\right|$$
$$= \left|\int_0^1\left[\frac{1}{2}\sum_{j,k=1}^r E\{(\sigma^U_{jk} - \sigma^W_{jk})\partial_{jk}m^t(\delta^{1/2}U + (1-\delta)^{1/2}W)\}\right]d\delta\right|$$
$$\le \frac{1}{2}\sum_{j,k=1}^r\int_0^1 E\{|(\sigma^U_{jk} - \sigma^W_{jk})\partial_{jk}m^t(\delta^{1/2}U + (1-\delta)^{1/2}W)|\}d\delta$$
$$\le \frac{1}{2}\max_{1\le j,k\le r}|\sigma^U_{jk} - \sigma^W_{jk}|\sum_{j,k=1}^r\int_0^1 E\{|\partial_{jk}m^t(\delta^{1/2}U + (1-\delta)^{1/2}W)|\chi(\delta)\}d\delta$$
$$\le c(\phi^2\log r)\Delta\int_0^1 E\{\chi(\delta)\}d\delta,$$

where the third equality is by Stein's identity and

$$\chi(\delta) = \begin{cases} 1 & \text{if } F^t_\beta(\delta^{1/2}U + (1-\delta)^{1/2}W) \in [0, \phi^{-1}] \\ 0 & \text{otherwise}\end{cases}$$

with F_β^t defined in the proof of Lemma S7. It can be shown that

$$\int_0^1 E\{\chi(\delta)\}d\delta \le \int_0^1 \sup_{w\in\mathbb{R}^r} \mathrm{pr}\big[\|W-w\|_\infty \le 1/\{\phi(1-\delta)^{1/2}\}\big]d\delta \le c(\log r)^{1/2}/(b\phi),$$

where the last inequality is by Lemma S18. Therefore,

$$|E\{m^t(U)\} - E\{m^t(W)\}| \le c\frac{\Delta\phi\log^{3/2}(r)}{b}. \tag{S67}$$

Combining (S66) and (S67) and choosing

$$\phi \asymp (\Delta b \log^{3/2} r)^{-1/2}\Bigg[E\bigg(\max_{1\le j\le r} W_j\bigg) + \bar\sigma^W\bigg\{1 \vee \log^{1/2}\bigg(\frac{b}{\Delta\log^{3/2} r}\bigg)\bigg\}\Bigg]^{1/2}$$

completes the proof. $\square$

LEMMA S17 (LOWER-TAIL BOUNDS). *Let $\xi = (\xi_1,\dots,\xi_p)$ be a centered Gaussian vector with a correlation matrix R and var$(\xi_j) = 1$ for $j = 1,\dots,p$. Fix two constants $\delta_0, \rho_0 \in (0,1)$, and suppose that $\max_{i\ne j} R_{ij} \le \rho_0$ holds. Then, there is a constant $C > 0$ depending only on (δ_0,ρ_0), such that*

$$\mathrm{pr}\bigg[\max_{j=1,\dots,p} \xi_j \le \delta_0\{2(1-\rho_0)\log p\}^{1/2}\bigg] \le Cp^{\frac{-(1-\rho_0)(1-\delta_0)^2}{\rho_0}}(\log p)^{\frac{1-\rho_0(2-\delta_0)-\delta_0}{2\rho_0}}.$$

Proof. See Theorem 2.1 in Lopes & Yao (2022). $\square$

LEMMA S18 (NAZAROV'S INEQUALITY). *Let $Z = (Z_1,\dots,Z_p)^\top$ be a centered Gaussian random vector in $\mathbb{R}^p$ such that $E(Z_j^2) \ge \underline{\sigma}^2$ for all $j = 1,\dots,p$ and some constant $\underline{\sigma} > 0$. Then for every $x \in \mathbb{R}^p$ and $\delta > 0$,*

$$\mathrm{pr}(Z \le x+\delta) - \mathrm{pr}(Z \le x) \le \frac{\delta}{\underline{\sigma}}\{(2\log p)^{1/2} + 2\}.$$

Proof. See Lemma A.1 in Chernozhukov et al. (2017a) and Theorem 1 in Chernozhukov et al. (2017b). $\square$

LEMMA S19 (ANTI-CONCENTRATION FOR GAUSSIAN MAXIMA). *Suppose $(Z_1,\dots,Z_p)^\top$ is a centered Gaussian random vector in $\mathbb{R}^p$ with $\sigma_j^2 = E(Z_j^2)$ for all $1 \le j \le p$. Let $\underline{\sigma} = \min_{1\le j\le p}\sigma_j$ and $\bar\sigma = \max_{1\le j\le p}\sigma_j$. Then for sufficiently large $c > 0$, for every $\delta > 0$,*

$$\sup_{z\in\mathbb{R}} \mathrm{pr}\bigg(\bigg|\max_{1\le j\le p} Z_j - z\bigg| \le \delta\bigg) \le c\frac{\delta}{\underline{\sigma}^2}\bigg(E\bigg(\max_{1\le j\le p} Z_j\bigg) + \bar\sigma[1 \vee \{\log(\underline{\sigma}/\delta)\}^{1/2}]\bigg).$$

Proof. The proof below resembles the proof of Theorem 3 in Chernozhukov et al. (2015), with modifications to slightly improve the dependence on $\underline{\sigma}$.

For every $z \in \mathbb{R}$ and $\delta > 0$, we have

$$\mathrm{pr}\bigg(\bigg|\max_{1\le j\le p} Z_j - z\bigg| \le \delta\bigg) \le 4\delta\{(1/\underline{\sigma} - 1/\bar\sigma)|z| + a_p + 1\}/\underline{\sigma},$$

where $a_p = E\{\max_{1\le j\le p}(Z_j/\sigma_j)\}$. By the Borell–Sudakov–Tsirel'son inequality (van der Vaart & Wellner, 1996, Proposition A.2.1), if $|z| \ge \delta + E(\max_{1\le j\le p} Z_j) + \bar\sigma\{2\log(\underline{\sigma}/\delta)\}^{1/2}$, we have

$$\operatorname{pr}\left(\left|\max_{1\le j\le p} Z_j - z\right| \le \delta\right) \le \operatorname{pr}\left(\max_{1\le j\le p} Z_j \ge |z| - \delta\right)$$
$$\le \operatorname{pr}\left[\max_{1\le j\le p} Z_j \ge E\left(\max_{1\le j\le p} Z_j\right) + \bar\sigma\{2\log(\underline{\sigma}/\delta)\}^{1/2}\right] \le \delta/\underline{\sigma}.$$

For $|z| \le \delta + E(\max_{1\le j\le p} Z_j) + \bar\sigma\{2\log(\underline{\sigma}/\delta)\}^{1/2}$ and $\delta \le \underline{\sigma}$, we have

$$\begin{aligned}&\operatorname{pr}\left(\left|\max_{1\le j\le p} Z_j - x\right| \le \delta\right)\\ &\le \frac{4\delta}{\underline{\sigma}}\left[\left(\frac{1}{\underline{\sigma}} - \frac{1}{\bar\sigma}\right)E\left(\max_{1\le j\le p} Z_j\right) + \left(\frac{\bar\sigma}{\underline{\sigma}} - 1\right)\{2\log(\underline{\sigma}/\delta)\}^{1/2} + a_p + 2 - \underline{\sigma}/\bar\sigma\right]\\ &\le c\frac{\delta}{\underline{\sigma}}\left(\frac{1}{\underline{\sigma}}E\left(\max_{1\le j\le p} Z_j\right) + \frac{\bar\sigma}{\underline{\sigma}}[1 \vee \{\log(\underline{\sigma}/\delta)\}^{1/2}]\right).\end{aligned}$$

If $\delta > \underline{\sigma}$, the inequality holds trivially. □

LEMMA S20 (TALAGRAND CONCENTRATION FOR EMPIRICAL PROCESSES). *Let $W_1, \dots, W_n$ be independent random vectors in $\mathbb{R}^p$. Let $\mathcal{H}$ be a pointwise measurable class of functions such that $\|h\|_\infty \le b$ and $E\{h(W)\} = 0$ for all $h \in \mathcal{H}$. The empirical process $\mathbb{W}_n h = n^{-1/2}\sum_{i=1}^n h(W_i)$ indexed by $h \in \mathcal{H}$ satisfies*

$$\operatorname{pr}\left\{\|\mathbb{W}_n\|_{\mathcal{H}} \ge (1+\delta)E(\|\mathbb{W}_n\|_{\mathcal{H}}) + c\sigma t^{1/2} + (c + c^2/\delta)\frac{bt}{n^{1/2}}\right\} \le e^{-t}, \text{ for all } t \ge 0 \text{ and } \delta > 0,$$

where $\sigma^2 = \sup_{h\in\mathcal{H}} n^{-1}\sum_{i=1}^n E\{h^2(W_i)\}$, and c is some universal positive constant.

Proof. See Inequality (3.86) of Wainwright (2019) and Theorem 8.7 in Sen (2018). □

LEMMA S21. *Let $W_1, \dots, W_n$ be independent random vectors in $\mathbb{R}^p$ such that $W_{ij} \ge 0$ for all $i = 1, \dots, n$ and $j = 1, \dots, p$. Define $U = \max_{j=1,\dots,p}\sum_{i=1}^n W_{ij}$ and $\Lambda = \max_{i=1,\dots,n}\max_{j=1,\dots,p} W_{ij}$. Then*

$$E(U) \le C\left\{\max_{j=1,\dots,p} E\left(\sum_{i=1}^n W_{ij}\right) + E(\Lambda)\log p\right\},$$

where C is a universal constant.

Proof. See Lemma 9 in Chernozhukov et al. (2015). □

LEMMA S22. *Assume the setting of Lemma S21. Then for every $\delta > 0, \beta \in (0, 1]$ and $t > 0$,*

$$\operatorname{pr}\{U \ge (1+\delta)E(U) + t\} \le 3\exp\left(-\left[t/\{C(\delta,\beta)\|\Lambda\|_{\psi_\beta}\}\right]^\beta\right),$$

where $C(\delta, \beta)$ is a constant depending only on δ and β.

Proof. See Lemma E.4. in Chernozhukov et al. (2017a). □

LEMMA S23. *For $a > 1$, $0 < \delta \leq 1$ and $b \geq e$, we have*

$$\int_0^\delta \{1 + a\log(b/z)\}^{1/2} dz \leq 2(2a)^{1/2}\delta\{\log(b/\delta)\}^{1/2}.$$

Proof. Note that

$$\int_0^\delta \{1 + a\log(b/z)\}^{1/2} dz \leq ba^{1/2}\int_{b/\delta}^\infty \frac{(1+\log z)^{1/2}}{z^2} dz.$$

Applying integration by parts yields

$$\begin{aligned}\int_{b/\delta}^\infty \frac{(1+\log z)^{1/2}}{z^2} dz &= -\frac{(1+\log z)^{1/2}}{z}\bigg|_{b/\delta}^\infty + \int_{b/\delta}^\infty \frac{1}{2z^2(1+\log z)^{1/2}} dz \\ &\leq \frac{\{1+\log(b/\delta)\}^{1/2}}{b/\delta} + \frac{1}{2}\int_{b/\delta}^\infty \frac{(1+\log z)^{1/2}}{z^2} dz, \quad \text{if} \quad b/\delta \geq 1.\end{aligned}$$

If $b/\delta \geq e$, then

$$\int_{b/\delta}^\infty \frac{(1+\log z)^{1/2}}{z^2} dz \leq \frac{2\{2\log(b/\delta)\}^{1/2}}{b/\delta}.$$

This completes the proof. □

REFERENCES

BISWAS, M. & GHOSH, A. K. (2014). A nonparametric two-sample test applicable to high dimensional data. *Journal of Multivariate Analysis* **123**, 160–171.

CHATTERJEE, S. & MECKES, E. (2008). Multivariate normal approximation using exchangeable pairs. *Alea* **4**, 257–283.

CHERNOZHUKOV, V., CHETVERIKOV, D. & KATO, K. (2014). Gaussian approximation of suprema of empirical processes. *The Annals of Statistics* **42**, 1564–1597.

CHERNOZHUKOV, V., CHETVERIKOV, D. & KATO, K. (2015). Comparison and anti-concentration bounds for maxima of Gaussian random vectors. *Probability Theory and Related Fields* **162**, 47–70.

CHERNOZHUKOV, V., CHETVERIKOV, D. & KATO, K. (2016). Empirical and multiplier bootstraps for suprema of empirical processes of increasing complexity, and related Gaussian couplings. *Stochastic Processes and their Applications* **126**, 3632–3651.

CHERNOZHUKOV, V., CHETVERIKOV, D. & KATO, K. (2017a). Central limit theorems and bootstrap in high dimensions. *The Annals of Probability* **45**, 2309–2352.

CHERNOZHUKOV, V., CHETVERIKOV, D. & KATO, K. (2017b). Detailed proof of Nazarov's inequality. *arXiv preprint arXiv:1711.10696* .

CHERNOZHUKOV, V., CHETVERIKOV, D., KATO, K. & KOIKE, Y. (2022). Improved central limit theorem and bootstrap approximations in high dimensions. *The Annals of Statistics* **50**, 2562–2586.

CHERNOZHUKOV, V., CHETVERIKOV, D. & KOIKE, Y. (2023). Nearly optimal central limit theorem and bootstrap approximations in high dimensions. *The Annals of Applied Probability* **33**, 2374–2425.

DONOHO, D. & JIN, J. (2004). Higher criticism for detecting sparse heterogeneous mixtures. *The Annals of Statistics* **32**, 962–994.

FANG, X. & KOIKE, Y. (2021). High-dimensional central limit theorems by Stein's method. *The Annals of Applied Probability* **31**, 1660–1686.

GAO, H. & SHAO, X. (2023). Two sample testing in high dimension via maximum mean discrepancy. *Journal of Machine Learning Research* **24**, 1–33.

HALL, P. & JIN, J. (2010). Innovated higher criticism for detecting sparse signals in correlated noise. *The Annals of Statistics* **38**, 1686–1732.

LIN, Z., LOPES, M. E. & MÜLLER, H.-G. (2023). High-dimensional manova via bootstrapping and its application to functional and sparse count data. *Journal of the American Statistical Association* **118**, 177–191.

LIU, H., WANG, H. & SONG, M. (2020). Projections onto the intersection of a one-norm ball or sphere and a two-norm ball or sphere. *Journal of Optimization Theory and Applications* **187**, 520–534.

LOPES, M. E., LIN, Z. & MÜLLER, H.-G. (2020). Bootstrapping max statistics in high dimensions: Near-parametric rates under weak variance decay and application to functional and multinomial data. *The Annals of Statistics* **48**, 1214–1229.

LOPES, M. E. & YAO, J. (2022). A sharp lower-tail bound for gaussian maxima with application to bootstrap methods in high dimensions. *Electronic Journal of Statistics* **16**, 58–83.

REINERT, G. & RÖLLIN, A. (2009). Multivariate normal approximation with Stein's method of exchangeable pairs under a general linearity condition. *The Annals of Probability* **37**, 2150–2173.

SADHU, R., GOLDFELD, Z. & KATO, K. (2022). Limit distribution theory for the smooth 1-Wasserstein distance with applications. *arXiv preprint arXiv:2107.13494* .

SEN, B. (2018). A gentle introduction to empirical process theory and applications. http://www.stat.columbia.edu/~bodhi/Talks/Emp-Proc-Lecture-Notes.pdf.

TSYBAKOV, A. B. (2009). Introduction to nonparametric estimation .

VAN DER VAART, A. W. & WELLNER, J. A. (1996). *Weak convergence and empirical processes: with applications to statistics*. Springer Science & Business Media.

VERSHYNIN, R. (2018). *High-dimensional probability: An introduction with applications in data science*. Cambridge University Press.

WAINWRIGHT, M. J. (2019). *High-dimensional statistics: A non-asymptotic viewpoint*. Cambridge University Press.

WANG, J., GAO, R. & XIE, Y. (2021). Two-sample test using projected Wasserstein distance. In *2021 IEEE International Symposium on Information Theory (ISIT)*. IEEE.

YAN, J. & ZHANG, X. (2023). Kernel two-sample tests in high dimensions: interplay between moment discrepancy and dimension-and-sample orders. *Biometrika* **110**, 411–430.

ZHOU, W.-X., ZHENG, C. & ZHANG, Z. (2017). Two-sample smooth tests for the equality of distributions. *Bernoulli* **23**, 951–989.

ZHU, C. & SHAO, X. (2021). Interpoint distance based two sample tests in high dimension. *Bernoulli* **27**, 1189–1211.